



The Independent Enterprise Network Infrastructure Handbook: Strategy, Architecture, and Execution.

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Project 1. Build a LAN with Switches and a Router.

Project Scope

You have been hired as a Junior Network technician for a small IT firm in the D.C. Metropolis. As the technician for the startup company, your initial responsibility is to design and configure the network infrastructure to support basic LAN IT operations for the organization. As part of your task, the network must meet the needs of at least 10 users with the users having access to resources hosted on the local Server and be able to print to the LAN printer. The organization has purchased a single block of Public IP addresses to enable LAN users to access the Internet.

Follow the instructions provided in the network documentation to set up the network according to the topology, and then perform the necessary configurations based on the stated objectives. Build strong Foundation for STEM Literacy; Increase Diversity, Equity, and Inclusion in the STEM workforce; Prepare the STEM workforce for the future:



When the configuration is complete, you will use various **show** commands to verify the configuration and use the **ping** command to verify basic connectivity between devices in the LAN and to the Web Server.

Network Requirements

1. Setup the LAN network with a single Cisco Router and five Cisco Layer 2 Switches and connect an appropriate number of devices using the right cable connections.
2. The LAN should be designed to accommodate ten (10) users, but with the potential to add more users.
3. The network should use the same Private IP subnet block for all users and devices.
4. Internal users should be able to access content on the Web Server hosted in the Cloud.
5. You are required to configure the external Web Server to allow only HTTPS request.
6. Internal users should have unrestricted access to local resources including the Printer and File Server.
7. Your organization's network infrastructure is built on the Cisco platform.
8. Client devices should be a mix of devices with support for only wired connection.

Report Contents.

The following items must be included in your report:

1. Project Scope
2. Network documentation.
3. Network Topology

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4. List of Devices names and IP address Table.
5. Configuration scripts.

Objectives

Part 1: Set up the LAN network as shown in the Topology.

Part 2: Configure the PCs and the servers with IP address and enable HTTPS on Web Server.

Part 3: Configure Basic Switch Settings on the Switches.

Part 4: Configure the switches virtual Interfaces for management.

Part 5: Configure router R1 with IP addresses for both the ingress and egress interfaces.

Part 6. Verify End-to-End Connectivity.

Network Topology

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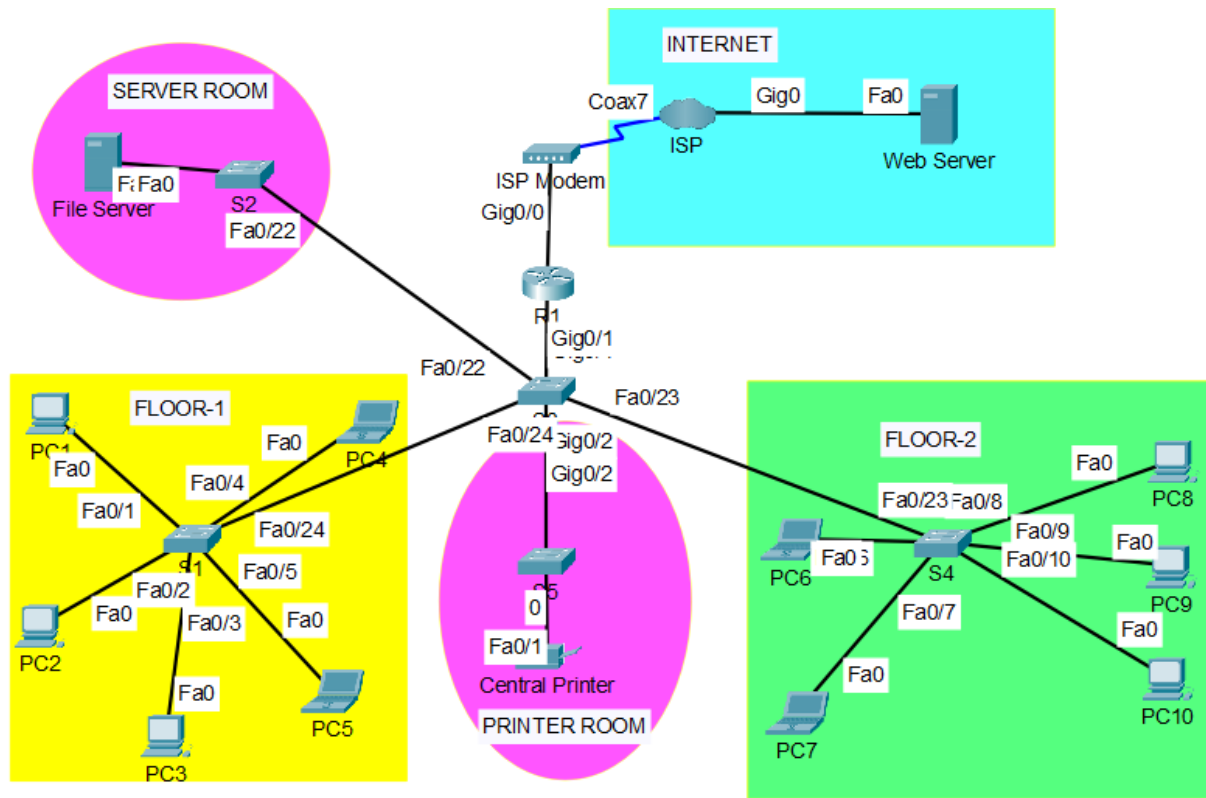


Figure 10.1: Build a LAN Network

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Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Connected To
R1	G0/0	209.165.200.2	255.255.255.240	209.165.200.1	ISP Modem Port 1
	G0/1	172.16.10.1	255.255.255.0	N/A	S3 Port G0/1
S1	VLAN 1	172.16.10.2	255.255.255.0	172.16.10.1	S3 Port Fa0/24
S2	VLAN 1	172.16.10.3	255.255.255.0	172.16.10.1	S3 Port Fa0/22
S3	VLAN 1	172.16.10.4	255.255.255.0	172.16.10.1	S1 Port Fa0/24
	VLAN 1	172.16.10.4	255.255.255.0	172.16.10.1	S2 Port Fa0/22
	VLAN 1	172.16.10.4	255.255.255.0	172.16.10.1	S4 Port Fa0/23
	VLAN 1	172.16.10.4	255.255.255.0	172.16.10.1	S5 Port Gig0/2
S4	VLAN 1	172.16.10.5	255.255.255.0	172.16.10.1	S3 Port Fa0/23
S5	VLAN 1	172.16.10.6	255.255.255.0	172.16.10.1	S3 Port Gig0/2
Web Server	NIC	209.165.200.14	255.255.255.240	209.165.200.1	ISP Port Gig0
ISP Modem	Port 1	N/A	N/A	N/A	R1 Port Gig0/0
	Coax0	N/A	N/A	N/A	ISP Cloud Coax7
ISP Cloud	Coax7	N/A	N/A	N/A	ISP Modem Coax0
	Gig0	N/A	N/A	N/A	Web Server Port Fa0
File Server	NIC	172.16.10.7	255.255.255.0	172.16.10.1	S2 Port Fa0/1
Printer	NIC	172.16.10.8	255.255.255.0	172.16.10.1	S5 Port Fa0/1
PC1	NIC	172.16.10.9	255.255.255.0	172.16.10.1	S1 Port Fa0/1
PC2	NIC	172.16.10.10	255.255.255.0	172.16.10.1	S1 Port Fa0/2
PC3	NIC	172.16.10.11	255.255.255.0	172.16.10.1	S1 Port Fa0/3
PC4	NIC	172.16.10.12	255.255.255.0	172.16.10.1	S1 Port Fa0/4
PC5	NIC	172.16.10.13	255.255.255.0	172.16.10.1	S1 Port Fa0/5
PC6	NIC	172.16.10.14	255.255.255.0	172.16.10.1	S4 Port Fa0/6
PC7	NIC	172.16.10.15	255.255.255.0	172.16.10.1	S4 Port Fa0/7
PC8	NIC	172.16.10.16	255.255.255.0	172.16.10.1	S4 Port Fa0/8
PC9	NIC	172.16.10.18	255.255.255.0	172.16.10.1	S4 Port Fa0/9
PC10	NIC	172.16.10.19	255.255.255.0	172.16.10.1	S4 Port Fa0/10

Figure 10.1: Addressing Table

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Instructions

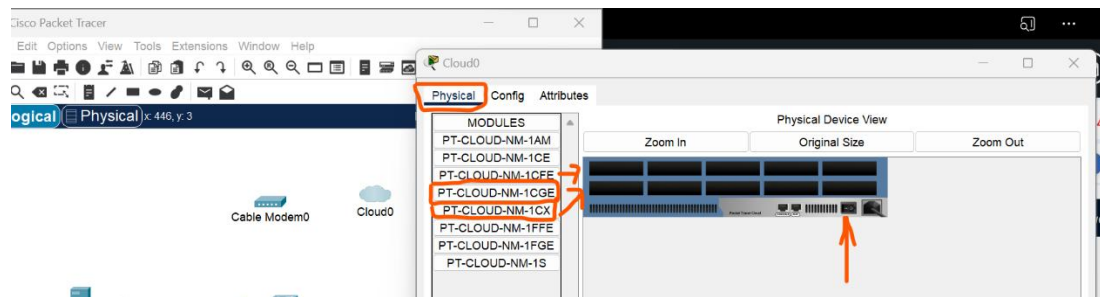
Part 1: Set up the LAN network as shown in the Topology.

Step 1: Select the appropriate devices, and cable connections and set up the network as shown in the topology.

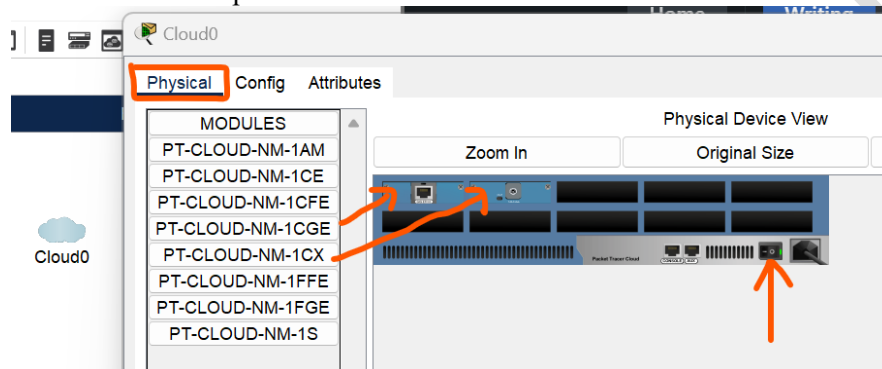
Step 2: Follow the instructions below to set up the Cloud device and Cable modem to establish a successful connection.

1. Click on Cloud, click on **Physical tab**, and power off the device.
2. Under the Physical tab, drag and drop the correct NIC modules to the available slots. (Read the modules description to help with your choice).
3. Also add **coaxial interface** to make provision for connecting to Cable modem. Then, power on the device.
4. Next, click on **Config tab**, and under the **Interface** option, select the correct interface connected to other device, and change the radio button to **Cable**.
5. Then under the **Connections**, select **Cable** option, use the drop-down arrow to select the correct **Coaxial port** that is connected to the Cable modem.
6. Finally, click the **Add** button to add the connection.

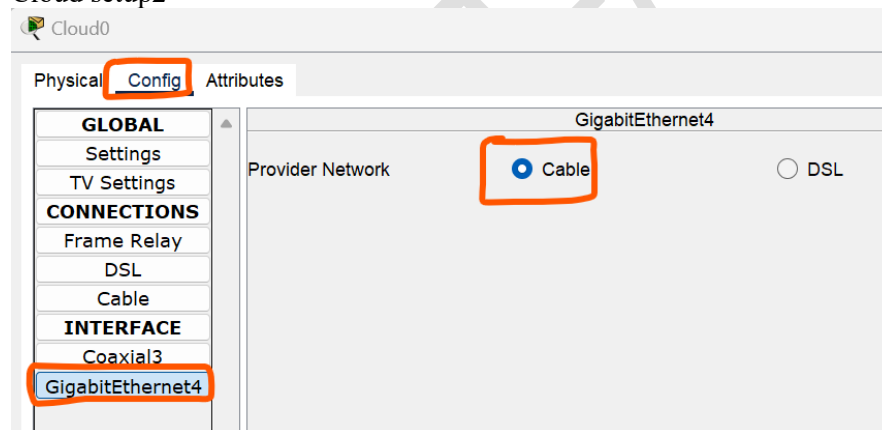
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Cloud modem setup1



Cloud setup2



Cloud setup3

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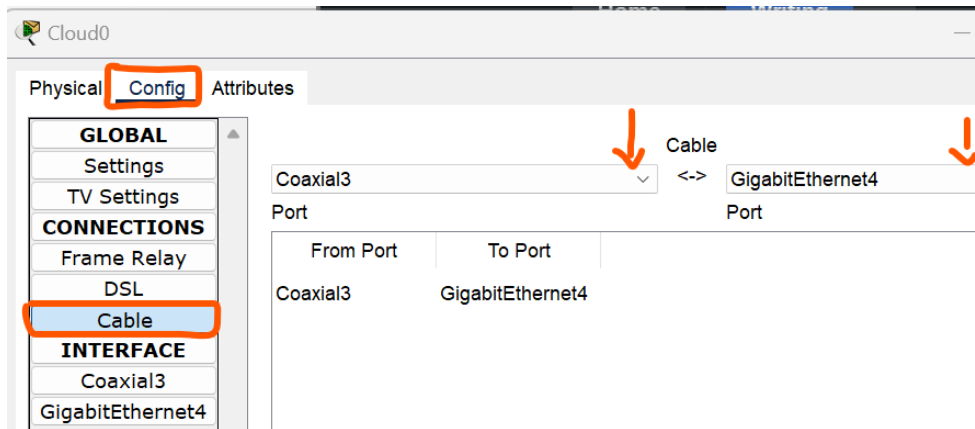
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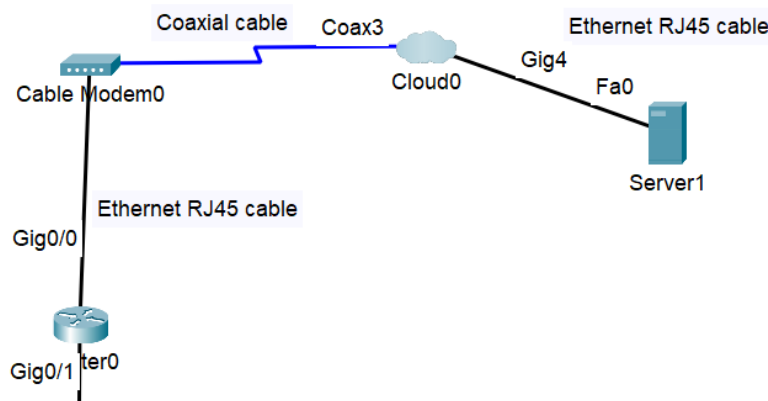
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Cloud setup4



Cloud setup5

Part 2: Configure the PCs and the servers with IP address.

Step 1. On PC1, open **Control Panel**. Select **Network and Sharing Center**. Select **Change Adapter Settings**. Right-click on **Ethernet** and select **Properties**. On the Properties page, click on **Internet Protocol Version 4 (TCP/IPv4)**, and select **Properties** tab. Click on the "Use the

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following IP address" radio button. Then enter the correct information for IP address, Subnet Mask and Default Gateway as shown in the Addressing Table. Then click OK to save settings.

Repeat this step to configure IP addresses on the other PCs and the File Server.

Note: If using Packet Tracer: On PC1 click **Desktop**, select **IP configuration**, select **Static**, and enter the information for IPv4 address, Subnet Mask and Default Gateway.

Note: Repeat this step to configure IP addresses on the other PCs, File Server and Web Server.

Step 2: If using Packet Tracer, perform the following on the Central Printer.

- Click the Central Printer, click on the **Config tab** and select the **Settings sub-tab**. Under **Global Settings**, select **Static** option, and enter the **Default Gateway** IP address.
- Under the **INTERFACE sub-tab**, select **FastEthernet**. Then select the Static option and enter the IP address and the Subnet Mask.

Step 3: Click on Web Server, and select the Services tab. Then select HTTP, turn off the HTTP radio button and make sure the HTTPS button is ON.

Part 3: Configure Basic Settings on the Switches.

Step 1: S1 configuration.

```
Switch>enable
```

```
Switch#config t
```

```
Switch(config)#hostname S1
```

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```
S1(config)#enable secret cisco1234
S1(config)#service password-encryption
S1(config)#line console 0
S1(config-line)#password console 1234
S1(config-line)#login
S1(config-line)#exit
S1(config)#line vty 0 15
S1(config-line)#password vty1234
S1(config-line)#login
S1(config-line)#exit
```

Step 2: S2 configuration.

```
Switch>
Switch>enable
Switch#config t
Switch(config)#hostname S2
S2(config)#enable secret cisco1234
S2(config)#service password-encryption
S2(config)#line cons 0
S2(config-line)#password console1234
S2(config-line)#login
```

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```
S2(config-line)#exit
```

```
S2(config)#line vty 0 15
```

```
S2(config-line)#password vty1234
```

```
S2(config-line)#login
```

```
S2(config-line)#exit
```

```
S2(config)#
```

Step 3: S3 configuration.

```
Switch>enable
```

```
Switch#config terminal
```

```
Switch(config)#hostname S3
```

```
S3(config)#enable secret cisco1234
```

```
S3(config)#service password-encryption
```

```
S3(config)#line console 0
```

```
S3(config-line)#password console1234
```

```
S3(config-line)#login
```

```
S3(config-line)#exit
```

```
S3(config)#line vty 0 15
```

```
S3(config-line)#password vty1234
```

```
S3(config-line)#login
```

```
S3(config-line)#exit
```

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Step 4: S4 configuration.

```
Switch>enable
Switch#config t
Switch(config)#hostname S4
S4(config)#enable secret cisco1234
S4(config)#service password-encryption
S4(config)#line console 0
S4(config-line)#password console1234
S4(config-line)#login
S4(config-line)#exit
S4(config)#line vty 0 15
S4(config-line)#password vty1234
S4(config-line)#login
S4(config-line)#exit
```

Step 5: S5 configuration.

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname S5
S5(config)#enable secret cisco1234
S5(config)#service password-encryption
```

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```
S5(config)#line cons 0
```

```
S5(config-line)#password console1234
```

```
S5(config-line)#login
```

```
S5(config-line)#exit
```

```
S5(config)#line vty 0 15
```

```
S5(config-line)#password vty1234
```

```
S5(config-line)#login
```

```
S5(config-line)#exit
```

Part 4: Configure the switches virtual Interfaces for management.

Step 1: Configure S1

```
S1(config)#int vlan 1
```

```
S1(config-if)#no shut
```

```
S1(config-if)#
```

```
S1(config-if)#ip address 172.16.10.2 255.255.255.0
```

```
S1(config-if)#exit
```

```
S1(config)#ip default-gateway 172.16.10.1
```

```
S1(config)#exit
```

```
S1#
```

```
S1#copy run start
```

```
Destination filename [startup-config]?
```

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Building configuration...

[OK]

S1#

Step 2: Configure S2

S2(config)#interface vlan 1

S2(config-if)#no shut

S2(config-if)#

S2(config-if)#ip address 172.16.10.3 255.255.255.0

S2(config-if)#exit

S2(config)#ip default-gateway 172.16.10.1

S2(config)#exit

S2#

S2#copy run start

Destination filename [startup-config]?

Building configuration...

[OK]

S2#

Step 3: Configure S3

S3(config)#int vlan 1

S3(config-if)#no shut

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```
S3(config-if)#
```

```
S3(config-if)#ip address 172.16.10.4 255.255.255.0
```

```
S3(config-if)#exit
```

```
S3(config)#ip default-gateway 172.16.10.1
```

```
S3(config)#exit
```

```
S3#
```

```
S3#copy run start
```

```
S3#
```

Step 4: Configure S4

```
S4(config)#int vlan 1
```

```
S4(config-if)#no shut
```

```
S4(config-if)#
```

```
S4(config-if)#ip address 172.16.10.5 255.255.255.0
```

```
S4(config-if)#exit
```

```
S4(config)#ip default-gateway 172.16.10.1
```

```
S4(config)#exit
```

```
S4#
```

```
S4#copy run start
```

```
S4#
```

Step 5: Configure S5

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```
S5(config)#int vlan 1
S5(config-if)#no shut
S5(config-if)#
S5(config-if)#ip address 172.16.10.6 255.255.255.0
S5(config-if)#exit
S5(config)#ip default-gateway 172.16.10.1
S5(config)#end
S5#
S5#copy run start
S5#
```

Part 5: Configure router R1 with IP addresses for both the ingress and egress interfaces.

```
Router>enable
Router#config t
Router(config)#hostname R1
R1(config)#enable secret cisco1234
R1(config)#service password-encryption
R1(config)#line console 0
R1(config-line)#password console1234
R1(config-line)#login
R1(config-line)#exit
```

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```
R1(config)#line vty 0 15
R1(config-line)#password vty1234
R1(config-line)#login
R1(config-line)#exit
R1(config)#interface g0/0
R1(config-if)#description CONNECTION TO ISP MODEM
R1(config-if)#ip address 209.165.200.2 255.255.255.240
R1(config-if)#no shut
R1(config-if)#
R1(config-if)#int g0/1
R1(config-if)#no shut
R1(config-if)#
R1(config-if)#description CONNECTION TO LAN Network
R1(config-if)#ip address 172.16.10.1 255.255.255.0
R1(config-if)#exit
R1(config)#exit
R1#
R1#copy run start
R1#
```

Part 6. Verify End-to-End Connectivity.

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Step 1: On PC 1, ping the IP address of PC2, PC5 and the File Server.

```

C:\>ping 172.16.10.10

Pinging 172.16.10.10 with 32 bytes of data:

Reply from 172.16.10.10: bytes=32 time<1ms TTL=128
Reply from 172.16.10.10: bytes=32 time<1ms TTL=128
Reply from 172.16.10.10: bytes=32 time<1ms TTL=128
Reply from 172.16.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.16.10.13

Pinging 172.16.10.13 with 32 bytes of data:

Reply from 172.16.10.13: bytes=32 time<1ms TTL=128
Reply from 172.16.10.13: bytes=32 time<1ms TTL=128
Reply from 172.16.10.13: bytes=32 time<1ms TTL=128
Reply from 172.16.10.13: bytes=32 time=39ms TTL=128

Ping statistics for 172.16.10.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 39ms, Average = 9ms

C:\>ping 172.16.10.7

Pinging 172.16.10.7 with 32 bytes of data:

Reply from 172.16.10.7: bytes=32 time<1ms TTL=128
Reply from 172.16.10.7: bytes=32 time<1ms TTL=128
Reply from 172.16.10.7: bytes=32 time<1ms TTL=128
Reply from 172.16.10.7: bytes=32 time<1ms TTL=128

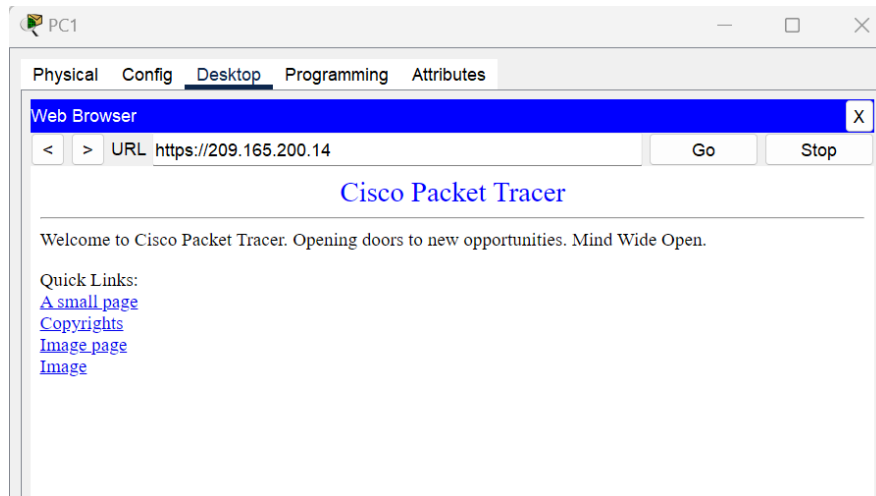
Ping statistics for 172.16.10.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
  
```

Step 2: On PC1, verify access to the Web Server page.

- Click on Desktop tab and select Web Browser Icon.
- On the URL page, enter the following: **https://209.165.200.14**

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Step 3: On PC10, ping the IP address of PC3, File Server and Central Printer

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```
PC10
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.10.11

Pinging 172.16.10.11 with 32 bytes of data:

Reply from 172.16.10.11: bytes=32 time=1ms TTL=128
Reply from 172.16.10.11: bytes=32 time<1ms TTL=128
Reply from 172.16.10.11: bytes=32 time<1ms TTL=128
Reply from 172.16.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 172.16.10.7

Pinging 172.16.10.7 with 32 bytes of data:

Reply from 172.16.10.7: bytes=32 time=1ms TTL=128
Reply from 172.16.10.7: bytes=32 time<1ms TTL=128
Reply from 172.16.10.7: bytes=32 time<1ms TTL=128
Reply from 172.16.10.7: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.10.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 172.16.10.8

Pinging 172.16.10.8 with 32 bytes of data:

Reply from 172.16.10.8: bytes=32 time<1ms TTL=128
Reply from 172.16.10.8: bytes=32 time<1ms TTL=128
Reply from 172.16.10.8: bytes=32 time<1ms TTL=128
Reply from 172.16.10.8: bytes=32 time<1ms TTL=128

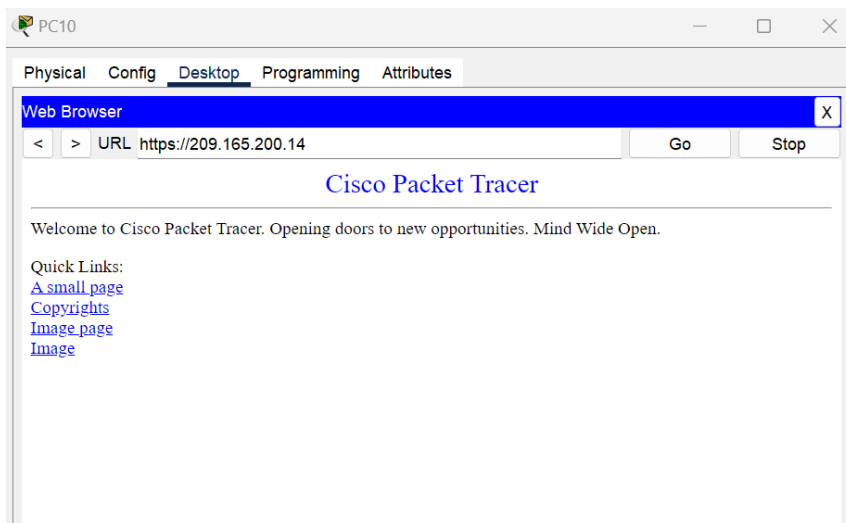
Ping statistics for 172.16.10.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Step 4: On PC10, verify access to the Web Server page.

- Click on Desktop tab and select Web Browser Icon.
- On the URL page, enter the following: **https://209.165.200.14**

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Project 2: Optimize performance for the LAN network with fault-tolerance, Layer 2 security and support for Remote Access.

Project Scope

You have been tasked to add more functionality to the LAN network. To achieve this goal, you will need to configure various Layer 2 features including Port Aggregation also known as EtherChannel on the switches to add fault-tolerant to the switched network to accommodate surge in traffic within the LAN due to users' constant requests to the File server and the Printer. Your

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organization wants you to implement additional password security to defeat/deter any attempted brute-force attacks on users' passwords. In addition, you must configure port security to control access to the LAN switches and prevent any form of LAN attacks. Lastly, you will configure a secure remote access for the newly hired network technician to be able to access the Router and LAN devices remotely.

Network Requirements

1. Setup the LAN network with one Cisco Router, one Multilayer switch and four Cisco Layer 2 Switches and connect an appropriate number of devices using the right cable connections. The ISP network topology is used to simulate access to the Internet.
2. The LAN should be designed to accommodate ten (10) users with the potential to add more users. The network should use the same Private IP subnet block for all users and devices and Internal users should be able to access content on the Web Server hosted in the Cloud using the secure protocol.
3. The LAN Edge router and switches must be configured with passwords security.
4. Edge router must be configured for secure remote access with accommodation for 2 remote workers.
5. All LAN switches must be configured with various layer 2 security features.
6. EtherChannel should be enabled on connections between switches to accommodate increase in LAN traffic.

Report Contents.

The following items must be included in your report:

Build strong Foundation for STEM Literacy; Increase Diversity, Equity, and Inclusion in the STEM workforce; Prepare the STEM workforce for the future:



1. Project Scope
2. Network documentation.
3. Network Topology
4. Devices names, IP addresses and Connection Ports.
5. Configuration scripts.

Objectives

Part 1: Set up the LAN network as shown in the Topology.

Part 2: Configure the PCs, the Printer, and the servers with IP address.

Part 3: Configure Basic Settings on the Switches.

Part 4: Configure Static Trunk ports and manually configure connections to endpoints as Static Access.

Part 5: Configure EtherChannel using IEEE 802.3ad LACP (Link Aggregation Protocol) and Cisco PAGP (Port Aggregation Protocol) on the physical connections between the switches.

Part 6: Configure Rapid Per-VLAN Spanning-Tree IEEE 802.1W on all switches.

Part 7: Configure the Distribution switch as the Root Bridge for VLAN 1 (default VLAN).

Part 8: Configure Port Security and additional Layer 2 security features on Switches and disable any unused ports.

Part 9: Configure the switches virtual Interfaces for remote management.

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Part 10. Configure basic settings on Edge router including password security settings.

Part 11: Configure IP address on Edge router and enable SSH remote access for two teleworkers.

Part 12: Verify end-to-end connectivity and establish SSH connection over the Internet from Remote1 and Remote 2 to Edge router.

Network Topology

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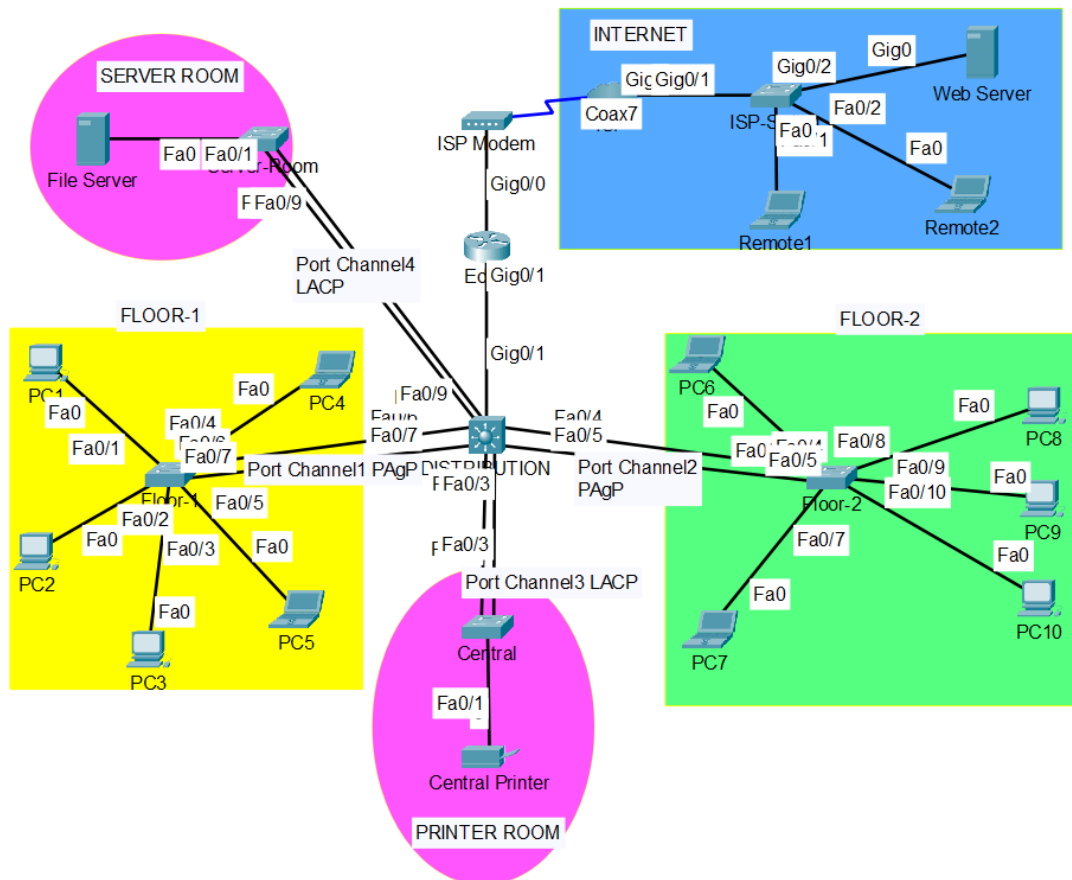


Figure 10.2: Lab 2 Network Topology

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Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Connected To
Edge	G0/0	209.165.200.1	255.255.255.240	N/A	ISP Modem Port 1
	G0/1	192.168.10.1	255.255.255.0	N/A	Distribution Port Gig0/1
Floor-1	VLAN 1	192.168.10.2	255.255.255.0	192.168.10.1	Distribution Port Fa0/6-7
Floor-2	VLAN 1	192.168.10.3	255.255.255.0	192.168.10.1	Distribution Port Fa0/4-5
Distribution	VLAN 1	192.168.10.4	255.255.255.0	192.168.10.1	Edge Port Gig0/1
	VLAN 1	192.168.10.4	255.255.255.0	192.168.10.1	Floor-1 Port Fa0/6-7
	VLAN 1	192.168.10.4	255.255.255.0	192.168.10.1	Floor-2 Port Fa0/4-5
	VLAN 1	192.168.10.4	255.255.255.0	192.168.10.1	Central Port Fa0/2-3
	VLAN 1	192.168.10.4	255.255.255.0	192.168.10.1	Server-Room Port Fa0/8-9
Server-Room	VLAN 1	192.168.10.5	255.255.255.0	192.168.10.1	Distribution Port Fa0/8-9
Central	VLAN 1	192.168.10.6	255.255.255.0	192.168.10.1	Distribution Port Fa0/2-3
Web Server	NIC	209.165.200.2	255.255.255.240	209.165.200.1	ISP Cloud Port Gig0
ISP Modem	Port 1	N/A	N/A	N/A	Edge Port Gig0/0
	Coax0	N/A	N/A	N/A	ISP Cloud Coax7
ISP Cloud	Coax7	N/A	N/A	N/A	ISP Modem Coax0
	Gig0	N/A	N/A	N/A	ISP-Switch Port Gig0/1

Table 10.2: Lab 2 Addressing Table 1

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Device	Interface	IP Address	Subnet Mask	Default Gateway	Connected To
Central Printer	NIC	192.168.10.8	255.255.255.0	192.168.10.1	Central Port Fa0/1
File Server	NIC	192.168.10.7	255.255.255.0	192.168.10.1	Server-Room Port Fa0/1
PC1	NIC	192.168.10.9	255.255.255.0	192.168.10.1	Floor-1 Port Fa0/1
PC2	NIC	192.168.10.10	255.255.255.0	192.168.10.1	Floor-1 Port Fa0/2
PC3	NIC	192.168.10.11	255.255.255.0	192.168.10.1	Floor-1 Port Fa0/3
PC4	NIC	192.168.10.12	255.255.255.0	192.168.10.1	Floor-1 Port Fa0/4
PC5	NIC	192.168.10.13	255.255.255.0	192.168.10.1	Floor-1 Port Fa0/5
PC6	NIC	192.168.10.14	255.255.255.0	192.168.10.1	Floor-2 Port Fa0/6
PC7	NIC	192.168.10.15	255.255.255.0	192.168.10.1	Floor-2 Port Fa0/7
PC8	NIC	192.168.10.16	255.255.255.0	192.168.10.1	Floor-2 Port Fa0/8
PC9	NIC	192.168.10.17	255.255.255.0	192.168.10.1	Floor-2 Port Fa0/9
PC10	NIC	192.168.10.18	255.255.255.0	192.168.10.1	Floor-2 Port Fa0/10
Remote1	NIC	209.165.200.3	255.255.255.240	209.165.200.1	ISP Cloud Port Gig1
Remote2	NIC	209.165.200.4	255.255.255.240	209.165.200.1	ISP Cloud Port Gig2
ISP-Switch	Gig0/1	N/A	N/A	N/A	ISP-Cloud Port Gig0
	Fa0/1	N/A	N/A	N/A	Remote1 Port Fa0
	Fa0/2	N/A	N/A	N/A	Remote2 Port Fa0
	Gig0/2	N/A	N/A	N/A	Web Server Port Gig0

Table 10.3: Lab 2 Addressing Table 2

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EtherChannel Table

Port Channel/Group	Devices	Port Connections	Type
1	Floor-1 to Distribution	Fa0/6 to Fa0/6	PAgP
		Fa0/7 to Fa0/7	
2	Floor-2 to Distribution	Fa0/4 to Fa0/4	PAgP
		Fa0/5 to Fa0/5	
3	Central to Distribution	Fa0/2 to Fa0/2	LACP
		Fa0/3 to Fa0/3	
4	Server-Room to Distribution	Fa0/8 to Fa0/8	LACP
		Fa0/9 to Fa0/9	

Table 10.4: EtherChannel Table

Instructions

Part 1: Set up the LAN network as shown in the Topology.

Part 2: Configure the PCs, the Printer and the servers with IP address.

Step 1. On PC1, open **Control Panel**. Select **Network and Sharing Center**. Select **Change Adapter Settings**. Right-click on **Ethernet** and select **Properties**. On the Properties page, click on **Internet Protocol Version 4 (TCP/IPv4)**, and select **Properties** tab. Click on the "Use the following IP address" radio button. Then enter the correct information for IP address, Subnet Mask and Default Gateway as shown in the Addressing Table. Then click OK to save settings.

Repeat this step to configure IP addresses on the other PCs, the File Server and Web Server.

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Note: If using Packet Tracer: On PC1 click **Desktop**, select **IP configuration**, select **Static**, and enter the information for IPv4 address, Subnet Mask and Default Gateway.

Note: Repeat this step to configure IP addresses on the other PCs, File Server and Web Server.

Step 2: If using Packet Tracer, perform the following on the Central Printer.

- Click the Central Printer, click on the **Config tab** and select the **Settings sub-tab**. Under **Global Settings**, select **Static** option, and enter the **Default Gateway** IP address.
- Under the **INTERFACE sub-tab**, select **FastEthernet**. Then select the Static option and enter the IP address and the Subnet Mask.

Step 3: Click on Web Server, and select the Services tab. Then select HTTP, turn off the HTTP radio button and make sure the HTTPS button is ON.

Part 3: Configure Basic Settings on the Switches.

Step 1: Floor-1 Configuration.

```
Switch>en
Switch#conf t
Switch(config)#hostname Floor-1
Floor-1(config)#no ip domain lookup
Floor-1(config)#enable secret cisco1234
Floor-1(config)#service password-encryption
Floor-1(config)#line console 0
```

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```
Floor-1(config-line)#password console1234
```

```
Floor-1(config-line)#login
```

```
Floor-1(config-line)#exit
```

```
Floor-1(config)#line vty 0 15
```

```
Floor-1(config-line)#password vty1234
```

```
Floor-1(config-line)#login
```

```
Floor-1(config-line)#exit
```

```
Floor-1(config)#line console 0
```

```
Floor-1(config-line)#exec-timeout 0 120
```

```
Floor-1(config-line)#exit
```

```
Floor-1(config)#line vty 0 15
```

```
Floor-1(config-line)#exec-timeout 2
```

```
Floor-1(config-line)#exit
```

```
Floor-1(config)#exit
```

```
Floor-1#
```

```
Floor-1#clock set 06:27:01 Dec 28 2023
```

```
Floor-1#copy run start
```

```
Floor-1#
```

Step 2: Floor-2 Configuration.

```
Switch>enable
```

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```
Switch#config t
```

```
Switch(config)#hostname Floor-2
```

```
Floor-2(config)#no ip domain lookup
```

```
Floor-2(config)#enable secret cisco1234
```

```
Floor-2(config)#service password-encryption
```

```
Floor-2(config)#line console 0
```

```
Floor-2(config-line)#password console1234
```

```
Floor-2(config-line)#login
```

```
Floor-2(config-line)#exec-timeout 0 120
```

```
Floor-2(config-line)#exit
```

```
Floor-2(config)#line vty 0 15
```

```
Floor-2(config-line)#password vty1234
```

```
Floor-2(config-line)#login
```

```
Floor-2(config-line)#exec-timeout 2
```

```
Floor-2(config-line)#exit
```

```
Floor-2(config)#exit
```

```
Floor-2#
```

```
Floor-2#clock set 06:27:01 Dec 28 2023
```

```
Floor-2#copy run start
```

```
Floor-2#
```

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Step 3: Central Configuration.

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname Central
Central(config)#no ip domain lookup
Central(config)#enable secret cisco1234
Central(config)#service password-encryption
Central(config)#line console 0
Central(config-line)#password console1234
Central(config-line)#login
Central(config-line)#exit
Central(config)#line vty 0 15
Central(config-line)#password vty1234
Central(config-line)#login
Central(config-line)#exec-timeout 2
Central(config-line)#exit
Central(config)#line console 0
Central(config-line)#exec-timeout 0 120
Central(config-line)#exit
Central(config)#exit
```

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Central#

Central#copy run start

Central#

Step 4: Distribution Configuration

Switch>enable

Switch#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#no ip domain lookup

Switch(config)#hostname DISTRIBUTION

DISTRIBUTION(config)#enable secret cisco1234

DISTRIBUTION(config)#service password-encryption

DISTRIBUTION(config)#line conso 0

DISTRIBUTION(config-line)#password console1234

DISTRIBUTION(config-line)#login

DISTRIBUTION(config-line)#exec 0 120

DISTRIBUTION(config-line)#exit

DISTRIBUTION(config)#line vty 0 15

DISTRIBUTION(config-line)#passw vty1234

DISTRIBUTION(config-line)#login

DISTRIBUTION(config-line)#exec 2

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```
DISTRIBUTION(config-line)#end
```

```
DISTRIBUTION#
```

```
DISTRIBUTION#clock set 06:43:10 Dec 28 2023
```

```
DISTRIBUTION#copy run start
```

```
DISTRIBUTION#
```

Step 5: Server-Room Configuration.

```
Switch>enable
```

```
Switch#conf t
```

```
Switch(config)#no ip domain look
```

```
Switch(config)#host Server-Room
```

```
Server-Room(config)#enable secret cisco1234
```

```
Server-Room(config)#service password-encryption
```

```
Server-Room(config)#line cons 0
```

```
Server-Room(config-line)#password console1234
```

```
Server-Room(config-line)#login
```

```
Server-Room(config-line)#exec 0 120
```

```
Server-Room(config-line)#exit
```

```
Server-Room(config)#line vty 0 15
```

```
Server-Room(config-line)#pass vty1234
```

```
Server-Room(config-line)#login
```

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```
Server-Room(config-line)#exec 2
```

```
Server-Room(config-line)#end
```

```
Server-Room#
```

```
Server-Room#clock set 06:46:00 Dec 28 2023
```

```
Server-Room#copy run start
```

```
Server-Room#
```

Part 4: Configure Static Trunk ports and manually configure connections to endpoints as Static Access.

Step 1: Floor-1 Configuration

```
loor-1#conf t
```

```
Floor-1(config)#interface range f0/6-7
```

```
Floor-1(config-if-range)#switchport mode trunk
```

```
Floor-1(config-if-range)#
```

```
Floor-1(config-if-range)#switchport nonegotiate
```

```
Floor-1(config-if-range)#int range f0/1-5
```

```
Floor-1(config-if-range)#switchport mode access
```

```
Floor-1(config-if-range)#end
```

```
Floor-1#
```

```
Floor-1#copy run start
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

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[OK]

Floor-1#

Step 2: Floor-2 Configuration.

Floor-2#conf t

Floor-2(config)#int range f0/4-5

Floor-2(config-if-range)#switchport mode trunk

Floor-2(config-if-range)#

Floor-2(config-if-range)#int range f0/6-10

Floor-2(config-if-range)#switchport mode access

Floor-2(config-if-range)#int range f0/4-5

Floor-2(config-if-range)#switchport nonegotiate

Floor-2(config-if-range)#end

Floor-2#

Floor-2#copy run start

Floor-2#

Step 3: Central Configuration.

Central#conf t

Central(config)#int range f0/2-3

Central(config-if-range)#switch mode trunk

Central(config-if-range)#

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```
Central(config-if-range)#switch nonegotiate
```

```
Central(config-if-range)#int f0/1
```

```
Central(config-if)#swi mode access
```

```
Central(config-if)#end
```

```
Central#
```

```
Central#copy run start
```

```
Central#
```

Step 4: Server-Room Configuration.

```
Server-Room#conf t
```

```
Server-Room(config)#int range f0/8-9
```

```
Server-Room(config-if-range)#swi mode trunk
```

```
Server-Room(config-if-range)#
```

```
Server-Room(config-if-range)#swi nonegotiate
```

```
Server-Room(config-if-range)#int fa0/1
```

```
Server-Room(config-if)#swi mode access
```

```
Server-Room(config-if)#end
```

```
Server-Room#
```

```
Server-Room#copy run start
```

```
Server-Room#
```

Step 5: Distribution Configuration

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Note: (On a Layer 3 Switch, the interfaces are set to trunk dynamically. To configure a Static Trunk link, you will first set the interface to dynamic desirable, then set it to static trunk. See the below:).

```
DISTRIBUTION#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
DISTRIBUTION(config)#int range f0/2-9
```

```
DISTRIBUTION(config-if-range)#swi mode trunk
```

Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk" mode.

```
<cr>
```

```
DISTRIBUTION(config-if-range)#switchport mode dynamic desirable
```

```
DISTRIBUTION(config-if-range)#
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up
```

```
DISTRIBUTION(config-if-range)#switchport mode trunk
```

```
DISTRIBUTION(config-if-range)#
```

```
DISTRIBUTION(config-if-range)#switchport nonegotiate
```

...Use the show ip interface fa0/9 switchport command, to verify that the interface is now set to static trunk...

```
DISTRIBUTION(config-if-range)#do show int f0/9 switchport
```

```
Name: Fa0/9
```

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Switchport: **Enabled**

Administrative Mode: **trunk**

Operational Mode: **trunk**

Administrative Trunking Encapsulation: negotiated

Operational Trunking Encapsulation: dot1q

Negotiation of Trunking: **Off**

Access Mode VLAN: 1 (default)

Trunking Native Mode VLAN: 1 (default)

Voice VLAN: none

DISTRIBUTION(config-if-range)#int g0/1

DISTRIBUTION(config-if)#swi mode dynamic desirable

DISTRIBUTION(config-if)#

DISTRIBUTION(config-if)#swi mode trunk

DISTRIBUTION(config-if)#

DISTRIBUTION(config-if)#switchport nonegotiate

DISTRIBUTION(config-if)#end

DISTRIBUTION#

DISTRIBUTION#copy run start

Destination filename [startup-config]?

Building configuration...

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[OK]

DISTRIBUTION#

Part 5: Configure EtherChannel using IEEE 802.3ad LACP (Link Aggregation Control Protocol) and Cisco PAgP (Port Aggregation Protocol) on the physical connections between the switches.

Step 1a: Cisco PAgP configuration between Distribution switch and Floor-1 switch.

- On Floor-1 and Distribution, add ports Fa0/6 and Fa0/7 to Port Channel 1 with the channel-group 1 mode desirable command. The mode desirable option enables the switch to actively negotiate to form a PAgP link.
- Note: Interfaces must be shut down before adding them to the channel group.
- Next, configure the logical interface to become a trunk by first entering the interface port-channel number command and then the switchport mode trunk command. Add this configuration to both switches.

...Part 1...

```
DISTRIBUTION#conf t
```

```
DISTRIBUTION(config)#int range f0/6-7
```

```
DISTRIBUTION(config-if-range)#shut
```

```
DISTRIBUTION(config-if-range)#
```

```
DISTRIBUTION(config-if-range)#channel-group 1 mode desirable
```

```
DISTRIBUTION(config-if-range)#
```

```
Creating a port-channel interface Port-channel 1 DISTRIBUTION(config-if-range)#no shutdown
```

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...Part 2...

```
DISTRIBUTION(config)#interface port-channel 1
```

```
DISTRIBUTION(config-if)#switchport mode trunk
```

Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk" mode.

```
DISTRIBUTION(config-if)#switchport mode dynamic desirable
```

```
DISTRIBUTION(config-if)#switchport mode trunk
```

```
DISTRIBUTION(config-if)#
```

...Use the "**show etherchannel summary**" command to verify that ports Fa0/6 and Fa0/7 are now operating as etherchannel bundle ports...

```
DISTRIBUTION#show etherchannel summary
```

Flags: D - down P - in port-channel

Number of channel-groups in use: 1

Number of aggregators: 1

Group Port-channel Protocol Ports

```
-----+-----+-----+-----
```

```
1 Po1(SU) PAgP Fa0/6(P) Fa0/7(P)
```

```
DISTRIBUTION#
```

Step 1b: Cisco PAgP configuration between Distribution switch and Floor-1 switch.

- On Floor-1 and Distribution, add ports Fa0/6 and Fa0/7 to Port Channel 1 with the channel-group 1 mode desirable command. The mode desirable option enables the switch to

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actively negotiate to form a PAgP link. Note: Interfaces must be shut down before adding them to the channel group.

- Next, configure the logical interface to become a trunk by first entering the interface port-channel number command and then the switchport mode trunk command. Add this configuration to both switches.

```
Floor-1#conf t
```

```
Floor-1(config)#int range f0/6-7
```

```
Floor-1(config-if-range)#shut
```

```
Floor-1(config-if-range)#
```

```
Floor-1(config-if-range)#channel-group 1 mode desirable
```

```
Floor-1(config-if-range)#
```

```
Creating a port-channel interface Port-channel 1
```

```
Floor-1(config-if-range)#no shut Floor-1(config-if-range)#
```

```
Floor-1(config-if-range)#int port-channel 1
```

```
Floor-1(config-if)#swi mode trunk
```

```
Floor-1(config-if)#end
```

```
Floor-1#
```

```
Floor-1#copy run start
```

```
Floor-1#
```

...Use the "**show etherchannel summary**" command to verify success configuration...

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Floor-1#show etherchannel summary

Number of channel-groups in use: 1

Number of aggregators: 1

Group Port-channel Protocol Ports

-----+-----+-----+-----

1 Po1(SU) PAgP Fa0/6(P) Fa0/7(P)

Floor-1#

Step 2a: Cisco PAgP configuration between Distribution switch and Floor-2 switch.

- On Floor-2 and Distribution switches, add ports Fa0/4 and Fa0/5 to Port Channel 2 with the channel-group 2 mode desirable command. The mode desirable option enables the switch to actively negotiate to form a PAgP link. Note: Interfaces must be shut down before adding them to the channel group.
- Next, configure the logical interface to become a trunk by first entering the interface port-channel number command and then the switchport mode trunk command. Add this configuration to both switches.

DISTRIBUTION#conf t

DISTRIBUTION(config)#int range f0/4-5

DISTRIBUTION(config-if-range)#shut

DISTRIBUTION(config-if-range)#

DISTRIBUTION(config-if-range)#channel-group 2 mode desirable

DISTRIBUTION(config-if-range)#

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Creating a port-channel interface Port-channel 2

```
DISTRIBUTION(config-if-range)#no shutdown
```

```
DISTRIBUTION(config-if-range)#int port-channel 2
```

```
DISTRIBUTION(config-if)#switchport mode dynamic desirable
```

```
DISTRIBUTION(config-if)#switchport mode trunk
```

```
DISTRIBUTION(config-if)#end
```

```
DISTRIBUTION#
```

```
DISTRIBUTION#copy run start
```

```
Destination filename [startup-config]?
```

Building configuration...

[OK]

```
DISTRIBUTION#show etherchannel summary
```

Flags: D - down P - in port-channel

Number of channel-groups in use: 2

Number of aggregators: 2

Group Port-channel Protocol Ports

-----+-----+-----+-----

1 Po1(SU) PAgP Fa0/6(P) Fa0/7(P)

2 Po2(SU) PAgP Fa0/4(P) Fa0/5(P)

```
DISTRIBUTION#
```

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Step 2b: Cisco PAgP configuration between Distribution switch and Floor-2 switch.

```
Floor-2#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Floor-2(config)#int range f0/4-5
```

```
Floor-2(config-if-range)#shut
```

```
Floor-2(config-if-range)#channel-group 2 mode desirable
```

```
Floor-2(config-if-range)#
```

Creating a port-channel interface Port-channel 2

```
Floor-2(config-if-range)#no shut
```

```
Floor-2(config-if-range)#int port-channel 2
```

```
Floor-2(config-if)#swi mode trunk
```

```
Floor-2(config-if)#end
```

```
Floor-2#
```

```
Floor-2#copy run start
```

```
Floor-2#show etherchannel summary
```

Flags: D - down P - in port-channel

I - stand-alone s - suspended

H - Hot-standby (LACP only)

R - Layer3 S - Layer2

Number of channel-groups in use: 1

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Number of aggregators: 1

Group Port-channel Protocol Ports

-----+-----+-----+-----

2 Po2(SU) PAgP Fa0/4(P) Fa0/5(P)

Floor-2#

Step 3a: IEEE LACP configuration between Distribution switch and Central switch.

- On the Central switch and Distribution switch, add ports Fa0/2 and Fa0/3 to Port Channel 3 with the channel-group 3 mode active command. The mode active option enables the switch to actively negotiate to form a link. Note: Interfaces must be shut down before adding them to the channel group.
- Next, configure the logical interface to become a trunk by first entering the interface port-channel number command and then the switchport mode trunk command. Add this configuration to both switches.

DISTRIBUTION#conf t

DISTRIBUTION(config)#int range f0/2-3

DISTRIBUTION(config-if-range)#shut

DISTRIBUTION(config-if-range)#

DISTRIBUTION(config-if-range)#channel-group 3 mode active

DISTRIBUTION(config-if-range)#

Creating a port-channel interface Port-channel 3

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```
DISTRIBUTION(config-if-range)#no shut
```

```
DISTRIBUTION(config-if-range)#
```

```
DISTRIBUTION(config-if-range)#int port-channel 3
```

```
DISTRIBUTION(config-if)#switchport mode dynamic desirable
```

```
DISTRIBUTION(config-if)#
```

```
DISTRIBUTION(config-if)#switchport mode trunk
```

```
DISTRIBUTION(config-if)#end
```

```
DISTRIBUTION#
```

```
DISTRIBUTION#copy run start
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

```
[OK]
```

```
DISTRIBUTION#
```

```
DISTRIBUTION#show etherchannel summary
```

```
Flags: D - down P - in port-channel
```

```
I - stand-alone s - suspended
```

```
H - Hot-standby (LACP only)
```

```
R - Layer3 S - Layer2
```

```
Number of channel-groups in use: 3
```

```
Number of aggregators: 3
```

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Group Port-channel Protocol Ports

-----+-----+-----+-----

1 Po1(SU) PAgP Fa0/6(P) Fa0/7(P)

2 Po2(SU) PAgP Fa0/4(P) Fa0/5(P)

3 Po3(SU) LACP Fa0/2(P) Fa0/3(P)

DISTRIBUTION#

Step 3b: IEEE LACP configuration between Distribution switch and Central switch.

Central#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Central(config)#int range f0/2-3

Central(config-if-range)#shut

Central(config-if-range)#

Central(config-if-range)#channel-group 3 mode active

Central(config-if-range)#

Creating a port-channel interface Port-channel 3

Central(config-if-range)#no shut

Central(config-if-range)#

Central(config-if-range)#int port-channel 3

Central(config-if)#swi mode trunk

Central(config-if)#end

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Central#

Central#copy run start

Destination filename [startup-config]?

Building configuration...

[OK]

Central#sho etherchannel summary

Flags: D - down P - in port-channel

I - stand-alone s - suspended

H - Hot-standby (LACP only)

R - Layer3 S - Layer2

Number of channel-groups in use: 1

Number of aggregators: 1

Group Port-channel Protocol Ports

-----+-----+-----+-----

3 Po3(SU) LACP Fa0/2(P) Fa0/3(P)

Central#

Step 4a: IEEE LACP configuration between Distribution switch and Server-Room switch.

- On Server-Room switch and Distribution switch, add ports Fa0/8 and Fa0/9 to Port Channel 4 with the channel-group 4 mode active command. The mode active option

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enables the switch to actively negotiate to form a LACP link. Note: Interfaces must be shut down before adding them to the channel group.

- Next, configure the logical interface to become a trunk by first entering the interface port-channel number command and then the switchport mode trunk command. Add this configuration to both switches.

```
DISTRIBUTION#conf t
DISTRIBUTION(config)#int range f0/8-9
DISTRIBUTION(config-if-range)#shut
DISTRIBUTION(config-if-range)#
DISTRIBUTION(config-if-range)#channel-group 4 mode active
DISTRIBUTION(config-if-range)#
Creating a port-channel interface Port-channel 4
DISTRIBUTION(config-if-range)#no shutdown
DISTRIBUTION(config-if-range)#
DISTRIBUTION(config-if-range)#int port-channel 4
DISTRIBUTION(config-if)#switchport mode dynamic desirable
DISTRIBUTION(config-if)#
DISTRIBUTION(config-if)#switchport mode trunk
DISTRIBUTION(config-if)#int port-channel 4
DISTRIBUTION(config-if)#end
```

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DISTRIBUTION#

DISTRIBUTION#copy run start

Destination filename [startup-config]?

Building configuration...

[OK]

DISTRIBUTION#show etherchannel summary

number of channel-groups in use: 4

Number of aggregators: 4

Group Port-channel Protocol Ports

-----+-----+-----+-----

1 Po1(SU) PAgP Fa0/6(P) Fa0/7(P)

2 Po2(SU) PAgP Fa0/4(P) Fa0/5(P)

3 Po3(SU) LACP Fa0/2(P) Fa0/3(P)

4 Po4(SU) LACP Fa0/8(P) Fa0/9(P)

DISTRIBUTION#

Step 4b: IEEE LACP configuration between Distribution switch and Server-Room switch.

Server-Room#conf t

Server-Room(config)#int range f0/8-9

Server-Room(config-if-range)#shut

Server-Room(config-if-range)#

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```
Server-Room(config-if-range)#channel-group 4 mode active
```

```
Server-Room(config-if-range)#
```

Creating a port-channel interface Port-channel 4

```
Server-Room(config-if-range)#no shut
```

```
Server-Room(config-if-range)#
```

```
Server-Room(config-if-range)#int port-channel 4
```

```
Server-Room(config-if)#switchport mode trunk
```

```
Server-Room(config-if)#end
```

```
Server-Room#
```

```
Server-Room#show etherchannel summary
```

...Output cut...

Number of channel-groups in use: 1

Number of aggregators: 1

Group Port-channel Protocol Ports

-----+-----+-----+-----

4 Po4(SU) LACP Fa0/8(P) Fa0/9(P)

```
Server-Room#
```

Part 6: Configure Rapid Per-VLAN Spanning-Tree IEEE 802.1W on all switches.

Step 1. Floor-1 Configuration.

```
Floor-1(config)#spanning-tree mode rapid-pvst
```

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```
Floor-1(config)#end
```

```
Floor-1#
```

...Use the "**show spanning-tree**" command to verify that Rapid PVST is enabled as shown below...

```
Floor-1#show spanning-tree
```

```
VLAN0001
```

```
Spanning tree enabled protocol rstp
```

```
Root ID Priority 32769
```

```
Address 0001.6408.81C1
```

```
Cost 24
```

```
Port 27(Port-channel1)
```

```
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
```

```
Address 0090.2B2A.A329
```

```
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Aging Time 20
```

```
Interface Role Sts Cost Prio.Nbr Type
```

```
-----
```

```
Fa0/2 Desg FWD 19 128.2 P2p
```

```
Fa0/4 Desg FWD 19 128.4 P2p
```

```
Fa0/3 Desg FWD 19 128.3 P2p
```

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Fa0/1 Desg FWD 19 128.1 P2p

Fa0/5 Desg FWD 19 128.5 P2p

Po1 Root LRN 12 128.27 Shr

Floor-1#

Step 2: Floor-2 Configuration.

Floor-2#conf t

Floor-2(config)#spanning-tree mode rapid-pvst

Floor-2(config)#end

Floor-2#

...Use the "**show spanning-tree**" command to verify that Rapid PVST is enabled...

Floor-2#show spanning-tree

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 32769

Address 0001.6408.81C1

Cost 24

Port 27(Port-channel2)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 00D0.9754.33DE

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Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Fa0/10 Desg FWD 19 128.10 P2p

Fa0/6 Desg FWD 19 128.6 P2p

Fa0/7 Desg FWD 19 128.7 P2p

Fa0/8 Desg FWD 19 128.8 P2p

Fa0/9 Desg FWD 19 128.9 P2p

Po2 Root LSN 12 128.27 Shr

Floor-2#

Step 3: Central Configuration.

Central#conf t

Central(config)#spanning-tree mode rapid-pvst

Central(config)#end

Central#

...Use the "**show spanning-tree**" command to verify that Rapid PVST is enabled as shown below...

Central#show spanning-tree

VLAN0001

Spanning tree enabled protocol rstp

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Root ID Priority 32769

Address 0001.6408.81C1

Cost 24

Port 27(Port-channel3)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0002.16A0.1755

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Fa0/1 Desg FWD 19 128.1 P2p

Po3 Root LSN 12 128.27 Shr

Central#

Step 4: Server-Room Configuration.

Server-Room#conf t

Server-Room(config)#spanning-tree mode rapid-pvst

Server-Room(config)#end

Server-Room#

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Note: From the output of the show spanning-tree command, we can tell that this switch is the current Root Switch (Bridge) for the VLAN 1.

Server-Room#show spanning-tree

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 32769

Address 0001.6408.81C1

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0001.6408.81C1

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Fa0/1 Desg FWD 19 128.1 P2p

Po4 Desg FWD 12 128.27 Shr

Server-Room#

Step 5: Distribution switch configuration.

DISTRIBUTION#show spanning-tree

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VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 32769

Address 0001.6408.81C1

Cost 12

Port 30(Port-channel4)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 000B.BE75.8E09

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Desg FWD 4 128.25 P2p

Po2 Desg FWD 12 128.28 Shr

Po4 Root FWD 12 128.30 Shr

Po3 Desg FWD 12 128.29 Shr

Po1 Desg FWD 12 128.27 Shr

Part 7: Configure the Distribution switch as the Root Bridge for VLAN 1 (default VLAN).

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- In **Part 6**, the output of **show spanning-tree** command shows that Server-Room switch is the current Root Switch (Bridge) for the LAN.
- The root switch is the logical center of the STP topology in a switched network. By carefully selecting which bridge becomes the root, we can control how traffic flows in the STP domain, as all other switches make forwarding decisions based on the perspective of the root switch.
- In this part, we will configure the Distribution switch to become the Root Switch for the LAN.
- The Distribution switch is a Layer 3 switch with more computing power than Layer 2 switches. In the topology, we can see that it is centrally located to receive and forward traffic to the upstream router. Thus, it is reasonable to make it the root bridge.
- To change the root bridge, you can use any of the following commands:

```
(config)#spanning-tree vlan vlan-id root primary
```

```
(config)#spanning-tree vlan 1 priority 4096
```

- **(bridge priority should be in increments of 4096)**
- The below command is used to change the Distribution switch to the Root switch.

```
DISTRIBUTION#conf t
```

```
DISTRIBUTION(config)#spanning-tree vlan 1 root primary
```

```
DISTRIBUTION(config)#end
```

```
DISTRIBUTION#
```

```
DISTRIBUTION#show spanning-tree
```

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VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 24577

Address 000B.BE75.8E09

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 24577 (priority 24576 sys-id-ext 1)

Address 000B.BE75.8E09

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Desg FWD 4 128.25 P2p

Po2 Desg FWD 12 128.28 Shr

Po4 Desg FWD 12 128.30 Shr

Po3 Desg FWD 12 128.29 Shr

Po1 Desg FWD 12 128.27 Shr

DISTRIBUTION#

Part 8: Configure Port Security and additional Layer 2 security features on Switches and disable any unused ports.

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Step 1: Configure PortFast on all access ports (ports connected to endpoints) on Floor-1 switch.

- PortFast enables a switch port to bypass STP convergence time and moves into forwarding state faster. STP convergence time may take between 30 to 50 seconds. This delay may affect some LAN functions, such as a client unable to receive IP address from a DHCP server due to timeout caused by STP.

```
Floor-1(config)#interface range f0/1-5
```

```
Floor-1(config-if-range)#spanning-tree portfast
```

%Warning: portfast should only be enabled on ports connected to a single host. Connecting hubs, concentrators, switches, bridges, etc... to this interface when portfast is enabled, can cause temporary bridging loops.

Use with CAUTION

...Use the "show spanning-tree inter fa0/1 detail" command to verify your configuration...

```
Floor-1#show spanning-tree inter fa0/1 detail
```

Port 1 (FastEthernet0/1) of VLAN0001 is designated forwarding

Port path cost 19, Port priority 128, Port Identifier 128.1

Designated root has priority 24577, address 000B.BE75.8E09

Designated bridge has priority 32769, address 0090.2B2A.A329

Designated port id is 128.1, designated path cost 19

Timers: message age 16, forward delay 0, hold 0

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Number of transitions to forwarding state: 1

The port is in the portfast mode

Link type is point-to-point by default

Floor-1#

Step 2: Configure BPDUGuard on all Access ports on Floor-1 switches.

- PortFast causes an interface to go into Forwarding state immediately. There is a risk that if two PortFast-enabled ports are inadvertently or maliciously connected, they will both come up as Forwarding ports, immediately creating a switching loop.
- BPDUGuard is a safety mechanism used to put a port which has already been configured as a PortFast into an error-disabled mode if a BPDU frame is received on that interface.
- BPDU packets are used for communication between switches only and should not be seen on an interface connected to end-user's device as threat actors can leverage on this to change the STP topology and carry out man-in-the-middle attack in the LAN.

```
Floor-1#conf t
```

```
Floor-1(config)#int range f0/1-5
```

```
Floor-1(config-if-range)#spanning-tree bpduguard enable
```

```
Floor-1(config-if-range)#end
```

Step 3: Configure Port Security on Floor-2.

- Statically configure the Switch to allow maximum of 10 MAC addresses to be learned on the ports.
- Configure the switch to store the MAC addresses of connected devices.

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- Set the violation security mode to shut down.

```
Floor-2#conf t
```

```
Floor-2(config)#int range f0/6-10
```

```
Floor-2(config-if-range)#switchport mode access
```

```
Floor-2(config-if-range)#switchport port-security
```

```
Floor-2(config-if-range)#switchport port-security maximum 10
```

```
<cr>
```

```
Floor-2(config-if-range)#switchport port-security mac-address sticky
```

```
Floor-2(config-if-range)#switchport port-security violation shutdown
```

```
Floor-2(config-if-range)#end
```

```
Floor-2#
```

...The show "show port-security" command shows that ports Fa0/6-10 are now enabled with port-security...

```
Floor-2#show port-security
```

```
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
```

```
(Count) (Count) (Count)
```

```
-----
```

```
Fa0/6 10 0 0 Shutdown
```

```
Fa0/7 10 0 0 Shutdown
```

```
Fa0/8 10 0 0 Shutdown
```

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Fa0/9 10 0 0 Shutdown

Fa0/10 10 0 0 Shutdown

Floor-2#

Step 4: Disable all unused ports on both Floor-1 and Floor-2 switches.

Floor-1 Configuration.

Floor-1#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Floor-1(config)#**int range f0/8-24,g0/1-2**

Floor-1(config-if-range)#shutdown

...The "**show ip interface brief**" command is used to verify the status of the interfaces...

Floor-1#sho ip interface brief

Interface IP-Address OK? Method Status Protocol

Port-channel1 unassigned YES manual up up

FastEthernet0/1 unassigned YES manual up up

FastEthernet0/2 unassigned YES manual up up

FastEthernet0/3 unassigned YES manual up up

FastEthernet0/4 unassigned YES manual up up

FastEthernet0/5 unassigned YES manual up up

FastEthernet0/6 unassigned YES manual up up

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FastEthernet0/7 unassigned YES manual up up

Floor-1#

Floor-2 Configuration.

Floor-2#config t

Floor-2(config)#int range f0/11-24,g0/1-2

Floor-2(config-if-range)#

Floor-2(config-if-range)#shutdown

...The "show ip interface brief" command is used to verify the status of the interfaces...

Interface IP-Address OK? Method Status Protocol

Port-channel2 unassigned YES manual up up

FastEthernet0/1 unassigned YES manual down down

FastEthernet0/2 unassigned YES manual down down

FastEthernet0/3 unassigned YES manual down down

FastEthernet0/4 unassigned YES manual up up

FastEthernet0/5 unassigned YES manual up up

FastEthernet0/6 unassigned YES manual up up

FastEthernet0/7 unassigned YES manual up up

FastEthernet0/8 unassigned YES manual up up

FastEthernet0/9 unassigned YES manual up up

FastEthernet0/10 unassigned YES manual up up

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Floor-2#

Part 9: Configure the switches virtual Interfaces for remote management.

Step 1: Floor-1 Configuration.

Floor-1#conf t

Floor-1(config)#int vlan 1

Floor-1(config-if)#ip address 192.168.10.2 255.255.255.0

Floor-1(config-if)#no shut

Floor-1(config-if)#

Floor-1(config-if)#exit

Floor-1(config)#ip default-gateway 192.168.10.1

Floor-1(config)#end

Floor-1#

Floor-1#copy run start

Destination filename [startup-config]?

Building configuration...

[OK]

Floor-1#show ip int vlan 1

Vlan1 is up, line protocol is up

Internet address is 192.168.10.2/24

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Broadcast address is 255.255.255.255

Address determined by setup command

MTU is 1500 bytes

....Output cut...

Floor-1#

Step 2: Floor-2 Configuration.

Floor-2#conf t

Floor-2(config)#int vlan 1

Floor-2(config-if)#ip address 192.168.10.3 255.255.255.0

Floor-2(config-if)#no shut

Floor-2(config-if)#

Floor-2(config-if)#exit

Floor-2(config)#ip default-gateway 192.168.10.1

Floor-2(config)#end

Floor-2#

Floor-2#copy run start

Destination filename [startup-config]?

Building configuration...

[OK]

Floor-2#

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Step 3: Distribution Configuration.

```
DISTRIBUTION#conf t
DISTRIBUTION(config)#int vlan 1
DISTRIBUTION(config-if)#ip address 192.168.10.4 255.255.255.0
DISTRIBUTION(config-if)#no shut
DISTRIBUTION(config-if)#
DISTRIBUTION(config-if)#exit
DISTRIBUTION(config)#ip default-gateway 192.168.10.1
DISTRIBUTION(config)#end
DISTRIBUTION#
DISTRIBUTION#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
DISTRIBUTION#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
DISTRIBUTION#
```

Step 4: Server-Room Configuration.

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```
Server-Room#conf t
```

```
Server-Room(config)#int vlan 1
```

```
Server-Room(config-if)#ip address 192.168.10.5 255.255.255.0
```

```
Server-Room(config-if)#no shut
```

```
Server-Room(config-if)#
```

```
Server-Room(config-if)#exit
```

```
Server-Room(config)#
```

```
Server-Room(config)#ip default-gateway 192.168.10.1
```

```
Server-Room(config)#end
```

```
Server-Room#
```

```
Server-Room#copy run start
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

```
[OK]
```

```
Server-Room#
```

Step 5: Central Configuration.

```
Central#conf t
```

```
Central(config)#int vlan 1
```

```
Central(config-if)#ip address 192.168.10.6 255.255.255.0
```

```
Central(config-if)#no shut
```

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```
Central(config-if)#  
Central(config-if)#exit  
Central(config)#ip default-gateway 192.168.10.1  
Central(config)#end  
Central#  
Central#copy run start  
Destination filename [startup-config]?  
Building configuration...  
[OK]  
Central#
```

Part 10. Configure basic settings on Edge router including password security settings.

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: n

Press RETURN to get started!

Router>enable

Router#config t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname Edge

Edge(config)#enable secret cisco1234

Edge(config)#service password-encryption

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```
Edge(config)#line con 0
Edge(config-line)#password console1234
Edge(config-line)#login
Edge(config-line)#exit
Edge(config)#line vty 0 15
Edge(config-line)#password vty1234
% Password too short - must be at least 8 characters. Password not configured.
Edge(config-line)#password vty123456
Edge(config-line)#login
Edge(config-line)#exit
Edge#
Edge#clock set 02:40:01 Dec 28 2023
Edge#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
Edge#sho ip int bri
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 209.165.200.1 YES manual up up
GigabitEthernet0/1 192.168.10.1 YES manual up up
```

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Vlan1 unassigned YES NVRAM administratively down down

Edge#

Step 3: Configure Secure passwords on Edge router.

Edge(config)#security passwords min-length 8

Edge(config)#login block-for 30 attempts 5 within 60

Step 4: Configure Secure Shell for Remote Users Access.

- Create first user as **Admin1** and password as **admin1pass1234**
- Create second user as **Admin2** and password as **admin2pass1234**
- Configure domain name as shown below.
- Generate encryption keys using key length of 1024 (longer key lengths are stronger than shorter keys).
- Enable SSH version 2.
- Only SSH connections should be allowed, and users will be authenticated using local database information.

Edge(config)#username Admin1 secret admin1pass1234

Edge(config)#username Admin2 secret admin2pass1234

Edge(config)#ip domain name www.itfirm.com

Edge(config)#crypto key generate rsa general-keys modulus 1024

The name for the keys will be: Edge.www.itfirm.com

% The key modulus size is 1024 bits

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```
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
```

```
*Mar 1 0:4:59.579: %SSH-5-ENABLED: SSH 1.99 has been enabled
```

```
Edge(config)#ip ssh version 2
```

```
Edge(config)#line vty 0 15
```

```
Edge(config-line)#transport input ssh
```

```
Edge(config-line)#login local
```

```
Edge(config-line)#exit
```

Part 11: Configure IP address on Edge router and enable SSH remote access for two teleworkers.

```
Edge(config)#int g0/0
```

```
Edge(config-if)#no shut
```

```
Edge(config-if)#
```

```
Edge(config-if)#description CONNECTION TO ISP MODEM
```

```
Edge(config-if)#ip address 209.165.200.1 255.255.255.240
```

```
Edge(config-if)#int g0/1
```

```
Edge(config-if)#no shut
```

```
Edge(config-if)#
```

```
Edge(config-if)#description CONNECTION TO LAN DISTRIBUTION
```

```
Edge(config-if)#ip address 192.168.10.1 255.255.255.0
```

```
Edge(config-if)#exit
```

```
Edge(config)#exit
```

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Part 12: Verify end-to-end connectivity and establish SSH connection over the Internet from Remote1 and Remote 2 to Edge router.

- On PC1, ping the IP address of PC8, File Server, Central Printer and access Web Server using HTTPS protocol.
- On Remote1 PC, establish a remote secure session to Edge router.
- On Remote2 PC, establish a remote secure session to Edge router.

Step 1a: On PC1, ping the IP address of PC8, File Server, Central Printer and access Web Server using HTTPS protocol.

...The below shows a successful communication between PC1 and PC8...

```
C:\>ping 192.168.10.16
```

```
Pinging 192.168.10.16 with 32 bytes of data:
```

```
Reply from 192.168.10.16: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.16: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.16: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.10.16: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.10.16:
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

...The below shows a successful communication between PC1 and File Server...

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```
C:\>ping 192.168.10.7
```

Pinging 192.168.10.7 with 32 bytes of data:

Reply from 192.168.10.7: bytes=32 time<1ms TTL=128

Reply from 192.168.10.7: bytes=32 time<1ms TTL=128

Reply from 192.168.10.7: bytes=32 time<1ms TTL=128

Reply from 192.168.10.7: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.7:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

...The below shows a successful communication between PC1 and Central Printer...

```
C:\>ping 192.168.10.7
```

Pinging 192.168.10.7 with 32 bytes of data:

Reply from 192.168.10.7: bytes=32 time=1ms TTL=128

Reply from 192.168.10.7: bytes=32 time<1ms TTL=128

Reply from 192.168.10.7: bytes=32 time<1ms TTL=128

Reply from 192.168.10.7: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.7:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

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Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

Step 1b: On PC1, access Web Server using HTTPS protocol.

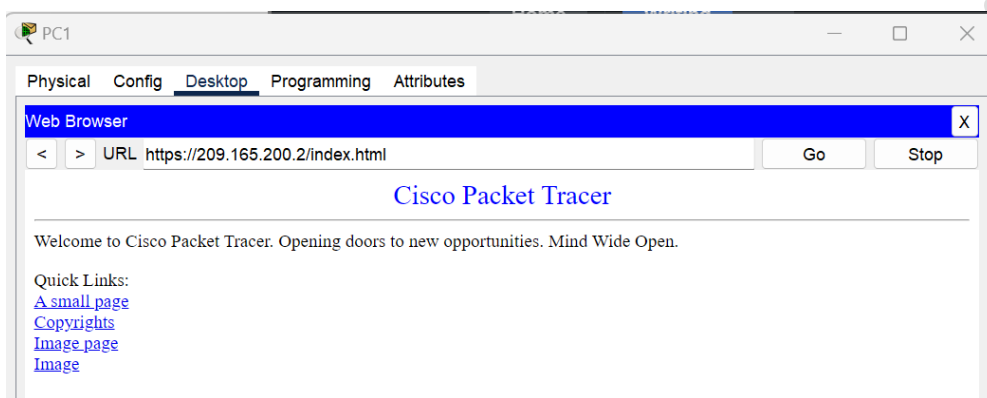


Figure 10.3: Web Server Page as displayed on PC1

Step 2: On Remote1 PC, establish a remote secure session to Edge router.

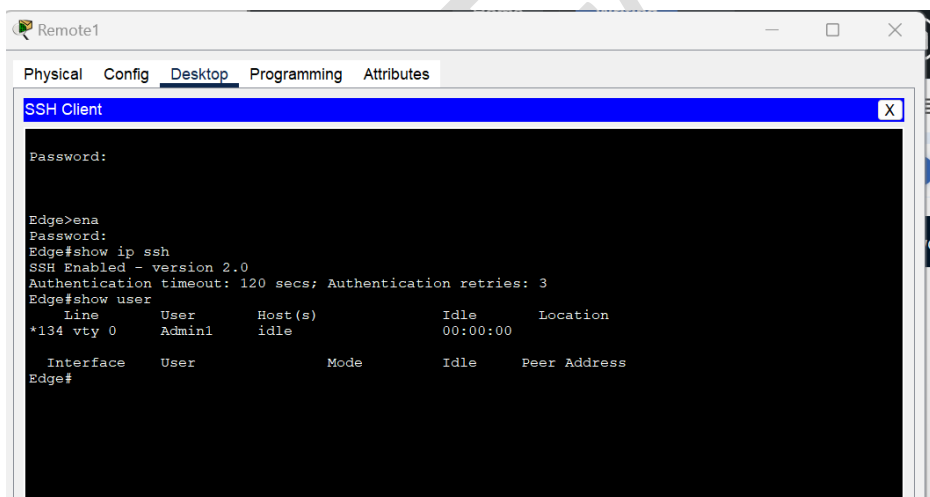


Figure 10.4: Remote1 User Session on Edge Router

Step 3: On Remote2 PC, establish a remote secure session to Edge router.

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```

Remote2
Physical Config Desktop Programming Attributes
SSH Client
Password:

Edge>enable
Password:
Edge#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
Edge#show user
  Line      User      Host(s)      Idle      Location
+*134 vty 0   Admin2     idle        00:00:00
  Interface  User      Mode        Idle      Peer Address
Edge#

```

Figure 10.5: Remote2 User Session on Edge Router

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Project 3: Configure a Wireless Network on a SOHO Router.

Scenario

As a newly hired wireless network engineer for a small IT firm located in a densely populated Urban area, you have been tasked to setup a basic WLAN for your organization to support its online-IT services delivery to remote users.

The nature of the business demands the network to be ready in two to three days and accommodate BYOD policy to allow employees to connect with their personal/company owned portable devices to allow for mobility and provide a degree of flexibility.

Topology

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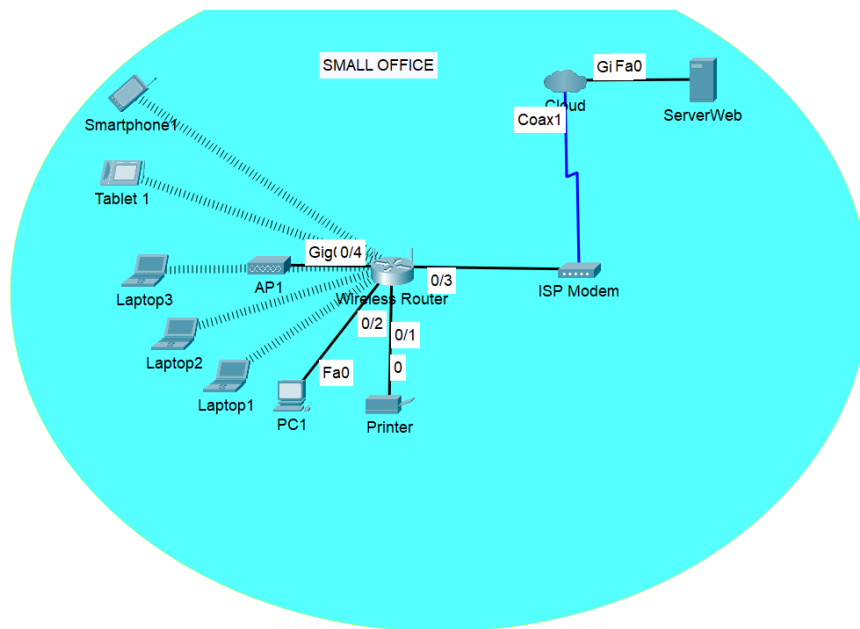


Figure 12.2: A Small Office Network

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Wireless Router	Eth3	209.165.200.1	255.255.255.240	209.165.200.3
Webserver	NIC	209.165.200.2	255.255.255.252	209.165.200.1
AP1	Gig0/0	DHCP	255.255.255.0	192.168.10.1
Printer	NIC	192.168.10.2	255.255.255.0	192.168.10.1
PC1	NIC	DHCP	255.255.255.0	192.168.10.1
Laptop1	Wireless	DHCP	255.255.255.0	192.168.10.1
Laptop2	Wireless	DHCP	255.255.255.0	192.168.10.1
Laptop3	Wireless	DHCP	255.255.255.0	192.168.10.1
Tablet1	Wireless	DHCP	255.255.255.0	192.168.10.1
Smartphone1	Wireless	DHCP	255.255.255.0	192.168.10.1

Table 12.2: Lab 1 Address Table

Objectives

- Setup the network topology as shown.

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- Connect to a wireless router.
- Configure the Wireless Settings on the Wireless Router.
- Connect a wireless client device to the wireless router.
- Add an AP to the network to extend wireless coverage.
- Update default router settings

Part 1: Connect to a wireless router.

Step 1: Connect PC1 to the Wireless Router using a straight-through cable the Ethernet port as shown in the topology.

Step 2: Configure PC1 to use DHCP address from the WR.

- To reach the **Wireless Router** management page, **PC1** must communicate on the network. A wireless router usually includes a DHCP server, and the DHCP server is usually enabled by default on the LAN. **PC1** will receive IP address information from the DHCP server on **WR**.
1. Click PC1, and select the **Desktop** tab.
 2. Click **IP Configuration** and select **DHCP**.
 3. Close the **IP Configuration** window.

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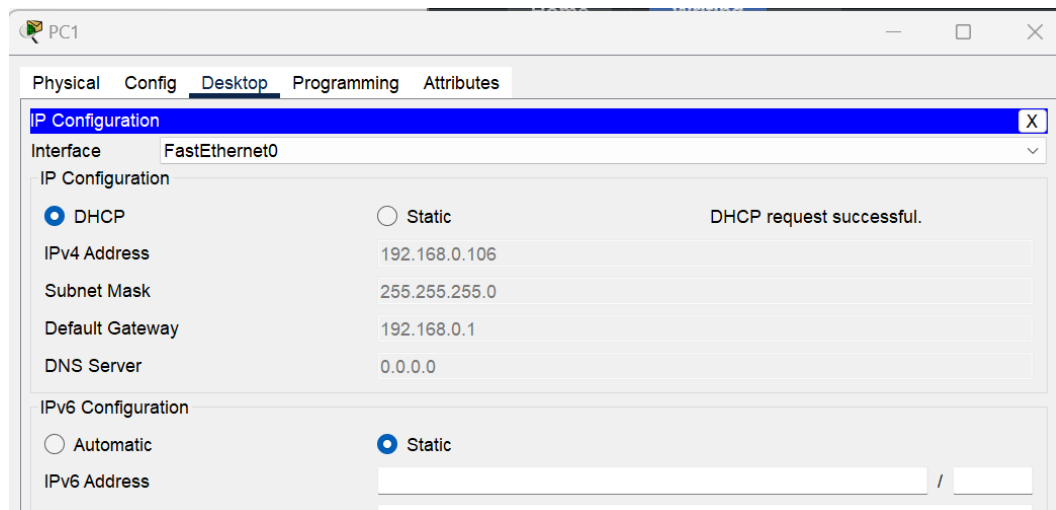


Figure 12.3: PC1 configuration

Step 3: Login to the WR through a Web browser.

1. In the **Desktop** tab on PC1, choose **Web Browser**.
2. Enter **192.168.0.1** in the URL field to open the web configuration page of the wireless router.
3. Use **admin** for both the username and password.

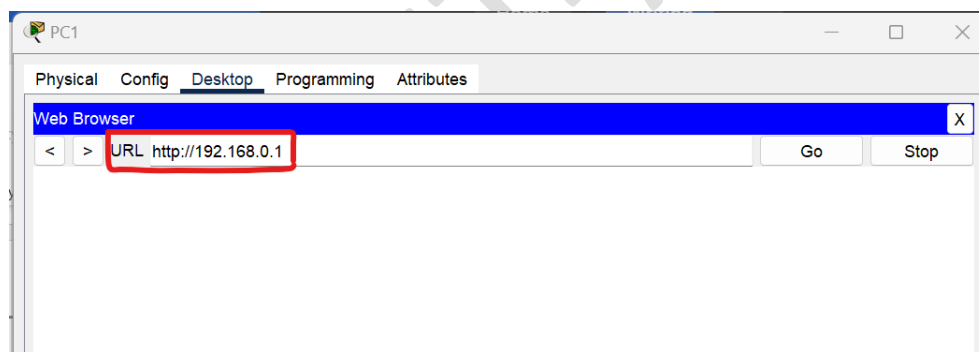


Figure 12.4: Web page on PC1

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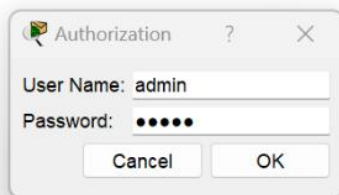


Figure 12.5: Login page

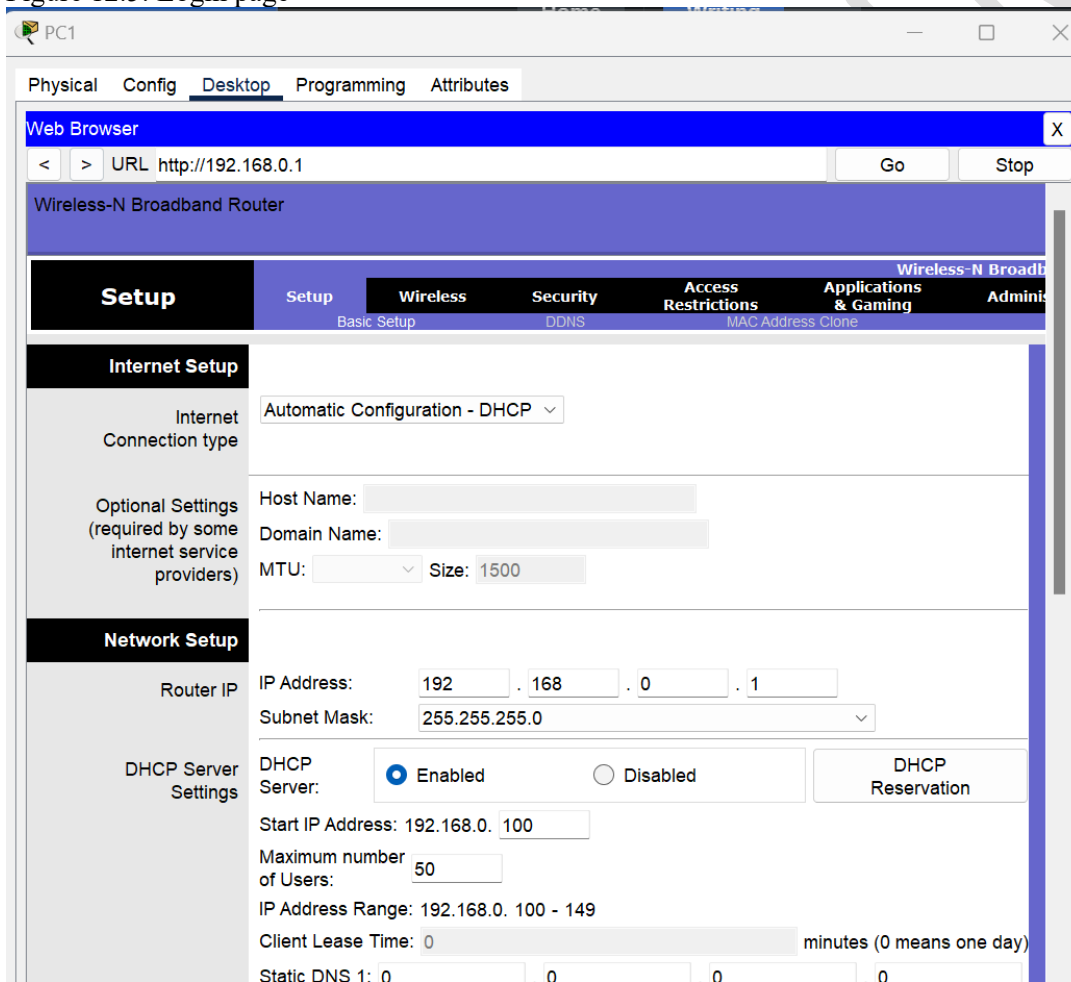


Figure 12.6: Wireless Router Network Setup page

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Step 4: Configure the Internet Port of WR.

The Wireless Router (WR) Internet Port must be configured to enable internal users access the internet.

Under the **Internet Setup** at the top of the **Basic Setup** page, change the Internet IP address method from **Automatic Configuration – DHCP** to **Static IP**.

Type the IP address to be assigned to the Internet interface as follows:

Internet IP Address: 209.165.200.1

Subnet Mask: 255.255.255.252

Default Gateway: 209.165.200.3

DNS Server: 209.165.200.3

Scroll down the page and click **Save Settings**.

Note: If you are doing this in Packet Tracer, you may get a **Request Timeout** message. Close the PC1 window and wait for the orange lights to turn into green triangles.

Part 2: Configure the Wireless Settings on the Wireless Router.

In this section, you will configure the wireless router to operate using the **2.4 GHz** frequency range.

Step 1: Configure the WR SSID (Service Set Identifier – Network Name)

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1. On the WR GUI, navigate to **Wireless > Basic Wireless Settings**.
2. Select Network Mode as: **Mixed**.
3. Change **Network Name (SSID)** to **Small-Office**.
4. Change the **Standard Channel** to **1 – 2.412GHz**.
5. Enable SSID Broadcast.
6. Disable both 5 GHz frequencies. Leave the rest of the settings unchanged.
7. Scroll to the bottom of the window and click **Save Settings**.

Step 2: Configure wireless security settings.

In this step, you configure the wireless security settings using **WPA2** security mode with encryption and passphrase.

1. Click on the **Wireless tab > Wireless Security**.
2. Under the **2.4 GHz Security Mode**, select **WPA2 Personal** for the Security Mode.
3. For the Encryption field, keep the default **AES** setting.
4. In the Passphrase field, enter **Cisco1234** as the passphrase.
5. Click **Save Settings**.
6. Verify that the settings in the **Basic Wireless Settings** and **Wireless Security** pages are correct and saved.

Step 3: Connect the Wireless Clients.

If you are doing this in Packet Tracer, you will need to change the Laptop Ethernet NIC to a wireless NIC. Click on Laptop **Physical** tab, click on Zoom In to see the buttons. Then click the **power button** to turn off the laptop. Click on the **NIC port**, drag it off to the empty spaces under the **Modules** tab. Then, under the **Modules tab** select **WPC300N** and drag it to the NIC port on Build strong Foundation for STEM Literacy; Increase Diversity, Equity, and Inclusion in the STEM workforce; Prepare the STEM workforce for the future:



the laptop. This will cause the Laptop NIC to become a wireless NIC. Finally, click on the power button to turn on the laptop.

You can perform these steps on Desktop PC or any Wired devices in Packet Tracer to make it become a Wireless client.

- Open **Laptop1**. Select **Desktop** tab. Click **PC Wireless**.
- Select the **Connect** tab. Click **Refresh** as necessary. Select the Wireless Network Name **Small-Office**.
- Enter the passphrase as **Cisco1234** in the pre-shared key field and click **Connect**. Close the PC Wireless window.
- Open a web browser and verify that you can navigate to the Webserver page.
- Repeat the above steps to connect **Laptop2** to the wireless network.

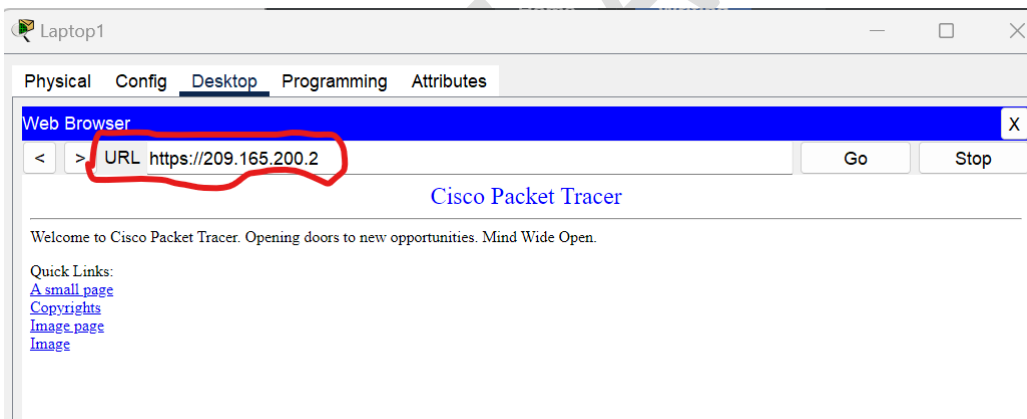


Figure 12.7: Webserver page on Laptop1

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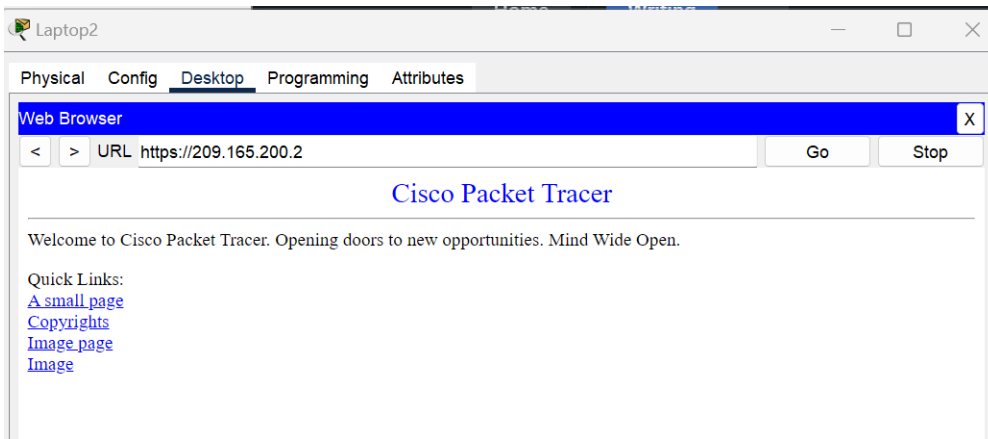


Figure 12.8: Webserver page on Laptop2

Part 3: Add an AP to the network to extend wireless coverage.

An access point (AP) is a device that extends the wireless local area network. An access point is connected to a wired router using an Ethernet cable to project the signal to a desired location.

Step 1: Configure the Wireless Access Point.

1. Connect the Access Point to the Wireless Router using a straight-through cable as shown in the topology.
2. Click **AP1**. Select the **Config** tab.
3. Under the INTERFACE heading, select **Port 1**.
4. In the SSID field, enter **Small-Office**.
5. Under the 2.4 GHz Channel, select **6**.
6. Select **WPA2-PSK**. Enter the passphrase **Cisco1234** in the Passphrase field.
7. Keep **AES** as the default Encryption Type.

Step 2: Connect the Wireless Clients to the AP1.

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- Open **Laptop3**. Select **Desktop** tab. Click **PC Wireless**.
- Select the **Connect** tab. Click **Refresh** as necessary. Select the Wireless Network Name **Small-Office with channel 6**.
- Enter the passphrase as **Cisco1234** in the pre-shared key field and click **Connect**. Close the PC Wireless window.
- On **Tablet1**, Select **Config** tab. Select **Wireless0** under **Interface**. Change the SSID to **Small-Office**. Select **WPA2-PSK** and enter password as **Cisco1234**.
- Repeat the above steps to connect **Smartpone1** to the wireless network.

Part 4: Update default router settings

Step 1: Change the WR Access Password.

1. On PC1, navigate to WR GUI interface at **192.168.0.1**.
2. Navigate to **Administration > Management** and change the current **Router Password** to **Class1234**.
3. Scroll to the bottom of the window and click **Save Settings**.
4. Use the username **admin** and the new password **Class1234** when prompted to log in to the wireless router. Click **OK** to continue.
5. Click **Continue** and move on to the next step.

Step 2: Change the DHCP address range in WR.

In this step, you will change the internal network address from **192.168.0.0/24** to **192.168.10.0/24**. When the LAN network address changes, the IP addresses on the devices in the LAN and WLAN must be renewed to receive new IP addresses before the lease is timed out.

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1. Navigate to **Setup > Basic Setup**.
2. Scroll down the page to **Network Setup**.
3. The IP address assigned to **Router IP** is 192.168.0.1. Change it to **192.168.10.1**. Verify that IP address still start at .100, and there are 50 available IP addresses in the DHCP pool.
4. Add **209.165.200.3** as the DNS server with the DHCP settings.
5. Scroll to the bottom of the window and click **Save Settings**.
6. Close the PC1 web browser.
7. **On PC1 Desktop** tab, click **Command Prompt**.
8. Type **ipconfig /renew** to force **PC1** re-acquire its IP information via DHCP.
9. Renew the IP address on other client devices and verify that you can still navigate to the Web server.

Project 4: Implement DHCPv4 in an Enterprise network.

Background

As the network technician for your company, you have been assigned the task of configuring a dedicated DHCP server to manage address allocation for LAN 1 and LAN 2. You will then configure a Cisco router to forward the DHCP requests from both LANs to the DHCP server located in a different network. In addition, you will configure Cisco router as a DHCP server. Lastly, you will configure another router as a DHCP client so that it receives an IP address from the ISP-Router network.

Objectives

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Part 1: Configure basic settings on all routers.

Part 2: Configure OSPF process 10 and Area 0 on all routers.

Part 3: Configure ISP router as a DHCP server.

Part 4: Configure Branch router as a DHCP Relay agent for LAN1 and LAN2 networks.

Part 5: Configure two DHCP pool on DHCP Server Computer, one for LAN1 and the other for LAN 2 using the information below.

LAN1: Pool name: (LAN1-Pool), Network (192.168.10.0/24), Default Gateway (192.168.10.1), DNS-Server (10.0.0.254), Domain (www.Corp.org).

LAN2: Pool name: (LAN2-Pool) Network (192.168.20.0/24), Default Gateway (192.168.20.1), DNS-Server (10.0.0.254), Domain (www.Corp.org).

Part 6: Configure the DHCP-CLIENT router as a DHCP client.

Part 5: Verify DHCP and Connectivity

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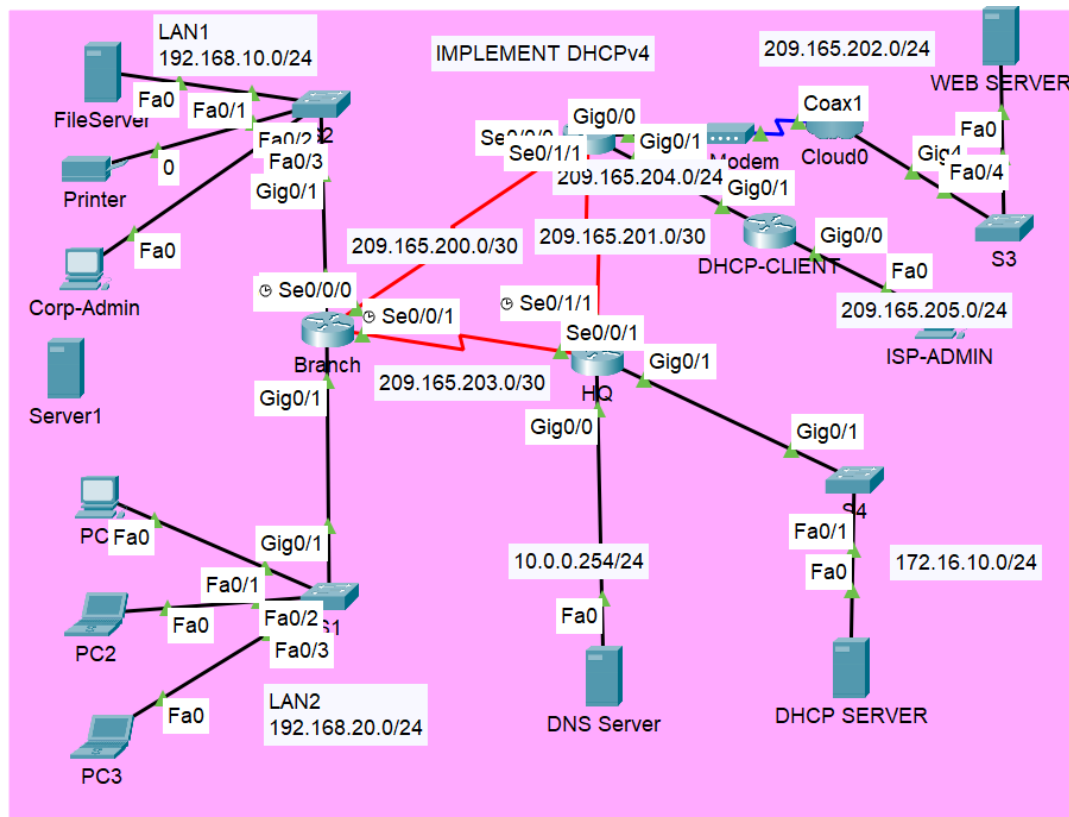


Figure 15.1: Implement DHCPv4

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Device	Interface	IP Address
ISP	S0/0/0	209.165.200.2/30
	S0/1/1	209.165.201.1/30
	G0/1	209.165.204.1/30
	G0/0	209.165.202.1/24
Branch	G0/0	192.168.10.1/24
	G0/1	192.168.20.1/24
	S0/0/0	209.165.200.1/30
	S0/0/1	209.165.203.1/30
HQ	S0/0/1	209.165.203.2/30
	S0/1/1	209.165.201.2/30
	G0/0	10.0.0.1/24
	G0/1	172.16.10.1/24
DHCP-CLIENT	G0/0	209.165.205.1/24
	G0/1	DHCP
DNS Server	NIC	10.0.0.254
DHCP Server	NIC	172.16.10.254
WEB SERVER	NIC	209.165.202.254/24

Figure 15.1: DHCP Addressing Table

Instructions:

Part 1: Configure basic settings on all routers.

Step 1: Branch router configuration

```
Router>enable
```

```
Router#confi t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Router(config)#hostname Branch
```

```
Branch(config)#int s0/0/0
```

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```
Branch(config-if)#no shut
Branch(config-if)#ip address 209.165.200.1 255.255.255.252
Branch(config-if)#int s0/0/1
Branch(config-if)#no shut
Branch(config-if)#ip address 209.165.203.1 255.255.255.252
Branch(config-if)#int g0/0
Branch(config-if)#no shut
Branch(config-if)#ip address 192.168.10.1 255.255.255.0
Branch(config-if)#int g0/1
Branch(config-if)#no shut
Branch(config-if)#ip address 192.168.20.1 255.255.255.0
Branch(config-if)#exit
```

Step 2: HQ router configuration.

```
Router>enable
Router#conf t
Router(config)#hostname HQ
HQ(config)#int g0/1
HQ(config-if)#no shut
HQ(config-if)#ip add 172.16.10.1 255.255.255.0
HQ(config-if)#int s0/0/1
```

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```
HQ(config-if)#no shut
```

```
HQ(config-if)#ip add 209.165.203.2 255.255.255.252
```

```
HQ(config-if)#int s0/1/1
```

```
HQ(config-if)#no shut
```

```
HQ(config-if)#ip address 209.165.201.2 255.255.255.252
```

```
HQ(config-if)#int g0/0
```

```
HQ(config-if)#no shut
```

```
HQ(config-if)#ip add 10.0.0.1 255.255.255.0
```

```
HQ(config-if)#
```

```
HQ(config-if)#exit
```

Step 3: ISP router configuration.

```
Router>enable
```

```
Router#config t
```

```
Router(config)#hostname ISP
```

```
ISP(config)#
```

```
ISP(config)#int s0/0/0
```

```
ISP(config-if)#no shut
```

```
ISP(config-if)#ip add 209.165.200.2 255.255.255.252
```

```
ISP(config-if)#int s0/1/1
```

```
ISP(config-if)#no shut
```

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```
ISP(config-if)#ip add 209.165.201.1 255.255.255.252
```

```
ISP(config-if)#int g0/1
```

```
ISP(config-if)#no shut
```

```
ISP(config-if)#ip add 209.165.204.1 255.255.255.0
```

```
ISP(config-if)#int g0/0
```

```
ISP(config-if)#no shut
```

```
ISP(config-if)#ip add 209.165.202.1 255.255.255.252
```

```
ISP(config-if)#exit
```

Step 4: DHCP Client router configuration.

```
Router>enable
```

```
Router#conf t
```

```
Router(config)#hostname DHCP-CLIENT
```

```
DHCP-CLIENT(config)#
```

```
DHCP-CLIENT(config)#int g0/1
```

```
DHCP-CLIENT(config-if)#no shut
```

```
DHCP-CLIENT(config-if)#ip address dhcp
```

```
DHCP-CLIENT(config-if)#int g0/0
```

```
DHCP-CLIENT(config-if)#no shut
```

```
DHCP-CLIENT(config-if)#ip address 209.165.205.1 255.255.255.0
```

```
DHCP-CLIENT(config-if)#exit
```

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Part 2: Configure OSPF process 10 and Area 0 on all routers.

Step 1: Branch router configuration.

```
Branch(config)#router ospf 10
Branch(config-router)#network 192.168.10.0 0.0.0.255 area 0
Branch(config-router)#network 192.168.20.0 0.0.0.255 area 0
Branch(config-router)#network 209.165.200.0 0.0.0.3 area 0
Branch(config-router)#network 209.165.203.0 0.0.0.3 area 0
Branch(config-router)#exit
Branch(config)#
```

Step 2: HQ router configuration.

```
HQ(config)#router ospf 10
HQ(config-router)#network 172.16.10.0 0.0.0.255 area 0
HQ(config-router)#network 10.0.0.0 0.0.0.255 area 0
HQ(config-router)#network 209.165.203.0 0.0.0.3 area 0
HQ(config-router)#network 209.165.201.0 0.0.0.3 area 0
HQ(config-router)#exit
```

Step 3: ISP router configuration.

```
ISP(config)#router ospf 10
ISP(config-router)#network 209.165.204.0 0.0.0.3 area 0
ISP(config-router)#network 209.165.201.0 0.0.0.3 area 0
```

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```
ISP(config-router)#network 209.165.200.0 0.0.0.3 area 0
ISP(config-router)#network 209.165.202.0 0.0.0.255 area 0
ISP(config-router)#exit
ISP(config)#
```

Step 4: DHCP-CLIENT router configuration.

```
DHCP-CLIENT(config)#router ospf 10
DHCP-CLIENT(config-router)#network 209.165.205.0 0.0.0.255 area 0
DHCP-CLIENT(config-router)#network 209.165.204.0 0.0.0.3 area 0
DHCP-CLIENT(config-router)#exit
DHCP-CLIENT(config)#
```

Part 3: Configure ISP router as a DHCP server.

```
ISP(config)#ip dhcp pool ISP-POOL
ISP(dhcp-config)#network 209.165.204.0 255.255.255.252
ISP(dhcp-config)#default-router 209.165.204.1
ISP(dhcp-config)#dns-server 10.0.0.254
ISP(dhcp-config)#domain www.corp.org
ISP(dhcp-config)#exit
ISP(config)#
```

Part 4: Configure Branch router as a DHCP Relay agent for LAN1 and LAN2 networks.

```
Branch(config)#int g0/0
```

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```
Branch(config-if)#ip helper-address 172.16.10.254
```

```
Branch(config-if)#int g0/1
```

```
Branch(config-if)#ip helper-address 172.16.10.254
```

```
Branch(config-if)#exit
```

```
Branch(config)#
```

Part 5: Configure two DHCP pool on DHCP Server Computer, one for LAN1 and the other for LAN 2 using the information below.

LAN1: Pool name: (LAN1-Pool), Network (192.168.10.0), Subnet Mask (255.255.255.0) Default Gateway (192.168.10.1), DNS-Server (10.0.0.254), Domain (www.Corp.org).

LAN2: Pool name: (LAN2-Pool) Network (192.168.20.0), Subnet Mask (255.255.255.0), Default Gateway (192.168.20.1), DNS-Server (10.0.0.254), Domain (www.Corp.org).

Step 1: On DHCP SERVER PC, go to Desktop, select Services tab, select DHCP services, and add the information as shown below.

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Figure 15.3: DHCP configuration.

Part 6: Configure the DHCP-CLIENT router as a DHCP client.

Step 1: See the configuration below on DHCP Client router.

```

DHCP-CLIENT(config)#int g0/1
DHCP-CLIENT(config-if)#no shut
DHCP-CLIENT(config-if)#ip address dhcp
DHCP-CLIENT(config-if)#
DHCP-CLIENT(config-if)#

```

Part 5: Verify Client devices in LAN 1 and LAN2 receives IP addresses from the DHCP SERVER and Connectivity.

Step 1: On Corp-Admin PC, go to Desktop. Select IP Configuration and click on DHCP option.

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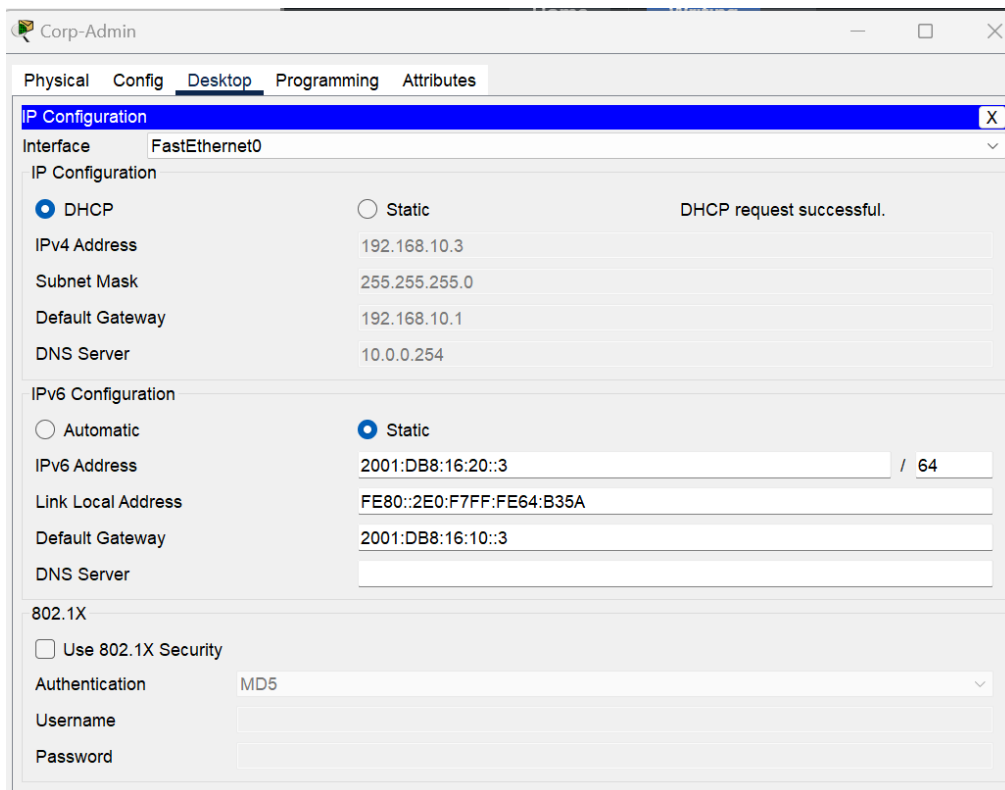


Figure 15.4: Corp Admin PC

Step 2: On PC, go to Desktop. Select IP Configuration and click on DHCP option.

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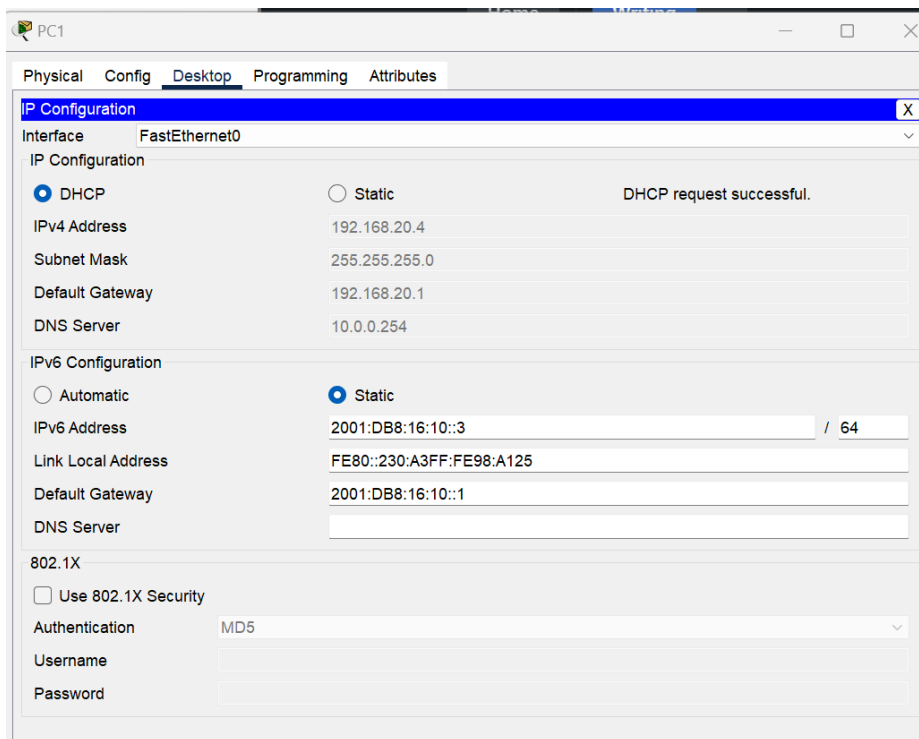


Figure 15.5: PC 1

Step 3: On Corp-Admin PC, ping the IP address of the WEB SERVER.

C:\>ping 209.165.202.254

Pinging 209.165.202.254 with 32 bytes of data:

Reply from 209.165.202.254: bytes=32 time=42ms TTL=126

Reply from 209.165.202.254: bytes=32 time=31ms TTL=126

Reply from 209.165.202.254: bytes=32 time=37ms TTL=126

Reply from 209.165.202.254: bytes=32 time=37ms TTL=126

Ping statistics for 209.165.202.254:

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Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 31ms, Maximum = 42ms, Average = 36ms

C:\>

Step 4: On PC1, ping the IP address of the WEB SERVER.

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 209.165.202.254

Pinging 209.165.202.254 with 32 bytes of data:

Reply from 209.165.202.254: bytes=32 time=2ms TTL=126
Reply from 209.165.202.254: bytes=32 time=46ms TTL=126
Reply from 209.165.202.254: bytes=32 time=2ms TTL=126
Reply from 209.165.202.254: bytes=32 time=19ms TTL=126

Ping statistics for 209.165.202.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 46ms, Average = 17ms

C:\>
  
```

Figure 15.6: PC1 successfully Ping of Web Server

Project 4 – Design and Implement NATs for an Enterprise Network.

Background

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An ISP has allocated three subnet blocks of public IP addresses for the use of NAT in a corporate network. The subnets are:

- 209.165.200.0/29 (to be used on HQ router)
- 209.165.201.0/29 (to be used on Data Center router)
- 209.165.202.0/29 (to be used on Remote router)

These addresses will be used to provide public IPv4 address for internet access to the company internal hosts in each LANs.

Scope

As the network administrator for this company, you will configure various types of NAT. You will test, view, and verify that the translations are taking place, and that internal devices are using the publicly configured IPv4 addresses for internet communication. Finally, you will configure all routers with OSPF process 10 into Area 0.

Objectives

Part 1: Configure IP addresses on Endpoints.

Part 2: Configure basic settings on all routers and enable the routers with OSPF process 10 into Area 0.

Part 3: Configure a Dynamic NAT on HQ router using a pool of three addresses: 209.165.200.3 – 209.165.200.5 with a netmask of 255.255.255.248.

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Part 4: Use the public IPv4 address 209.165.201.3 with netmask 255.255.255.248 to configure a PAT on DATA-CENTER router for the use of internal users.

Part 5: Configure a Static NAT on REMOTE router using the public IP 209.165.202.3 with a netmask 255.255.255.248 for the use of a Remote teleworker.

Part 6: Verify that NAT translations are working as expected.

Topology

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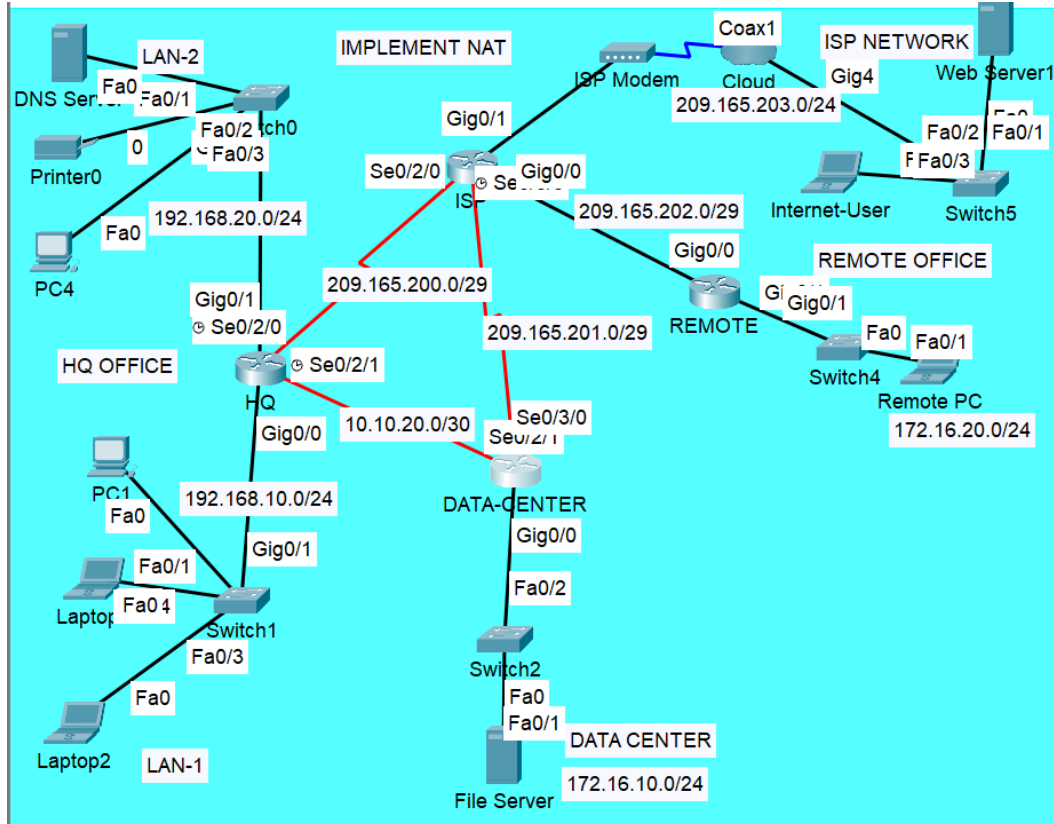
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Addressing Table

Device	Interface	IP Address
HQ	S0/2/0	209.165.200.1/29
	S0/2/1	10.10.20.1/30
	G0/1	192.168.20.1/24
	G0/0	192.168.10.1/24
ISP	S0/2/0	209.165.200.2/29
	S0/3/0	209.165.201.1/29
	G0/0	209.165.202.1/29
	G0/1	209.165.203.1/29
DATA CENTER	S0/2/1	10.10.20.2/30
	S0/3/0	209.165.201.2/29
	G0/0	172.16.10.1/24
REMOTE	G0/0	209.165.202.2/29
	G0/1	172.16.20.1/24
File Server	NIC	172.16.10.254/24
Web Server	NIC	209.165.203.2/24
PC1	NIC	192.168.10.3/24
PC4	NIC	192.168.20.3/24
REMOTE-PC	NIC	172.16.20.3/24
Internet-User	NIC	209.165.203.3/24

Table 17.1: NAT Addressing Table

Instructions.

Part 1: Configure IP addresses on Endpoints.

Requirements: Assign IPv4 addresses to End devices based on the information in the Addressing table.

Part 2: Configure basic settings on all routers and enable the routers with OSPF process 10 into Area 0.

Step 1: HQ router configuration.

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```
Router>enable
Router#config t
Router(config)#hostname HQ
HQ(config)#int s0/2/0
HQ(config-if)#no shut
HQ(config-if)#ip address 209.165.200.1 255.255.255.248
HQ(config-if)#int s0/2/1
HQ(config-if)#no shut
HQ(config-if)#ip address 10.10.20.1 255.255.255.252
HQ(config-if)#int g0/0
HQ(config-if)#no shut
HQ(config-if)#ip address 192.168.10.1 255.255.255.0
HQ(config-if)#int g0/1
HQ(config-if)#no shut
HQ(config-if)#ip address 192.168.20.1 255.255.255.0
HQ(config-if)#exit
HQ(config)#
HQ(config)#router ospf 10
HQ(config-router)#network 209.165.200.0 0.0.0.7 area 0
HQ(config-router)#network 10.10.20.0 0.0.0.3 area 0
```

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```
HQ(config-router)#network 192.168.10.0 0.0.0.255 area 0
```

```
HQ(config-router)#network 192.168.20.0 0.0.0.255 area 0
```

```
HQ(config-router)#exit
```

```
HQ(config)#
```

```
HQ(config)#end
```

```
HQ#
```

```
HQ#copy run start
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

```
[OK]
```

```
HQ#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
```

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
```

```
C 10.10.20.0/30 is directly connected, Serial0/2/1
```

```
L 10.10.20.1/32 is directly connected, Serial0/2/1
```

```
172.16.0.0/24 is subnetted, 1 subnets
```

```
O 172.16.10.0/24 [110/65] via 10.10.20.2, 00:06:14, Serial0/2/1
```

```
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
```

```
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0
```

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L 192.168.10.1/32 is directly connected, GigabitEthernet0/0

192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.20.0/24 is directly connected, GigabitEthernet0/1

L 192.168.20.1/32 is directly connected, GigabitEthernet0/1

209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks

C 209.165.200.0/29 is directly connected, Serial0/2/0

L 209.165.200.1/32 is directly connected, Serial0/2/0

209.165.201.0/29 is subnetted, 1 subnets

O 209.165.201.0/29 [110/128] via 10.10.20.2, 00:04:42, Serial0/2/1

[110/128] via 209.165.200.2, 00:04:42, Serial0/2/0

209.165.202.0/29 is subnetted, 1 subnets

O 209.165.202.0/29 [110/65] via 209.165.200.2, 00:02:58, Serial0/2/0

209.165.203.0/29 is subnetted, 1 subnets

O 209.165.203.0/29 [110/65] via 209.165.200.2, 00:04:42, Serial0/2/0

Step 2: ISP router configuration.

```
Router>enable
```

```
Router#configure terminal
```

```
Router(config)#hostname ISP
```

```
ISP(config)#int s0/2/0
```

```
ISP(config-if)#no shut
```

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```
ISP(config-if)#ip address 209.165.200.2 255.255.255.248
```

```
ISP(config-if)#int s0/3/0
```

```
ISP(config-if)#no shut
```

```
ISP(config-if)#ip address 209.165.201.1 255.255.255.248
```

```
ISP(config-if)#int g0/0
```

```
ISP(config-if)#no shut
```

```
ISP(config-if)#ip address 209.165.202.1 255.255.255.248
```

```
ISP(config-if)#int g0/1
```

```
ISP(config-if)#no shut
```

```
ISP(config-if)#ip address 209.165.203.1 255.255.255.248
```

```
ISP(config-if)#exit
```

```
ISP(config)#router ospf 10
```

```
ISP(config-router)#network 209.165.200.0 0.0.0.7 area 0
```

```
ISP(config-router)#network 209.165.201.0 0.0.0.7 area 0
```

```
ISP(config-router)#network 209.165.202.0 0.0.0.7 area 0
```

```
ISP(config-router)#network 209.165.203.0 0.0.0.7 area 0
```

```
ISP(config-router)#exit
```

```
ISP(config)#end
```

```
ISP#
```

```
ISP#copy run start
```

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Destination filename [startup-config]?

Building configuration...

[OK]

ISP#

Step 3: DATA-CENTER configuration.

Router>enable

Router#configure terminal

Router(config)#hostname DATA-CENTER

DATA-CENTER(config)#int g0/0

DATA-CENTER(config-if)#no shut

DATA-CENTER(config-if)#ip address 172.16.10.1 255.255.255.0

DATA-CENTER(config-if)#int s0/2/1

DATA-CENTER(config-if)#no shut

DATA-CENTER(config-if)#ip address 10.10.20.2 255.255.255.252

DATA-CENTER(config-if)#int s0/3/0

DATA-CENTER(config-if)#no shut

DATA-CENTER(config-if)#ip address 209.165.201.2 255.255.255.248

DATA-CENTER(config-if)#exit

DATA-CENTER(config)#router ospf 10

DATA-CENTER(config-router)#network 172.16.10.0 0.0.0.255 area 0

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```
DATA-CENTER(config-router)#network 10.10.20.0 0.0.0.3 area 0
DATA-CENTER(config-router)#network 209.165.201.0 0.0.0.7 area 0
DATA-CENTER(config-router)#exit
DATA-CENTER(config)#end
DATA-CENTER#
DATA-CENTER#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
DATA-CENTER#
```

Step 4: REMOTE router configuration.

```
Router>enable
Router#config t
Router(config)#hostname REMOTE
REMOTE(config)#int g0/1
REMOTE(config-if)#no shut
REMOTE(config-if)#ip address 172.16.20.1 255.255.255.0
REMOTE(config-if)#int g0/0
REMOTE(config-if)#no shut
REMOTE(config-if)#ip address 209.165.202.2 255.255.255.248
```

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```
REMOTE(config-if)#exit
```

```
REMOTE(config)#router ospf 10
```

```
REMOTE(config-router)#network 176.16.20.0 0.0.0.255 area 0
```

```
REMOTE(config-router)#network 209.165.202.0 0.0.0.7 area 0
```

```
REMOTE(config-router)#exit
```

```
REMOTE(config)#end
```

```
REMOTE#
```

```
REMOTE#copy run start
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

```
[OK]
```

```
REMOTE#
```

```
REMOTE#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
Gateway of last resort is not set
```

```
10.0.0.0/30 is subnetted, 1 subnets
```

```
O 10.10.20.0/30 [110/129] via 209.165.202.1, 00:07:15, GigabitEthernet0/0
```

```
172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
```

```
O 172.16.10.0/24 [110/66] via 209.165.202.1, 00:07:15, GigabitEthernet0/0
```

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C 172.16.20.0/24 is directly connected, GigabitEthernet0/1

L 172.16.20.1/32 is directly connected, GigabitEthernet0/1

O 192.168.10.0/24 [110/66] via 209.165.202.1, 00:07:15, GigabitEthernet0/0

O 192.168.20.0/24 [110/66] via 209.165.202.1, 00:07:15, GigabitEthernet0/0

209.165.200.0/29 is subnetted, 1 subnets

O 209.165.200.0/29 [110/65] via 209.165.202.1, 00:07:15, GigabitEthernet0/0

209.165.201.0/29 is subnetted, 1 subnets

O 209.165.201.0/29 [110/65] via 209.165.202.1, 00:07:15, GigabitEthernet0/0

209.165.202.0/24 is variably subnetted, 2 subnets, 2 masks

C 209.165.202.0/29 is directly connected, GigabitEthernet0/0

L 209.165.202.2/32 is directly connected, GigabitEthernet0/0

209.165.203.0/29 is subnetted, 1 subnets

O 209.165.203.0/29 [110/2] via 209.165.202.1, 00:07:15, GigabitEthernet0/0

REMOTE#conf t

REMOTE(config)#int g0/1

REMOTE(config-if)#no shut

REMOTE(config-if)#ip address 172.16.20.1 255.255.255.0

REMOTE(config-if)#end

REMOTE#conf t

REMOTE(config)#router ospf 10

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```
REMOTE(config-router)#network 172.16.20.0 0.0.0.255 area 0
```

```
REMOTE(config-router)#
```

```
REMOTE(config-router)#end
```

```
REMOTE#
```

Part 3: Configure a Dynamic NAT on HQ router using a pool of three addresses: 209.165.200.3 – 209.165.200.5 with a netmask of 255.255.255.248.

Step 1: Configure a simple access list that defines what hosts are going to be allowed for translation. In this case, all devices on the HQ LAN-1 are eligible for translation.

```
HQ#configure terminal
```

```
HQ(config)#
```

```
HQ(config)#access-list 10 permit 192.168.10.0 0.0.0.255
```

Step 2: Create the NAT pool and give it a name and a range of addresses to use.

```
HQ(config)#ip nat pool HQ-Pool 209.165.200.3 209.165.200.5 netmask 255.255.255.248
```

Step 3: Configure the translation, associating the ACL and Pool to the translation process.

```
HQ(config)#ip nat inside source list 10 pool HQ-Pool
```

Step 4: Define the inside interface and the outside interface.

```
HQ(config)#int g0/0
```

```
HQ(config-if)#ip nat inside
```

```
HQ(config-if)#int s0/2/0
```

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```
HQ(config-if)#ip nat outside
```

```
HQ(config-if)#end
```

```
HQ#
```

Step 5: Test and Verify the configuration.

```
HQ#show ip nat statistics
```

```
Total translations: 8 (0 static, 8 dynamic, 8 extended)
```

```
Outside Interfaces: Serial0/2/0
```

```
Inside Interfaces: GigabitEthernet0/0
```

```
Hits: 8 Misses: 32
```

```
Expired translations: 8
```

```
Dynamic mappings:
```

```
-- Inside Source
```

```
access-list 10 pool HQ-Pool refCount 8
```

```
pool HQ-Pool: netmask 255.255.255.248
```

```
start 209.165.200.3 end 209.165.200.5
```

```
type generic, total addresses 3 , allocated 1 (33%), misses 0
```

```
HQ#show ip nat translation
```

```
Pro Inside global Inside local Outside local Outside global
```

```
icmp 209.165.200.3:65 192.168.10.3:65 209.165.203.2:65 209.165.203.2:65
```

```
icmp 209.165.200.3:66 192.168.10.3:66 209.165.203.2:66 209.165.203.2:66
```

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```
icmp 209.165.200.3:67 192.168.10.3:67 209.165.203.2:67 209.165.203.2:67
```

```
icmp 209.165.200.3:68 192.168.10.3:68 209.165.203.2:68 209.165.203.2:68
```

```
icmp 209.165.200.3:69 192.168.10.3:69 209.165.203.3:69 209.165.203.3:69
```

```
icmp 209.165.200.3:70 192.168.10.3:70 209.165.203.3:70 209.165.203.3:70
```

```
icmp 209.165.200.3:71 192.168.10.3:71 209.165.203.3:71 209.165.203.3:71
```

```
icmp 209.165.200.3:72 192.168.10.3:72 209.165.203.3:72 209.165.203.3:72
```

```
HQ#
```

Part 4: Use the public IPv4 address 209.165.201.3 with netmask 255.255.255.248 to configure a PAT on DATA-CENTER router for the use of internal users.

Step 1: Configure a simple access list that defines what hosts are going to be allowed for translation. In this case, all devices on the DATA-CENTER LAN are eligible for translation.

```
DATA-CENTER>enable
```

```
DATA-CENTER#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
DATA-CENTER(config)#access-list 15 permit 172.16.10.0 0.0.0.255
```

```
DATA-CENTER(config)#
```

Step 2: Create the NAT pool and give it a name and a range of addresses to use.

```
DATA-CENTER(config)#ip nat pool DATA-CENTER-POOL 209.165.201.3 209.165.201.3 netmask 255.255.255.248
```

```
DATA-CENTER(config)#
```

Step 3: Configure the translation, associating the ACL and Pool to the translation process, and add the **OVERLOAD** keyword to the command.

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```
DATA-CENTER(config)#ip nat inside source list 15 pool DATA-CENTER-POOL overload
```

```
DATA-CENTER(config)#end
```

```
DATA-CENTER#
```

Step 4: Define the inside interface and the outside interface.

```
DATA-CENTER(config)#int g0/0
```

```
DATA-CENTER(config-if)#ip nat inside
```

```
DATA-CENTER(config-if)#int s0/3/0
```

```
DATA-CENTER(config-if)#ip nat outside
```

```
DATA-CENTER(config-if)#end
```

```
DATA-CENTER#
```

Step 5: Test and Verify the configuration.

```
DATA-CENTER#show ip nat statistics
```

```
Total translations: 0 (0 static, 0 dynamic, 0 extended)
```

```
Outside Interfaces: Serial0/3/0
```

```
Inside Interfaces: GigabitEthernet0/0
```

```
Hits: 16 Misses: 16
```

```
Expired translations: 16
```

```
Dynamic mappings:
```

```
-- Inside Source
```

```
access-list 15 pool DATA-CENTER-POOL refCount 0
```

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pool DATA-CENTER-POOL: netmask 255.255.255.248

start 209.165.201.3 end 209.165.201.3

type generic, total addresses 1 , allocated 0 (0%), misses 0

DATA-CENTER#

DATA-CENTER#show ip nat translation

Pro Inside global Inside local Outside local Outside global

icmp 209.165.201.3:33 172.16.10.254:33 209.165.203.3:33 209.165.203.3:33

icmp 209.165.201.3:34 172.16.10.254:34 209.165.203.3:34 209.165.203.3:34

icmp 209.165.201.3:35 172.16.10.254:35 209.165.203.3:35 209.165.203.3:35

icmp 209.165.201.3:36 172.16.10.254:36 209.165.203.3:36 209.165.203.3:36

icmp 209.165.201.3:37 172.16.10.254:37 209.165.203.2:37 209.165.203.2:37

icmp 209.165.201.3:38 172.16.10.254:38 209.165.203.2:38 209.165.203.2:38

icmp 209.165.201.3:39 172.16.10.254:39 209.165.203.2:39 209.165.203.2:39

icmp 209.165.201.3:40 172.16.10.254:40 209.165.203.2:40 209.165.203.2:40

DATA-CENTER#

Part 5: Configure a Static NAT on REMOTE router using the public IP 209.165.202.3 with a netmask 255.255.255.248 for the use of a Remote teleworker.

Step 1: Configure a static mapping between 172.16.20.3 and 209.165.202.2 using the following command:

REMOTE#configure terminal

REMOTE(config)#ip nat inside source static 172.16.20.3 209.165.202.2

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```
REMOTE(config)#
```

Step 2: Define the inside interface and the outside interface.

```
REMOTE(config)#int g0/1
```

```
REMOTE(config-if)#ip nat inside
```

```
REMOTE(config-if)#int g0/0
```

```
REMOTE(config-if)#ip nat outside
```

```
REMOTE(config-if)#end
```

```
REMOTE#
```

Step 3: Test and verify the configuration.

```
REMOTE#show ip nat translation
```

```
Pro Inside global Inside local Outside local Outside global
```

```
icmp 209.165.202.2:41 172.16.20.3:41 209.165.203.2:41 209.165.203.2:41
```

```
icmp 209.165.202.2:42 172.16.20.3:42 209.165.203.2:42 209.165.203.2:42
```

```
icmp 209.165.202.2:43 172.16.20.3:43 209.165.203.2:43 209.165.203.2:43
```

```
icmp 209.165.202.2:44 172.16.20.3:44 209.165.203.2:44 209.165.203.2:44
```

```
icmp 209.165.202.2:45 172.16.20.3:45 192.168.20.3:45 192.168.20.3:45
```

```
icmp 209.165.202.2:46 172.16.20.3:46 192.168.20.3:46 192.168.20.3:46
```

```
icmp 209.165.202.2:47 172.16.20.3:47 192.168.20.3:47 192.168.20.3:47
```

```
icmp 209.165.202.2:48 172.16.20.3:48 192.168.20.3:48 192.168.20.3:48
```

```
--- 209.165.202.2 172.16.20.3 --- ---
```

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```
REMOTE#show ip nat statistics
```

Total translations: 9 (1 static, 8 dynamic, 8 extended)

Outside Interfaces: GigabitEthernet0/0

Inside Interfaces: GigabitEthernet0/1

Hits: 8 Misses: 8

Expired translations: 0

Dynamic mappings:

```
REMOTE#
```

Part 6: Verify that NAT translations are working as expected.

Step 1: On PC1, ping the IP address of Webserver.

```
C:\>ping 209.165.203.2
```

Pinging 209.165.203.2 with 32 bytes of data:

Reply from 209.165.203.2: bytes=32 time=2ms TTL=126

Reply from 209.165.203.2: bytes=32 time=16ms TTL=126

Reply from 209.165.203.2: bytes=32 time=32ms TTL=126

Reply from 209.165.203.2: bytes=32 time=43ms TTL=126

Ping statistics for 209.165.203.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 43ms, Average = 23ms

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C:\>

Step 2: On HQ router, use the **show ip nat statistics** and **show ip nat translations** commands to verify real-time translations of Private IP address of PC1 to the Public IP address configured in the Pool.

HQ#show ip nat statistics

Total translations: 1 (0 static, **1 dynamic**, 1 extended)

Outside Interfaces: Serial0/2/0

Inside Interfaces: GigabitEthernet0/0

Hits: 15 Misses: 365

Expired translations: 29

Dynamic mappings:

-- Inside Source

access-list 10 pool HQ-Pool refCount 1

pool HQ-Pool: netmask 255.255.255.248

start 209.165.200.3 end 209.165.200.5

type generic, total addresses 3 , allocated 0 (0%), misses 0

HQ#show ip nat translation

Pro Inside global Inside local Outside local Outside global

icmp **209.165.200.3:87 192.168.10.3:87 209.165.203.2:87 209.165.203.2:87**

icmp 209.165.200.3:88 192.168.10.3:88 209.165.203.2:88 209.165.203.2:88

icmp 209.165.200.3:89 192.168.10.3:89 209.165.203.2:89 209.165.203.2:89

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```
icmp 209.165.200.3:90 192.168.10.3:90 209.165.203.2:90 209.165.203.2:90
```

HQ#

Step 3: Go to the Desktop of File Server, ping the IP address of Internet-User and Webserver.

C:\>ping 209.165.203.2

Pinging 209.165.203.2 with 32 bytes of data:

Reply from 209.165.203.2: bytes=32 time=35ms TTL=126

Reply from 209.165.203.2: bytes=32 time=12ms TTL=126

Reply from 209.165.203.2: bytes=32 time=27ms TTL=126

Reply from 209.165.203.2: bytes=32 time=27ms TTL=126

Ping statistics for 209.165.203.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 12ms, Maximum = 35ms, Average = 25ms

C:\>ping 209.165.203.3

Pinging 209.165.203.3 with 32 bytes of data:

Reply from 209.165.203.3: bytes=32 time=2ms TTL=126

Reply from 209.165.203.3: bytes=32 time=2ms TTL=126

Reply from 209.165.203.3: bytes=32 time=21ms TTL=126

Reply from 209.165.203.3: bytes=32 time=26ms TTL=126

Ping statistics for 209.165.203.3:

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Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 26ms, Average = 12ms

C:\>

Step 4: On DATA-CENTER router, use the **show ip nat statistics** and **show ip nat translations** commands to verify real-time translations of Private IP address of File Server to the Public IP address configured in the PAT Pool.

DATA-CENTER#show ip nat statistics

Total translations: 0 (0 static, 0 dynamic, 0 extended)

Outside Interfaces: Serial0/3/0

Inside Interfaces: GigabitEthernet0/0

Hits: 31 Misses: 288

Expired translations: 36

Dynamic mappings:

-- Inside Source

access-list 15 pool DATA-CENTER-POOL refCount 0

pool DATA-CENTER-POOL: netmask 255.255.255.248

start 209.165.201.3 end 209.165.201.3

type generic, total addresses 1 , allocated 0 (0%), misses 0

DATA-CENTER#show ip nat translation

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Pro Inside global Inside local Outside local Outside global

icmp 209.165.201.3:73 172.16.10.254:73 209.165.203.3:73 209.165.203.3:73

icmp 209.165.201.3:74 172.16.10.254:74 209.165.203.3:74 209.165.203.3:74

icmp 209.165.201.3:75 172.16.10.254:75 209.165.203.3:75 209.165.203.3:75

icmp 209.165.201.3:76 172.16.10.254:76 209.165.203.3:76 209.165.203.3:76

icmp 209.165.201.3:77 172.16.10.254:77 209.165.203.2:77 209.165.203.2:77

icmp 209.165.201.3:78 172.16.10.254:78 209.165.203.2:78 209.165.203.2:78

icmp 209.165.201.3:79 172.16.10.254:79 209.165.203.2:79 209.165.203.2:79

icmp 209.165.201.3:80 172.16.10.254:80 209.165.203.2:80 209.165.203.2:80

DATA-CENTER#

Step 5: Go to the Desktop of Remote PC, ping the IP address of Webserver and File Server.

Pinging 172.16.10.254 with 32 bytes of data:

Reply from 209.165.201.3: bytes=32 time=16ms TTL=125

Reply from 209.165.201.3: bytes=32 time=1ms TTL=125

Reply from 209.165.201.3: bytes=32 time=1ms TTL=125

Reply from 209.165.201.3: bytes=32 time=1ms TTL=125

Ping statistics for 172.16.10.254:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 16ms, Average = 4ms

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```
C:\>ping 209.165.203.2
```

Pinging 209.165.203.2 with 32 bytes of data:

Reply from 209.165.203.2: bytes=32 time=1ms TTL=126

Reply from 209.165.203.2: bytes=32 time=2ms TTL=126

Reply from 209.165.203.2: bytes=32 time=1ms TTL=126

Reply from 209.165.203.2: bytes=32 time=1ms TTL=126

Ping statistics for 209.165.203.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 2ms, Average = 1ms

```
C:\>
```

Step 6: On REMOTE router, use the **show ip nat statistics** and **show ip nat translations** commands to verify real-time translations of Private IP address of Remote PC to the Public IP address configured in under the Static NAT.

```
REMOTE#show ip nat statistics
```

Total translations: 13 (1 static, 12 dynamic, 12 extended)

Outside Interfaces: GigabitEthernet0/0

Inside Interfaces: GigabitEthernet0/1

Hits: 25 Misses: 43

Expired translations: 21

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Dynamic mappings:

REMOTE#show ip nat translation

Pro Inside global Inside local Outside local Outside global

icmp 209.165.202.2:57 172.16.20.3:57 172.16.10.254:57 172.16.10.254:57

icmp 209.165.202.2:57 172.16.20.3:57 209.165.201.3:57 209.165.201.3:57

icmp 209.165.202.2:58 172.16.20.3:58 172.16.10.254:58 172.16.10.254:58

icmp 209.165.202.2:58 172.16.20.3:58 209.165.201.3:58 209.165.201.3:58

icmp 209.165.202.2:59 172.16.20.3:59 172.16.10.254:59 172.16.10.254:59

icmp 209.165.202.2:59 172.16.20.3:59 209.165.201.3:59 209.165.201.3:59

icmp 209.165.202.2:60 172.16.20.3:60 172.16.10.254:60 172.16.10.254:60

icmp 209.165.202.2:64 172.16.20.3:64 209.165.203.2:64 209.165.203.2:64

--- 209.165.202.2 172.16.20.3 --- ---

Project 6 – Optimize the Enterprise Network with Features such as Syslog, NTP and SSH Operations.

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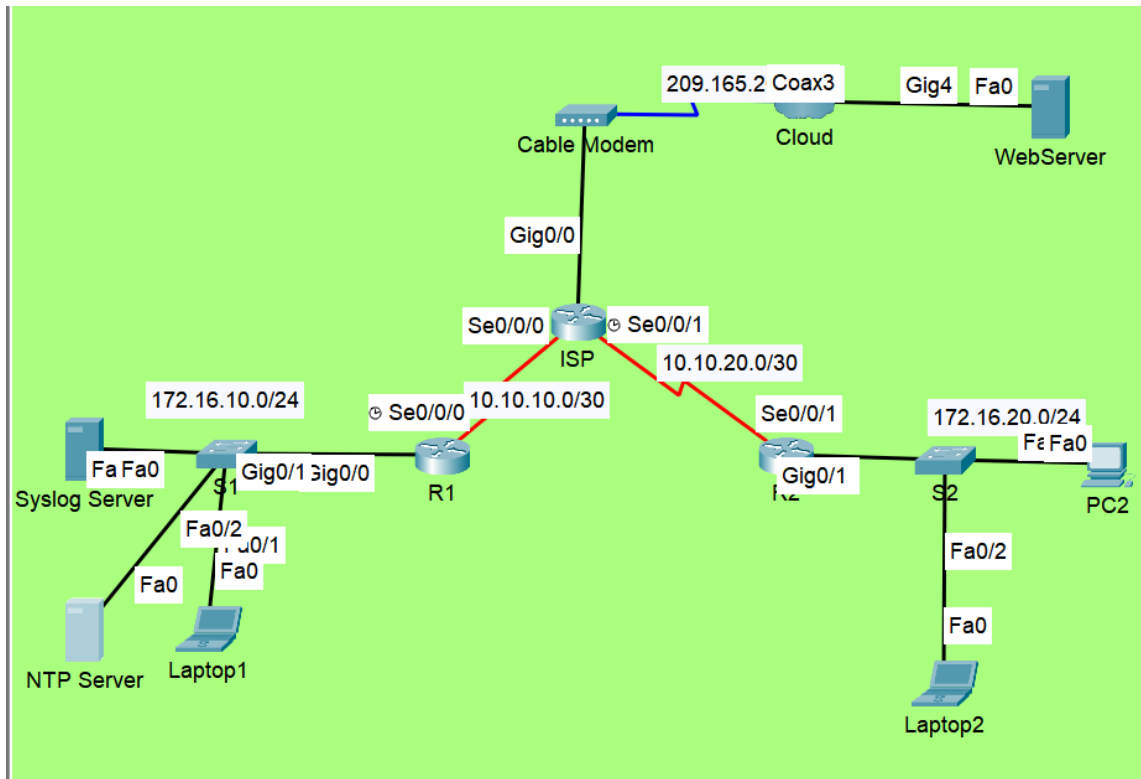


Figure 18.1: Lab 1 Topology

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Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	172.16.10.1	255.255.255.0	n/a
	S0/0/0	10.10.10.1	255.255.255.252	n/a
R2	G0/1	172.16.20.1	255.255.255.0	n/a
	S0/0/1	10.10.20.1	255.255.255.252	n/a
ISP	S0/0/0	10.10.10.2	255.255.255.252	n/a
	S0/0/1	10.10.20.2	255.255.255.252	n/a
	G0/0	209.165.200.1	255.255.255.0	n/a
NTP Server	NIC	172.16.10.254	255.255.255.0	172.16.10.1
Syslog Server	NIC	172.16.10.253	255.255.255.0	172.16.10.1
Laptop1	NIC	172.16.10.3	255.255.255.0	172.16.10.1
Laptop2	NIC	172.16.20.3	255.255.255.0	172.16.20.1
Web Server	NIC	209.165.200.254	255.255.255.0	209.165.200.1

Table 18.1: Lab 1 - Addressing Table

Background

In this activity, you will configure features such as OSPF, NTP, enable Syslog server to provide messaging logging.

Logging protocol – allows for the monitoring of network devices by sending event messages generated by network device to a logging server. These messages contain information such as timestamps, device ID and IP address, event severity rating, and event-specific information.

Network Time Protocol (NTP) – is an internet protocol used to synchronize time between devices on the network to ensure that all devices maintain the same clock. Accurate time for all devices on

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a computer network is important for many reasons; a discrepancy of even a fraction of a second can cause problems.

The Syslog Server will provide message logging in this activity. You will configure the routers (R1 and R2) to identify the Syslog server that will receive logging messages.

You will need to configure timestamp service for logging on the routers. Displaying the correct time and date in Syslog messages is vital when using Syslog to monitor a network.

Objectives

- Configure endpoints.
- Configure basic settings on all routers and enable OSPF process 1, Area 0.
- Configure the NTP Server.
- Configure the routers to synchronize clock with the NTP server.
- Enable the Syslog service on the Syslog server.
- Configure routers to log messages to the syslog server.
- Configure R2 to support SSH connections.
- Establish SSH connection from Laptop1 and Laptop2 to R2.
- Verify end-to-end connectivity.

Instructions:

Part 1: Assign IP addresses on all endpoints.

Part 2: Configure basic settings on all routers and enable OSPF process 1, Area 0.
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Step 1: R1 configuration

```
Router>enable
Router#configure terminal
Router(config)#hostname R1
R1(config)#int g0/0
R1(config-if)#no shut
R1(config-if)#ip address 172.16.10.1 255.255.255.0
R1(config-if)#int s0/0/0
R1(config-if)#no shut
R1(config-if)#ip address 10.10.10.1 255.255.255.252
R1(config-if)#exit
R1(config)#router ospf 1
R1(config-router)#network 172.16.10.0 0.0.0.255 area 0
R1(config-router)#network 10.10.10.0 0.0.0.3 area 0
R1(config-router)#exit
R1(config)#exit
R1#copy run start
R1#
```

Step 2: R2 configuration.

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```
Router>enable
Router#configure terminal
Router(config)#hostname R2
R2(config)#int g0/1
R2(config-if)#no shut
R2(config-if)#ip address 172.16.20.1 255.255.255.0
R2(config-if)#int s0/0/1
R2(config-if)#no shut
R2(config-if)#ip address 10.10.20.1 255.255.255.252
R2(config-if)#exit
R2(config)#router ospf 1
R2(config-router)#network 172.16.20.0 0.0.0.255 area 0
R2(config-router)#network 10.10.20.0 0.0.0.3 area 0
R2(config-router)#exit
R2(config)#end
R2#copy run start
R2#
```

Step 3: ISP configuration.

```
ISP>enable
ISP#configure terminal
```

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Enter configuration commands, one per line. End with CNTL/Z.

```
ISP(config)#hostname ISP
```

```
ISP(config)#interface GigabitEthernet0/0
```

```
ISP(config-if)# ip address 209.165.200.1 255.255.255.0
```

```
ISP(config-if)#no shutdown
```

```
ISP(config-if)#interface Serial0/0/0
```

```
ISP(config-if)#no shutdown
```

```
ISP(config-if)# ip address 10.10.10.2 255.255.255.252
```

```
ISP(config-if)#interface Serial0/0/1
```

```
ISP(config-if)# ip address 10.10.20.2 255.255.255.252
```

```
ISP(config-if)#no shutdown
```

```
ISP(config-if)#
```

```
ISP(config-if)#exit
```

```
ISP(config)#
```

```
ISP(config)#router ospf 1
```

```
ISP(config-router)# network 209.165.200.0 0.0.0.255 area 0
```

```
ISP(config-router)# network 10.10.10.0 0.0.0.3 area 0
```

```
ISP(config-router)# network 10.10.20.0 0.0.0.3 area 0
```

```
ISP(config-router)#
```

```
ISP(config-router)#exit
```

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```
ISP(config)#ntp server 172.16.10.254
ISP(config)#logging host 172.16.10.253
ISP(config)#service timestamps log datetime msec
ISP(config)#
ISP(config)#ntp authentication-key 1 md5 0802455D0A165445415F 7
ISP(config)#ntp authenticate
ISP(config)#ntp trusted-key 1
ISP(config)#ntp server 172.16.10.254
ISP(config)#ntp update-calendar
ISP(config)#
ISP(config)#end
```

Part 3: Configure the NTP Server.

Step 1: On NTP server, go to Desktop. Under the **Services** tab, click on NTP and make sure it is turned ON.

Step 2: Click **Enable** under Authentication. Use key **1** and password **Cisco1234** for authentication.

See the image below:

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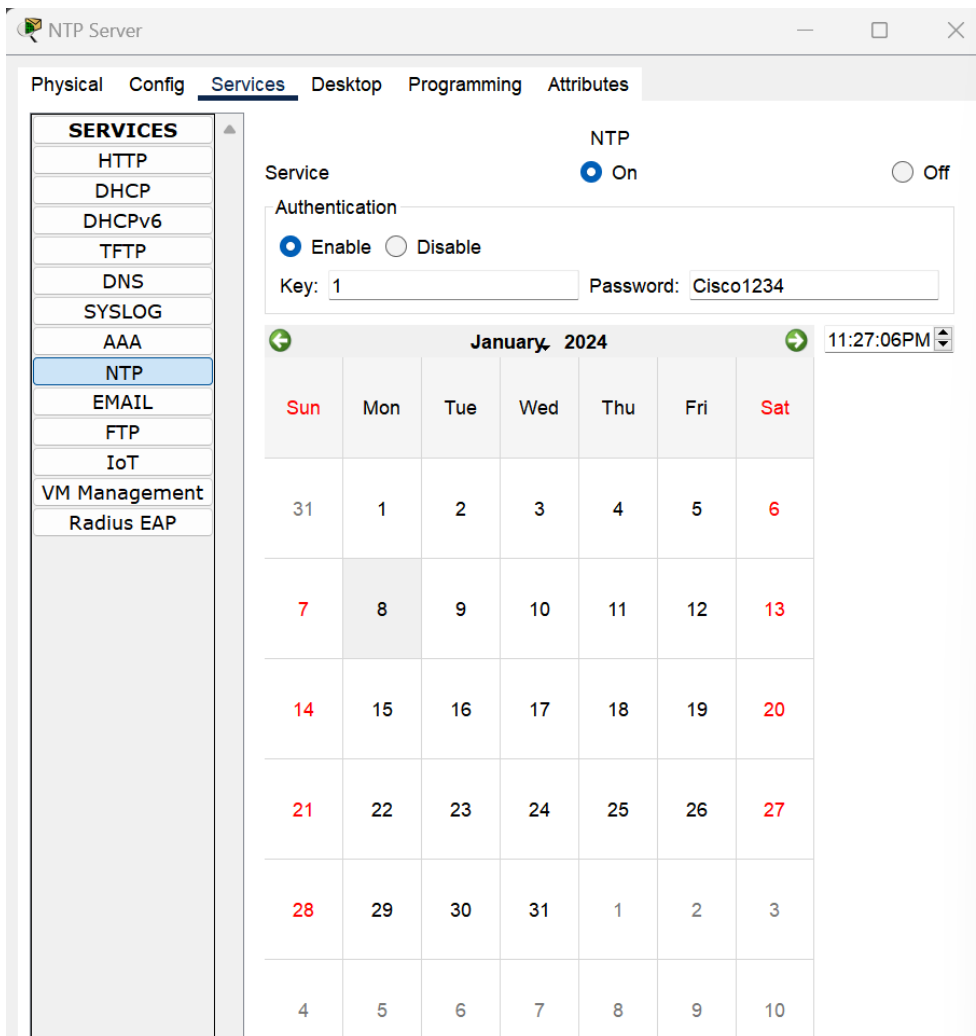


Figure 18.2: NTP Server configuration.

Part 4a: Configure the routers to synchronize clock with the NTP server.

Step 1: R1 configuration

```
R1>enable
```

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```
R1#configure terminal
```

```
R1(config)#ntp server 172.16.10.254
```

```
R1(config)#end
```

Step 2: R2 configuration.

```
R2>enable
```

```
R2#configure terminal
```

```
R2(config)#ntp server 172.16.10.254
```

```
R2(config)#
```

Step 3: ISP configuration.

```
ISP>enable
```

```
ISP#configure terminal
```

```
ISP(config)#ntp server 172.16.10.254
```

```
ISP(config)#
```

Part 4b: Configure NTP authentication on the routers.

Configure NTP authentication on **R1**, **R2**, and **ISP** using key **1** and password **Cisco1234**.

Step 1: R1 configuration.

```
R1#conf t
```

```
R1(config)#ntp authenticate
```

```
R1(config)#ntp trusted-key 1
```

```
R1(config)#ntp authentication-key 1 md5 Cisco1234
```

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```
R1(config)#
```

```
R1#show ntp status
```

Clock is synchronized, stratum 2, reference is 172.16.10.254

nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24

reference time is E920D15A.0000020D (0:7:54.525 UTC Wed Jan 10 2024)

clock offset is 1.00 msec, root delay is 0.00 msec

root dispersion is 10.69 msec, peer dispersion is 0.12 msec.

loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system poll interval is 4, last update was 5 sec ago.

```
R1#
```

Step 2: R2 configuration.

```
R2#conf t
```

```
R2(config)#ntp authenticate
```

```
R2(config)#ntp trusted-key 1
```

```
R2(config)#ntp authentication-key 1 md5 Cisco1234
```

```
R2(config)#
```

```
R2(config)#end
```

```
R2#
```

```
R2#show ntp status
```

Clock is synchronized, stratum 2, reference is 172.16.10.254

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nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24

reference time is E920D16F.0000007C (0:8:15.124 UTC Wed Jan 10 2024)

clock offset is 0.00 msec, root delay is 9.00 msec

root dispersion is 10.03 msec, peer dispersion is 0.12 msec.

loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system poll interval is 4, last update was 5 sec ago.

R2#

Step 3: ISP configuration.

```
ISP(config)#ntp authenticate
```

```
ISP(config)#ntp trusted-key 1
```

```
ISP(config)#ntp authentication-key 1 md5 Cisco1234
```

```
ISP(config)#end
```

ISP#

ISP#show ntp status

Clock is synchronized, stratum 2, reference is 172.16.10.254

nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24

reference time is E920D1EB.00000099 (0:10:19.153 UTC Wed Jan 10 2024)

clock offset is 1.00 msec, root delay is 2.00 msec

root dispersion is 10.93 msec, peer dispersion is 0.12 msec.

ISP#

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Part 4c: Configure routers to update hardware clock.

Configure **R1**, **R2**, and **ISP** to periodically update the hardware clock with the time learned from NTP.

Step 1: R1 configuration.

```
R1(config)# ntp update-calendar
```

```
R1(config)#end
```

```
R1#
```

```
R1#show clock
```

```
0:14:59.37 UTC Wed Jan 10 2024
```

```
R1#
```

Step 2: R2 configuration.

```
R2(config)# ntp update-calendar
```

```
R2(config)#end
```

```
R2#
```

```
R2#show clock
```

```
0:16:8.818 UTC Wed Jan 10 2024
```

```
R2#
```

Step 3: ISP configuration.

```
ISP#configure terminal
```

```
ISP(config)# ntp update-calendar
```

```
ISP(config)#end
```

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ISP#

ISP#show clock

0:16:40.358 UTC Wed Jan 10 2024

ISP#

Part 5: Enable the Syslog service on the Syslog server.

Step 1: On the Desktop of Syslog server, select Services tab, click on **SYSLOG** option, and ensure it is turned ON. See the image below.

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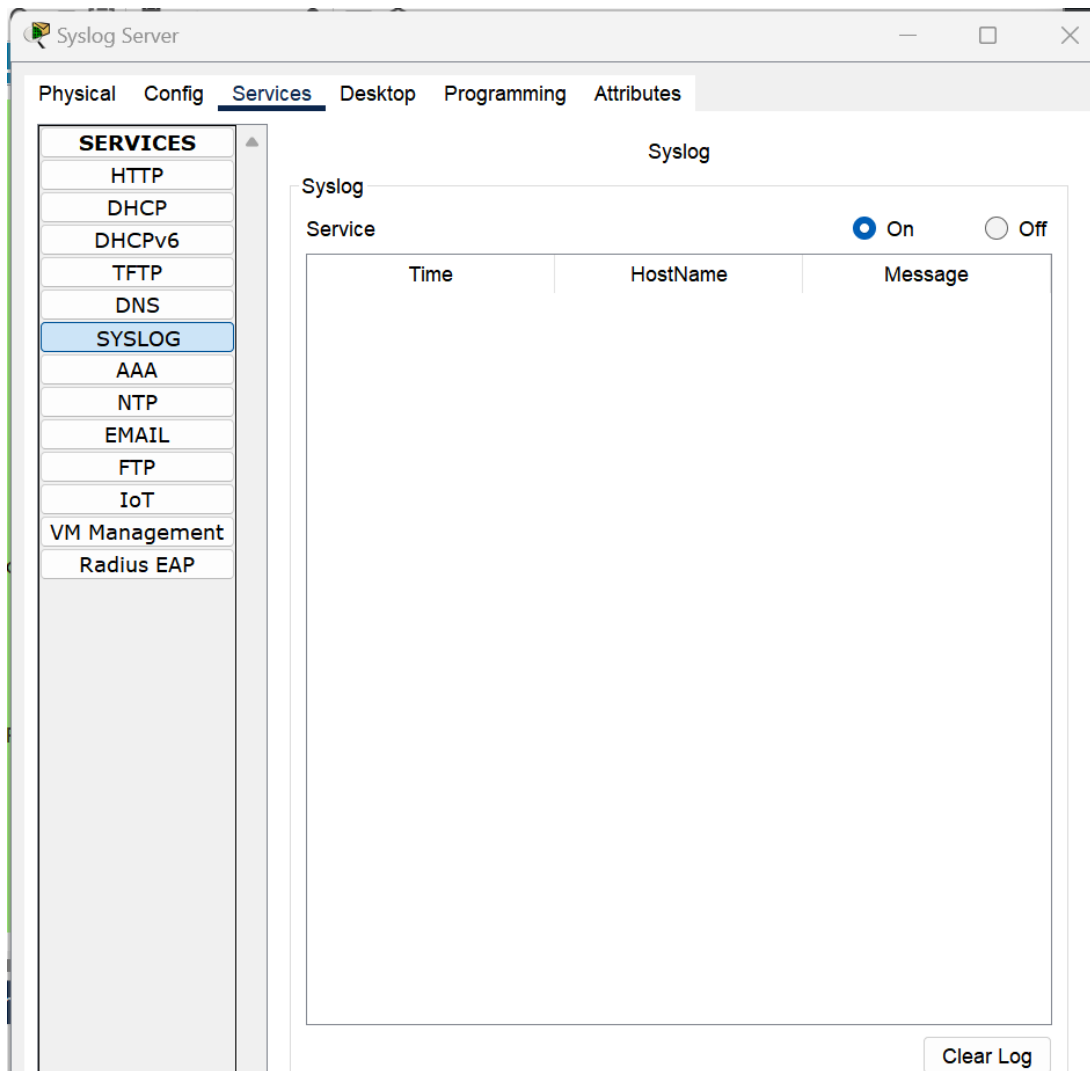


Figure 18.3: Lab 1 - Syslog server.

Part 6: Configure routers to log messages to the syslog server.

Step 1: Configure routers to timestamp log messages.

R1 configuration.

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```
R1>enable
```

```
R1#configure terminal
```

```
R1(config)#service timestamps log datetime msec
```

```
R1(config)#
```

R2 configuration.

```
R2>enable
```

```
R2#configure terminal
```

```
R2(config)#service timestamps log datetime msec
```

```
R2(config)#
```

ISP configuration.

```
ISP>enable
```

```
ISP#configure terminal
```

```
ISP(config)#service timestamps log datetime msec
```

```
ISP(config)#
```

Step 2: Configure Routers to Log Messages to the Syslog Server

Requirement: Configure the routers to identify the remote host (Syslog Server) that will receive logging messages.

R1 configuration.

```
R1(config)#logging host 172.16.10.253
```

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```
R1(config)#end
```

```
R1#
```

R2 configuration.

```
R2(config)#logging host 172.16.10.253
```

```
R2(config)#end
```

```
R2#
```

ISP configuration.

```
ISP(config)#logging host 172.16.10.253
```

```
ISP(config)#end
```

```
ISP#
```

Step 4: Verify logging configuration.

- Use the command **show logging** to verify logging has been enabled.

R1 configuration.

```
R1#show logging
```

Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,

Logging to 172.16.10.253 (udp port 514, audit disabled,

authentication disabled, encryption disabled, link up),

2 message lines logged,

R2 configuration.

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R2#show logging

Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,

Trap logging: level informational, 15 message lines logged

Logging to 172.16.10.253 (udp port 514, audit disabled,

authentication disabled, encryption disabled, link up),

2 message lines logged,

ISP configuration.

ISP#show logging

Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,

Trap logging: level informational, 21 message lines logged

Logging to 172.16.10.253 (udp port 514, audit disabled,

authentication disabled, encryption disabled, link up),

2 message lines logged,

Step 5: Verify that logs are sent to the Syslog Server.

- From the **Services** tab of the **Syslog Server**'s dialogue box, select the **Syslog** services button. Observe the logging messages received from the routers. See image below.

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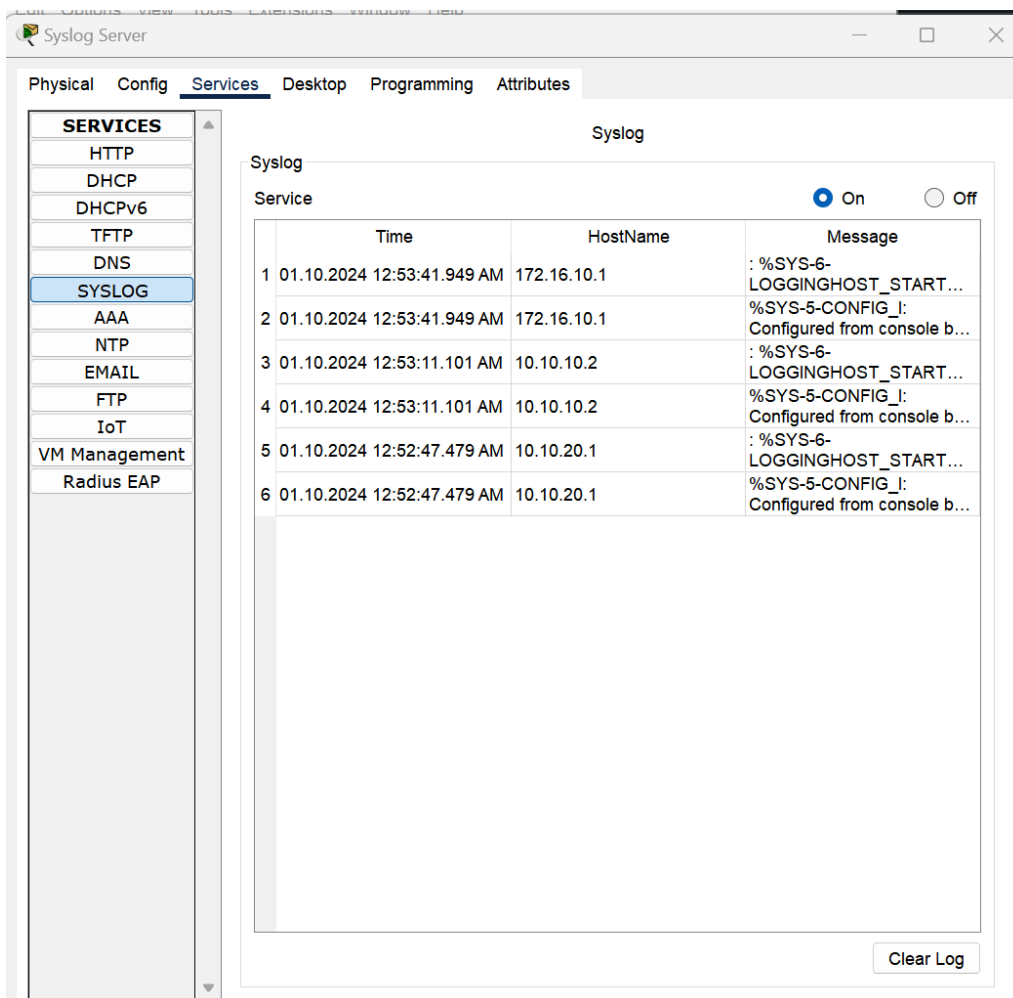


Figure 18.4: Logging activities verification on Syslog server

Part 7: Configure R2 to support SSH connections.

Requirements:

1. Configure privilege mode password using **Cisco1234**.
2. Create a user with username **Admin1** and secret **admin123**.

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3. Create a second user with username **Admin2** and secret **admin234**.
4. Configure domain name as **www.corp.net**.
5. Generate crypto RSA key using **1024** bit-key.
6. Enable SSH version 2.
7. Configure line **vty 0 15** to accept on SSH connection and enable **local authentication**.
8. Use **Cisco1234** as passwords for Line VTY and Line Console.

R2 configuration.

```
R2#configure terminal
```

```
R2(Config)#enable secret Cisco1234
```

```
R2(config)#username Admin1 secret admin123
```

```
R2(config)#username Admin2 secret admin234
```

```
R2(config)#ip domain-name www.corp.net
```

```
R2(config)#crypto key generate rsa general-key modulus 1024
```

The name for the keys will be: **R2.www.corp.net**

% The key modulus size is 1024 bits

% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

*Jan 10 1:17:17.815: %SSH-5-ENABLED: SSH 1.99 has been enabled

```
R2(config)#ip ssh version 2
```

```
R2(config)#line vty 0 15
```

```
R2(config-line)#password Cisco1234
```

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```
R2(config-line)#login local
```

```
R2(config-line)#transport input ssh
```

```
R2(config-line)#exit
```

```
R2(config)#line console 0
```

```
R2(config-line)#password Cisco1234
```

```
R2(config-line)#login
```

```
R2(config)#end
```

```
R2#copy run start
```

```
R2#
```

Part 8: Establish SSH connection from Laptop1 and Laptop2 to R2.

Step 1: On Laptop1, establish SSH connection to R2 IP address 10.10.20.1.

Note: Username is **Admin1**, and password is **admin123**.

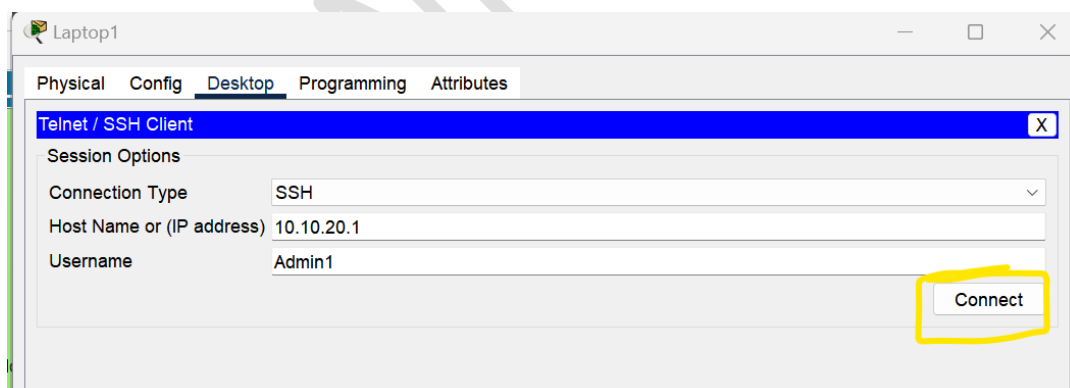


Figure 18.5: SSH client setup on Laptop1

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```

Laptop1
Physical Config Desktop Programming Attributes
SSH Client
Password:

R2>enable
Password:
R2#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R2#show ssh
Connection      Version Mode Encryption      Hmac      State      Username
0                1.99  IN  aes128-cbc      hmac-sha1  Session Started  Admin1
0                1.99  OUT  aes128-cbc      hmac-sha1  Session Started  Admin1
1                1.99  IN  aes128-cbc      hmac-sha1  Session Started  Admin1
1                1.99  OUT  aes128-cbc      hmac-sha1  Session Started  Admin1
%No SSHv1 server connections running.
R2#show user
      Line      User      Host(s)      Idle      Location
0 con 0
134 vty 0      Admin1      idle         00:01:17
*135 vty 1      Admin1      idle         00:06:13
00:00:00

Interface      User      Mode      Idle      Peer Address
R2#
  
```

Figure 18.6: Successful SSH connection to R1

Step 2: On Laptop2, establish SSH connection to R2 IP address 10.10.20.1.

Note: Username is **Admin2** and password is **admin234**.

The screenshot shows the "Telnet / SSH Client" window on Laptop2. The "Session Options" section is visible, with the following fields:

- Connection Type: SSH (selected from a dropdown menu)
- Host Name or (IP address): 10.10.20.1
- Username: Admin2

A "Connect" button is located at the bottom right of the window.

Figure 18.7: SSH client setup on Laptop2

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```

Laptop2
Physical Config Desktop Programming Attributes
SSH Client
Password:

R2>enable
Password:
R2#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R2#show ssh
Connection      Version Mode Encryption      Hmac      State      Username
0               1.99  IN   aes128-cbc      hmac-sha1  Session Started  Admin1
0               1.99  OUT  aes128-cbc      hmac-sha1  Session Started  Admin1
1               1.99  IN   aes128-cbc      hmac-sha1  Session Started  Admin2
1               1.99  OUT  aes128-cbc      hmac-sha1  Session Started  Admin2
%No SSHv1 server connections running.
R2#show user
Line           User      Host(s)      Idle      Location
0 con 0
134 vty 0      Admin1     idle         00:04:30
*135 vty 1     Admin2     idle         00:00:00

Interface      User      Mode      Idle      Peer Address
R2#

```

Figure 18.8: Successful SSH connection to R2

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Project 7 – Implement Enterprise WLAN Network

You will apply your WLAN skills and knowledge by configuring a home wireless router and an enterprise WLC. You will implement both WPA2-PSK and WPA2-Enterprise security. Finally, you will connect hosts to each WLAN and verify connectivity.

An organization is centralizing control of their wireless LAN by replacing their standalone access points with lightweight access points (LAP) and a wireless LAN controller (WLC). You will be leading this project, and you want to become familiar with the WLC and any potential challenges that may occur during the project. You will configure a WLC by adding a new wireless network and securing it with WPA-2 PSK security. To test the configuration, you will connect a laptop to the WLAN and ping devices on the network.

You have already configured and tested the WLC with an existing WLAN. You configured WPA2-PSK for that WLAN because it was to be used in a smaller business. You have been asked to configure and test a WLC topology that will be used in a larger enterprise. You know that WPA2-PSK does not scale well and is not appropriate to use in an enterprise network. This new topology will use a RADIUS server and WPA2-Enterprise to authenticate WLAN users. This allows administration of the user accounts from a central location and provides enhanced security and transparency because each account has its own username and password. In addition, user activity is logged on the server. In this lab, you will create a new VLAN interface, use that interface to create a new WLAN, and secure that WLAN with WPA2-Enterprise. You will also configure the WLC to use the enterprise RADIUS server to authenticate users. In addition, you will configure the WLC to use an SNMP server.

Project Scope

Build strong Foundation for STEM Literacy; Increase Diversity, Equity, and Inclusion in the STEM workforce; Prepare the STEM workforce for the future:



Part 1: ISP-RTA & ISP-RTB:

1. configure interfaces with ipv4 addresses.
2. configure OSPF Area 0 to provide routing services.
3. configure DHCP service on ISP-RTA for the interface connected to Home Wireless router.

Part 2: Remote User Home Wireless Network:

1. configure the Internet interface to obtain DHCP address from ISP-RTA.
2. configure the LAN network to provide DHCP services to LAN users.
3. configure WPA2-AES pre-share key for LAN wireless clients.
4. change the default administrator password.
5. connect wireless clients to the network.

Part 3: Corporate Network Configuration.

Step 1: Corp-R1 configuration:

1. configure interfaces with ipv4 addresses.

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2. configure a default static route so that internal Corp users can send and receive traffic from external networks.
3. configure DHCP service for VLAN 10 (WLAN-10) users.
4. configure DHCP service for VLAN 20 (WLAN-20) users.
5. configure DHCP service for VLAN 100 (management/native VLAN) users.
6. configure sub-interfaces on the physical interface G0/0/0 to provide inter-VLAN routing.

Part 4: Setup CorpRadius-Server

1. configure the Radius server to provide centralized authentication services to VLAN 20 (WLAN-20) network.

Part 5: WebServer

1. enable HTTPS service on the server.

Part 6: CorpWLC

1. configure interfaces on the WLC to support the two WLANs

Note: Make sure the physical interface (port) number also correspond to the interface number selected during configuration under the Controller tab.

2. configure DHCP scope for the wireless management network.

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3. configure the WLC to redirect authentication requests to the external RADIUS server address.

4. create the WLANs

Part 7: Wireless Clients

1. Perform the correct configurations on wireless clients to connect to the wireless network through the Corp-LAP1

Topology

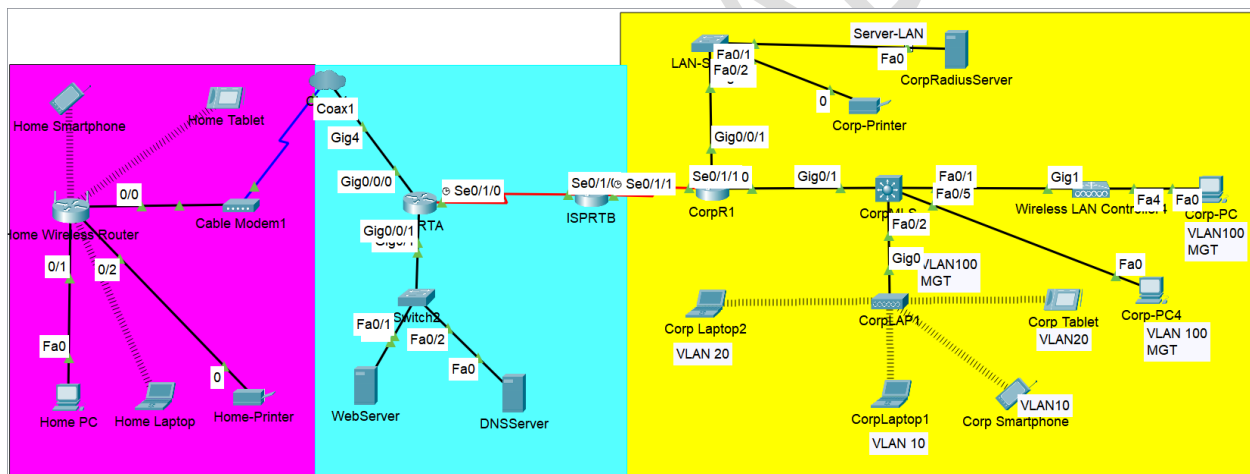


Figure 18.2: WLAN for Enterprise

Addressing Table.

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Addressing Table

Device	Interface	IP Address
Home Wireless Router	Internet	DHCP
	LAN	192.168.5.0/24
CorpR1	G0/0/0.10	192.168.10.1/24
	G0/0/0.20	192.168.20.1/24
	G0/0/0.100	192.168.100.1/24
	G0/0/1	192.168.30.1/24
	S0/1/1	209.165.200.225/30
ISP-RTB	S0/1/1	209.165.200.226/30
	S0/1/0	209.165.201.225/30
ISP-RTA	S0/1/0	209.165.201.226/30
	G0/0/0	209.165.202.225/30
	G0/0/1	209.165.203.1/24
Corp-WLC	Management	192.168.100.254/24
Corp-LAP1	G0/0	DHCP
Corp-MLS	VLAN 100 (Management/Native)	192.168.100.100/24
Corp-Radius-Server	Fa0/0	192.168.30.252/24
Corp-Printer	Fa0/0	192.168.30.253/24
Corp-PC	Fa0/0	DHCP
Corp-Laptop1	Wireless	DHCP

Table 18.1a: Addressing Table

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Device	Interface	IP Address
Corp-Laptop1	Wireless	DHCP
Corp-Laptop2	Wireless	DHCP
Corp Smartphone	Wireless	DHCP
Corp Tablet	Wireless	DHCP
Corp IP Phone	DHCP	DHCP
Home Smartphone	Wireless	DHCP
Home Tablet	Wireless	DHCP
Home PC	NIC	DHCP
Home Laptop	Wireless	DHCP
Home Printer	NIC	DHCP
DNS-Server	NIC	209.165.203.2/24
ISP-Web Server	NIC	209.165.203.11/24

Table 18.1b: Addressing Table

WLAN Information

WLAN	SSID	Authentication	Username	Password
Home Network	HomeSSID	WPA2-Personal	N/A	Pass1234
WLAN VLAN 10	SSID-10	WPA-2 Personal	N/A	MySecret
WLAN VLAN 20	SSID-20	WPA-2 Enterprise	User1	User1Pass

Table 18.1c: WLAN Information.

Instructions

Part 1: Configure ISP-RTA & ISP-RTB:

1. configure interfaces with ipv4 addresses.
2. configure OSPF Area 0 to provide routing services.
3. configure DHCP service on ISP-RTA for the interface connected to

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Home Wireless router.

Step 1: Below is the output of ISP-RTA router configuration.

```
ISP-RTA>en
ISP-RTA#show run
hostname ISP-RTA
ip dhcp excluded-address 209.165.203.1 209.165.203.10
!
ip dhcp pool LAN-POOL
network 209.165.203.0 255.255.255.0
default-router 209.165.203.1
dns-server 209.165.203.2
domain-name www.isp.com
ip dhcp pool ISP-POOL-TO-HOME-ROUTER
network 209.165.202.224 255.255.255.252
default-router 209.165.202.225
dns-server 209.165.203.2
interface GigabitEthernet0/0/0
description CONNECTION TO REMOTE WIRELESS ROUTER
ip address 209.165.202.225 255.255.255.252
```

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```
interface GigabitEthernet0/0/1
description CONNECTION TO WEB-SERVER-LAN
ip address 209.165.203.1 255.255.255.0
interface GigabitEthernet0/0/2
interface Serial0/1/0
description CONNECTION TO ISP-RTB
ip address 209.165.201.226 255.255.255.252
clock rate 2000000
!
router ospf 1
router-id 1.1.1.1
redistribute static
network 209.165.202.224 0.0.0.3 area 0
network 209.165.203.0 0.0.0.255 area 0
network 209.165.201.224 0.0.0.3 area 0
network 192.168.5.0 0.0.0.255 area 0
default-information originate
ip route 0.0.0.0 0.0.0.0 209.165.202.226
ISP-RTA#
```

Step 2: Below is the output of ISP-RTB router configuration.
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ISP-RTB#show run

Building configuration...

Current configuration : 1200 bytes

hostname ISP-RTB

interface GigabitEthernet0/0/0

no ip address

!

interface GigabitEthernet0/0/1

!

interface GigabitEthernet0/0/2

no ip address

interface Serial0/1/0

description CONNECTION TO ISP-RTA

ip address 209.165.201.225 255.255.255.252

!

interface Serial0/1/1

description CONNECTION TO CORP-R1

ip address 209.165.200.226 255.255.255.252

clock rate 2000000!

!

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```
router ospf 1
router-id 2.2.2.2
log-adjacency-changes
redistribute static
network 209.165.200.224 0.0.0.3 area 0
network 209.165.201.224 0.0.0.3 area 0
default-information originate
ISP-RTB#
```

Part 2: Configure Remote User Home Wireless Network

Requirement:

Step 1: On the Desktop of Home PC, open a web browser and type 192.168.0.1. On the login page, enter username admin and password as admin. Then click on Setup tab.

Step 2: Configure the Internet interface to obtain DHCP address from ISP-RTA.

Step 3: Configure the LAN network to provide DHCP services to only 20 LAN users. See details below.

a. Network address: 192.168.5.0

b. Subnet 255.255.255.0

c. Enable DHCP server.

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- d. Start IP: 192.168.5
- e. DNS server: 209.165.203.2
- f. Scroll down and click on Save Settings.

The screenshot shows the "Home Wireless Router" configuration page. The "Config" tab is selected, and the "Setup" sub-tab is active. The "Internet Setup" section shows "Automatic Configuration - DHCP" selected. The "Network Setup" section shows the "Router IP" configuration with IP Address: 192.168.5.1 and Subnet Mask: 255.255.255.192. The "DHCP Server Settings" section shows the DHCP Server is "Enabled" with a Start IP Address of 192.168.5.3, Maximum number of Users: 20, IP Address Range: 192.168.5.3 - 22, Client Lease Time: 0 minutes, and Static DNS 1: 209.165.203.2.

Figure 18.9: Home Wireless Router setup

Step 4: Configure WPA2-AES pre-share key for LAN wireless clients.

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1. Click on Wireless tab and under 2.4 GHz, click Enabled button.
2. Enter Network Name (SSID) as: HomeSSID.
3. Select Standard Channel as 6 – 2.437 GHz.
4. Disable both 5 GHz 1 and 2.
5. Scroll down to Save Settings.
6. Then click on Wireless Security tab under the Security tab.
7. Select 2.4 GHz and select Security Mode as: WPA2 Personal.
8. Select Encryption as AES.
9. Enter Passphrase as: Pass1234
10. Scroll down to Save Settings

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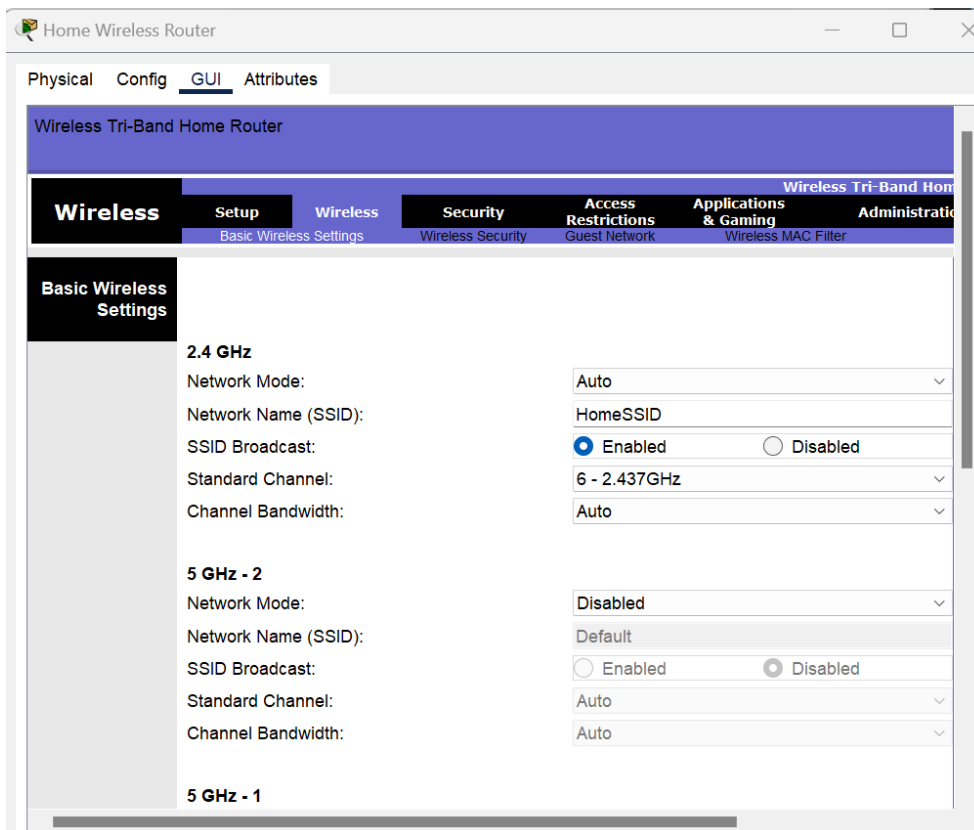


Figure 18.10: Home Wireless Router setup

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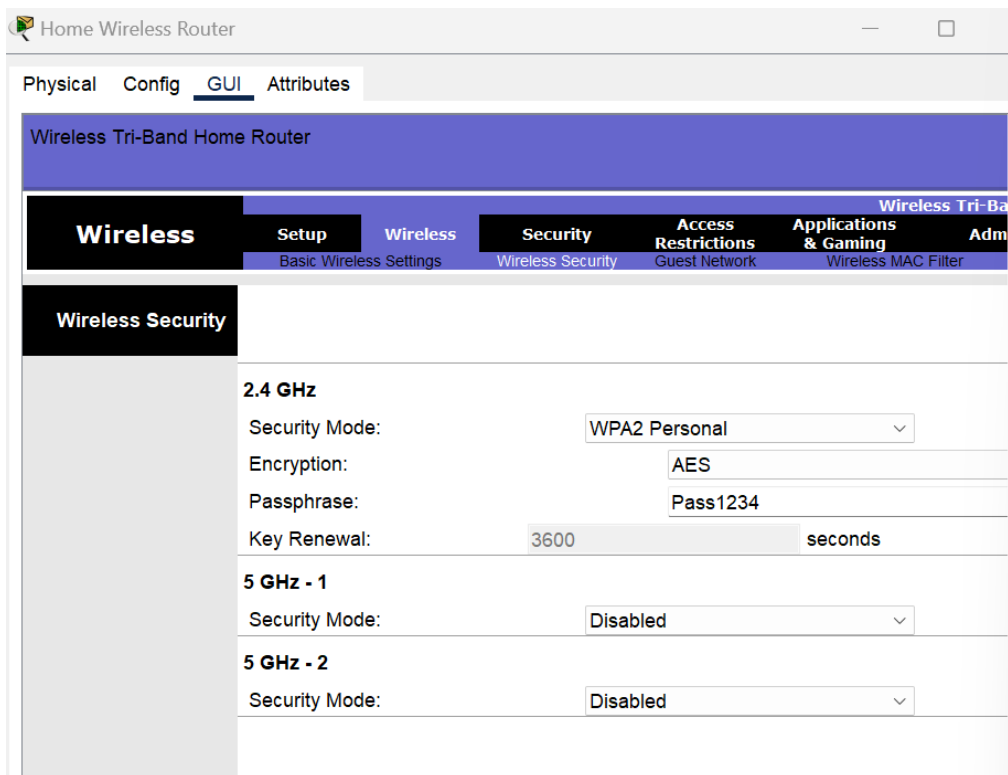


Figure 18.22: Home Laptop setup

Step 5: Change the default administrator password.

1. Click on Administration tab.
2. Change the Router Password to: **Cisco1234** and confirm the password.
3. Scroll down to Save Settings.

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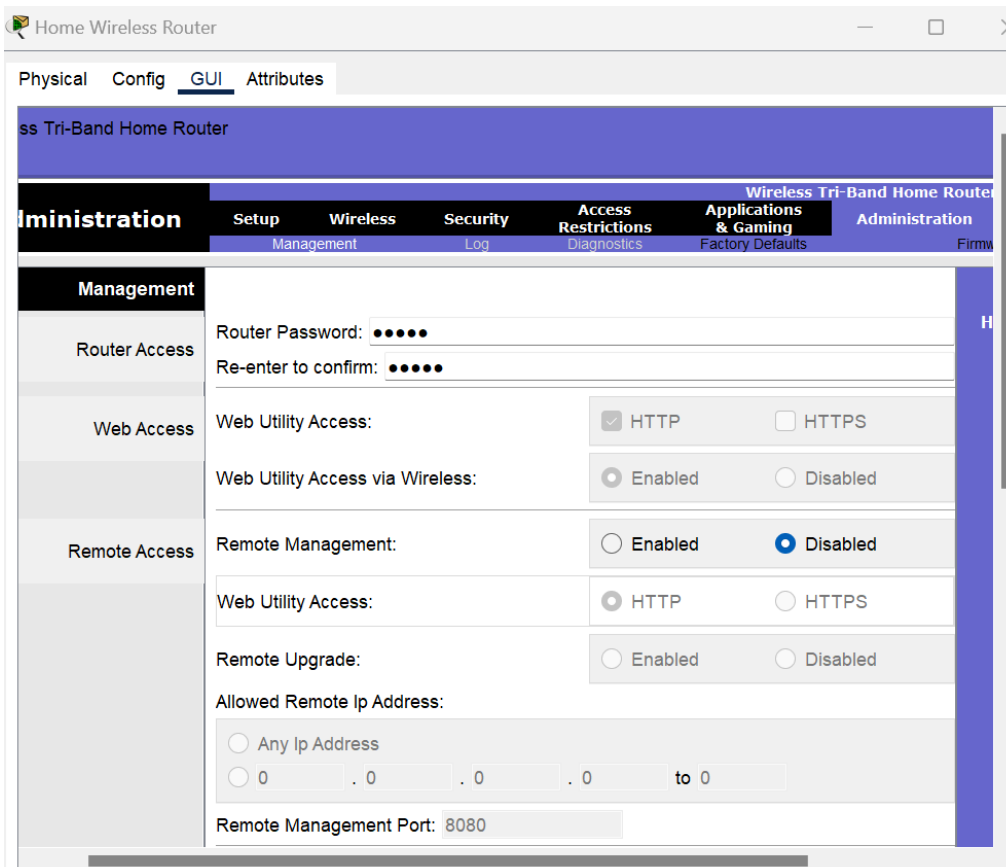


Figure 18.11: Home Wireless Router setup.

Step 6: Connect wireless clients to the network.

1. On the Desktop of Home Laptop, click on PC Wireless.
2. On the PC Wireless page, click on Connect tab and click on refresh button.
3. You should see HomeSSID with Channel 6, click on it and select Connect button.
4. Under the WPA2-Personal authentication, enter the Pre-shared key as **Pass1234** and click connect. If you click on Link Information tab, you should see a confirmation of successful connection.

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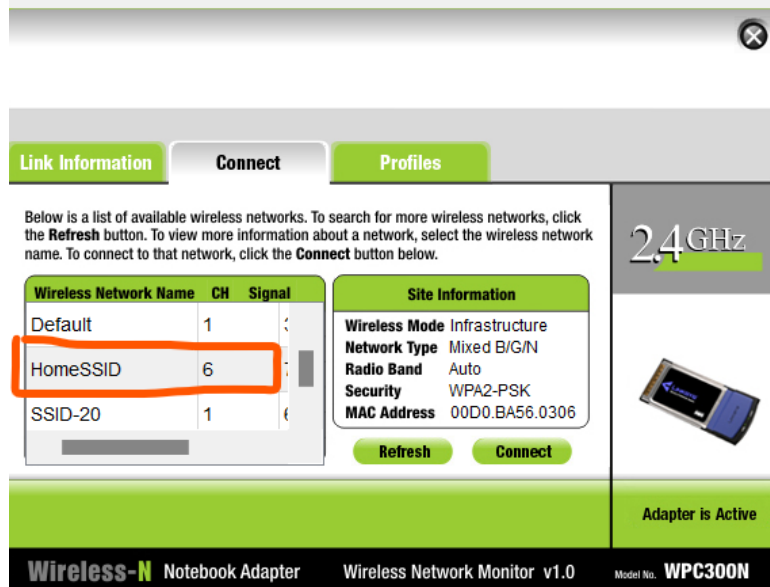


Figure 18.11: Home Laptop setup

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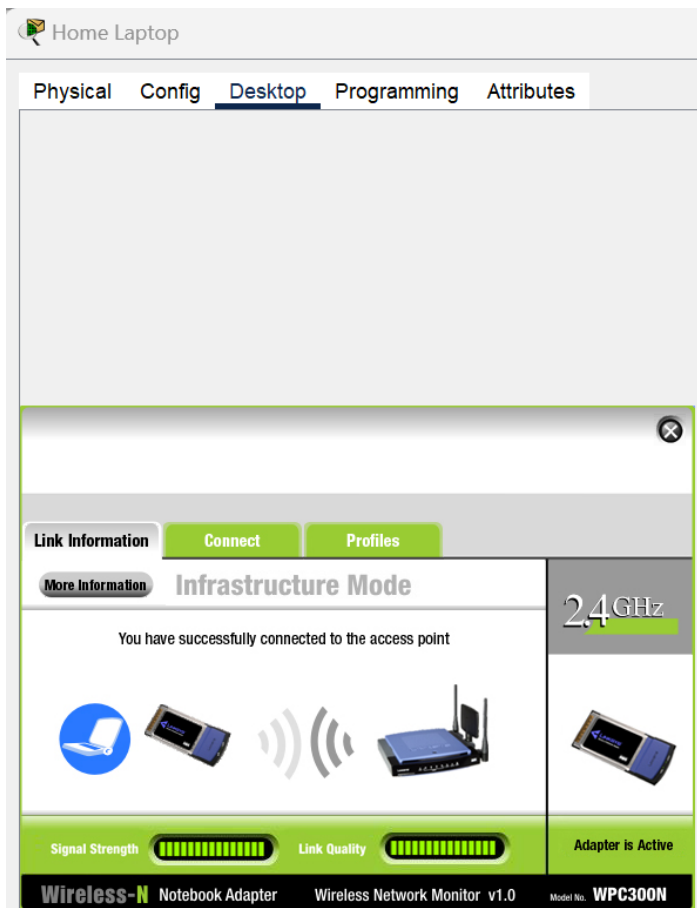


Figure 18.12: Home Laptop setup

Step 7: Verify connectivity by accessing the web page on Web Server from Home Laptop.

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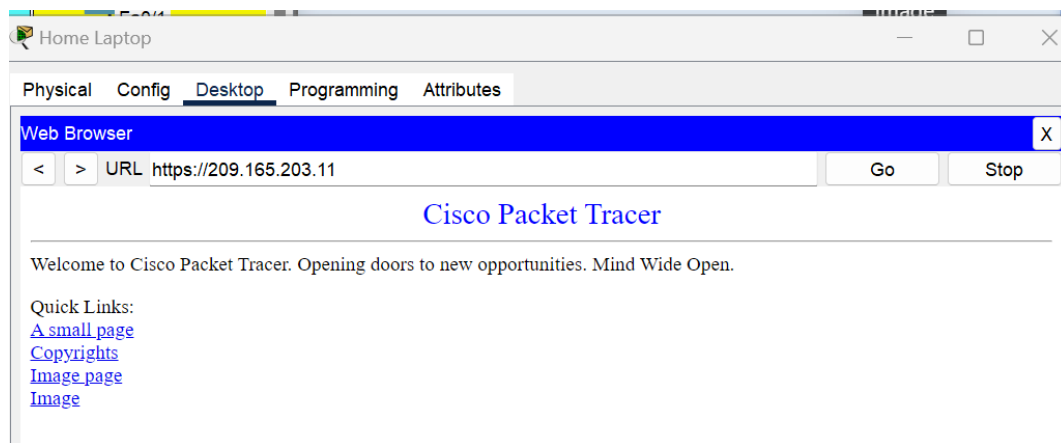


Figure 18.12: Home Laptop

Part 3: Corporate Network Configuration.

Step 1: Corp-R1 configuration:

1. configure interfaces with ipv4 addresses.
2. configure a default static route so that internal Corp users can send and receive traffic from external networks.
3. configure DHCP service for VLAN 10 (WLAN-10) users.
4. configure DHCP service for VLAN 20 (WLAN-20) users.
5. configure DHCP service for VLAN 100 (management/native VLAN) users.
6. configure sub-interfaces on the physical interface G0/0/0 to provide inter-VLAN routing.

```
CorpRouter#show run
```

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```
hostname CorpRouter

ip dhcp pool WLAN10

network 192.168.10.0 255.255.255.0

default-router 192.168.10.1

dns-server 209.165.203.2

ip dhcp pool WLAN20

network 192.168.20.0 255.255.255.0

default-router 192.168.20.1

dns-server 209.165.203.2

interface GigabitEthernet0/0/0

description CONNECTION TO WLAN-NETWORK

no ip address

duplex auto

speed auto

!

interface GigabitEthernet0/0/0.10

encapsulation dot1Q 10

ip address 192.168.10.1 255.255.255.0

!

interface GigabitEthernet0/0/0.20
```

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```
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
!
interface GigabitEthernet0/0/0.100
encapsulation dot1Q 100 native
ip address 192.168.100.1 255.255.255.0
!
interface GigabitEthernet0/0/1
ip address 192.168.30.1 255.255.255.0
duplex auto
speed auto
interface Serial0/1/1
description CONNECTION TO ISP-RTB
ip address 209.165.200.225 255.255.255.252
!
router ospf 1
router-id 3.3.3.3
log-adjacency-changes
network 209.165.200.224 0.0.0.3 area 0
network 192.168.30.0 0.0.0.255 area 0
```

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```
network 192.168.10.0 0.0.0.255 area 0  
network 192.168.20.0 0.0.0.255 area 0  
network 192.168.100.0 0.0.0.255 area 0  
end  
CorpRouter#
```

Step 2: Configure Corp-MLS switch as shown in the out of the **show run** command below:

CorpMLS#show run

```
hostname CorpMLS  
ip routing  
interface FastEthernet0/1  
switchport trunk native vlan 100  
switchport trunk allowed vlan 10,20,30,100  
switchport trunk encapsulation dot1q  
switchport mode trunk  
switchport nonegotiate  
!  
interface FastEthernet0/2  
switchport trunk native vlan 100  
switchport trunk allowed vlan 10,20,30,100  
switchport trunk encapsulation dot1q
```

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CISCO

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```
switchport mode trunk
```

```
switchport nonegotiate
```

```
!
```

```
interface FastEthernet0/3
```

```
!
```

```
interface FastEthernet0/4
```

```
switchport access vlan 100
```

```
switchport mode access
```

```
switchport nonegotiate
```

```
!
```

```
interface FastEthernet0/5
```

```
switchport access vlan 100
```

```
switchport mode access
```

```
interface GigabitEthernet0/1
```

```
switchport trunk native vlan 100
```

```
switchport trunk allowed vlan 10,20,30,100
```

```
switchport trunk encapsulation dot1q
```

```
switchport mode trunk
```

```
!
```

```
!
```

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```
interface Vlan100
mac-address 0001.97da.0a01
ip address 192.168.100.100 255.255.255.0
!
ip default-gateway 192.168.100.1
ip classless
ip route 0.0.0.0 0.0.0.0 192.168.100.1
end
CorpMLS#
```

Part 4: Setup CorpRadius-Server

Requirements: Configure the Radius server to provide centralized authentication services to VLAN 20 (WLAN-20) network.

1. On the Desktop of CorpRadiusServer, go to Services tab and select AAA under SERVICES.
2. Enable AAA services.
3. Under the Network Configuration, enter Client Name as: **CorpWLC** and Client IP as: **192.168.100.254**.
4. Enter Secret as: **Pass1234** and Server Type as Radius.
5. Click the Add button.
6. Scroll down to the User Setup page.
7. Enter Username as **User1** and Password as **User1Pass**.

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8. See the image below.

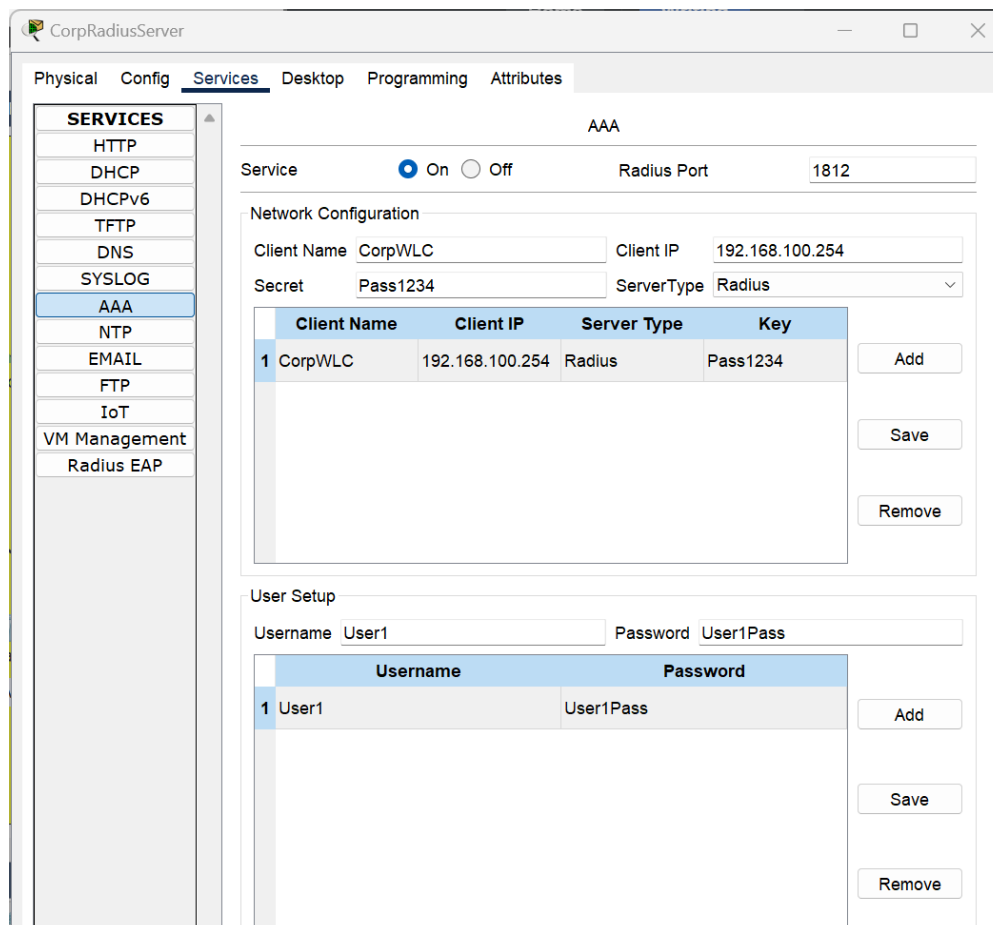


Figure 18.13: Radius Server setup

Part 5: WebServer

1. On WebServer computer, click on Services tab and select HTTP under SERVICES tab.
2. Enable HTTPS service on the server.

Part 6: CorpWLC

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Requirements.

1. Perform initial settings on the WLAN Controller.
2. Configure interfaces on the WLC to support the two WLANs. **Note:** Make sure the physical interface (port) number also corresponds to the interface number selected during configuration under the Controller tab.
3. Configure DHCP scope for the wireless management network.
4. Configure the WLC to redirect authentication requests to the external RADIUS server address.
5. Create the WLANs

Part1: Perform initial settings on the WLAN Controller.

Before you configure the required settings on WLAN, you will need to login to the WLAN and change the default settings.

Follow the steps below to complete these initial settings.

Step 1: Connect Corp-PC to the Wireless LAN Controller using straight-through cable.

Step 2: From the Desktop of Corp-PC click on IP Configuration and enter the IP address as 192.168.1.2 and Subnet Mask as: 255.255.255.240.

Step 3: From the Command prompt of Corp-PC, ping the IP address of the Wireless Controller: **192.168.1.1.**

Step 4: From the Desktop of Corp-PC click on Web Browser and enter 192.168.1.1 into the URL. Press enters.

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Step 5: On the Login page, create username as **admin** and password as **Class123**. Then click the Start button.

Step 6: On the Setup page, enter the following information:

- System Name: Wireless LAN Controller
- Management IP Address: **192.168.100.254**.
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.100.1
- Leave the Management VLAN ID as **0**.

Step 7: On the next page, enter the following:

- Network Name: Management
- Security: WPA2 Personal
- Passphrase Class123

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Physical Config Desktop Programming Attributes

Web Browser

< > URL http://192.168.1.1 Go

Cisco 3500 Series Wireless LAN Controller

1 Set Up Your Controller

System Name Wireless LAN Controller.. ?

Country United States (US) ?

Date & Time 01/09/2024 17:32:34

Timezone Eastern Time (US and Canada) ?

NTP Server (optional) ?

Management IP Address 192.168.100.254 ?

Subnet Mask 255.255.255.0

Default Gateway 192.168.100.1

Management VLAN ID 0 ?

Back Next

2 Configure Wireless Network ?

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Web Browser

URL http://192.168.1.1 Go

Cisco 3500 Series Wireless LAN Controller

- 1 Set Up Your Controller ✓
- 2 Create Your Wireless Networks

☒ Employee Network

Network Name

Security

Passphrase

Confirm Passphrase

VLAN

DHCP Server Address

☐ Guest Network

Back Next

Step 8: Click next and leave everything as it is.

Step 9: On the next page, confirm that everything looks good and then scroll down to Apply button. Wait for two to three minutes to allow the settings to be changed.

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Web Browser

URL: <http://192.168.1.1> Go

Cisco 3500 Series Wireless LAN Controller

Please confirm settings and apply

1 Controller Settings

Username	admin
System Name	Wireless LAN Controller..
Country	United States (US)
Date & Time	01/09/2024 17:36:42
Timezone	Eastern Time (US and Canada)
NTP Server	-
Management IP Address	192.168.100.254
Management IP Subnet	255.255.255.0
Management IP Gateway	192.168.100.1
Management VLAN ID	0

2 Wireless Network Settings

✓ Employee Network

Network Name	Management
Security	WPA2 Personal
Passphrase	*****
Employee VLAN	Management VLAN
DHCP Server Address	-

✗ Guest Network

3 Advanced Settings

Step 10: You can now close the Browser page to return back to Desktop.

Part 2: Configure interfaces on the WLC to support the two WLANs. **Note:** Make sure the physical interface (port) number also correspond to the interface number selected during configuration under the Controller tab.

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Step 1: Change the IP address of the Corp-PC to **192.168.100.10**, to match the subnet range of IP of the WLAN Controller. See image below.

The screenshot shows the "IP Configuration" window for the "FastEthernet0" interface. The "Static" option is selected under "IP Configuration". The fields are filled with the following values:

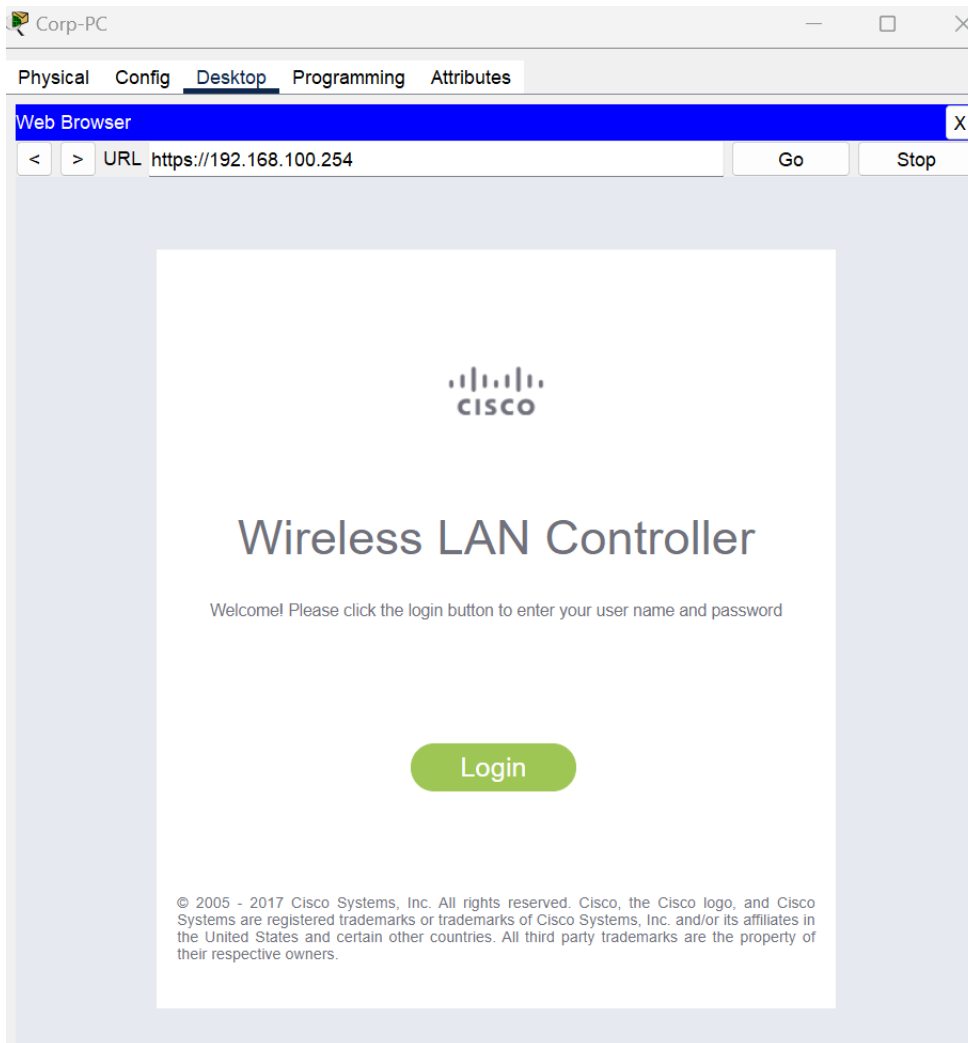
Field	Value
IPv4 Address	192.168.100.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.100.1
DNS Server	0.0.0.0

Step 2: From the Corp-PC Desktop, open the Browser and enter the below information:

https://192.168.100.254

Then click Login and use **admin** as the username and **Class123** as the password.

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Step 3: Configure an interface for the first WLAN (WLAN 10).

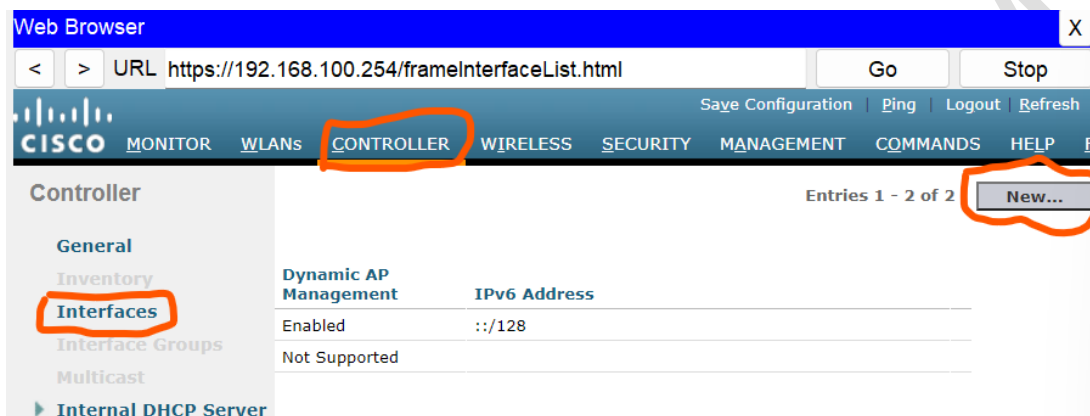
Note: Each WLAN requires a virtual interface on the WLC. These interfaces are known as dynamic interfaces. The virtual interface is assigned a VLAN ID and traffic that uses the interface will be tagged as VLAN traffic. This is why connections between the APs, the WLC, and the router

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are over trunk ports. For the traffic from multiple WLANs to be transported through the network, traffic for the WLAN VLANs must be trunked.

1. Click on CONTROLLER tab. Select Interfaces and click on New. See image below.



On the next screen, enter Interface Name as WLAN 10 and VLAN Identifier:10

Then click Apply.



On the Interfaces Edit page, enter the below information. Then click **Apply** to save changes. See image for explanation.

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Port Number: **1**

VLAN Identifier: **10**

Interface IP Address: **192.168.10.254**

Netmask: **255.255.255.0**

Gateway: **Corp-Router G0/0/0.10 address (192.168.10.1)**

Primary DHCP Server: **Gateway address (192.168.10.1)**

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Web Browser

URL <https://192.168.100.254/frameInterfaceEdit.html> Go Stop

Save Configuration Ping Logout Refresh

CISCO MONITOR WLANs CONTROLLER WIRELESS SECURITY MANAGEMENT COMMANDS HELP

Controller

- General
- Inventory
- Interfaces
- Interface Groups
- Multicast
- Internal DHCP Server
- Mobility Management
- Ports
- NTP
- CDP
- Tunneling
- IPv6
- mDNS
- Advanced

General Information

Interface Name: WLAN 10

MAC Address: 00:03:E4:95:87:9D

Configuration

Guest Lan ☐

Quarantine ☐

Quarantine Vlan Id: 0

NAS-ID:

Physical Information

Port Number: 1

Backup Port: 0

Active Port: 0

Enable Dynamic AP Management ☐

Interface Address

VLAN Identifier: 10

IP Address: 192.168.10.254

Netmask: 255.255.255.0

Gateway: 192.168.10.1

DHCP Information

Primary DHCP Server: 192.168.10.1

Step 4: Configure an interface for the second **WLAN20**.

On the Interfaces Edit page, click the Back button to return back to Interfaces. You should now see the newly created **WLAN 10** Interface.

If you want to make changes, just click on WLAN 10 and make necessary changes. Remember to click the Apply button to save changes.

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Click the Interfaces and click New to create a second interface.

On the next screen, enter Interface Name as **WLAN 20** and VLAN Identifier: **20**.

Then click Apply.

A screenshot of a web browser window displaying the Cisco Controller configuration page. The browser's address bar shows the URL 'https://192.168.100.254/frameInterfaceCreate.html'. The page has a Cisco logo and navigation tabs for MONITOR, WLANS, CONTROLLER, WIRELESS, SECURITY, MANAGEMENT, COMMANDS, and HELP. On the left, a sidebar shows 'Controller' with sub-links for General, Inventory, and Interfaces. The main content area is titled 'Interfaces > New' and contains two input fields: 'Interface Name' with the value 'WLAN 20' and 'VLAN Id' with the value '20'. There are '< BACK' and 'Apply' buttons at the bottom right of the form.

On the Interfaces Edit page, enter the below information. Then click **Apply** to save changes. See image for explanation.

Port Number: **1**

VLAN Identifier: **20**

Interface IP Address: **192.168.20.254**

Netmask: **255.255.255.0**

Gateway: **Corp-Router G0/0/0.20 address (192.168.20.1)**

Primary DHCP Server: **Gateway address (192.168.20.1)**

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Save Configuration | Ping | Logout | Refresh

MONITOR | WLANS | **CONTROLLER** | WIRELESS | SECURITY | MANAGEMENT | COMMANDS | HELP

Controller

General
Inventory
Interfaces
Interface Groups
Multicast
Internal DHCP Server
Mobility Management
Ports
NTP
CDP
Tunneling
IPv6
mDNS
Advanced

Interfaces > Edit

< BACK | Apply

General Information

Interface Name	WLAN 20
MAC Address	00:02:4A:29:CA:92

Configuration

Guest Lan	☐
Quarantine	☐
Quarantine Vlan Id	0
NAS-ID	

Physical Information

Port Number	1
Backup Port	0
Active Port	1
Enable Dynamic AP Management	☐

Interface Address

VLAN Identifier	20
IP Address	192.168.20.254
Netmask	255.255.255.0
Gateway	192.168.20.1

DHCP Information

Primary DHCP Server	192.168.20.1
---------------------	--------------

Note: On the Interfaces Edit page, click the Back button to return back to Interfaces. You should now see the newly created **WLAN 10** and **WLAN 20** Interfaces.

If you want to make changes, just click on WLAN 10 or WLAN 20 and make the necessary changes. Remember to click the Apply button to save changes.

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Web Browser

URL: <https://192.168.100.254/frameInterfaceList.html> Go Stop

CISCO MONITOR WLANS **CONTROLLER** WIRELESS SECURITY MANAGEMENT COMMANDS HELP

Controller

- General
- Inventory
- Interfaces**
- Interface Groups
- Multicast
- Internal DHCP Server
- Mobility Management

Interfaces

Interface Name	VLAN Identifier	IP Address	Interface Type
WLAN 10	10	192.168.10.254	Dynamic
WLAN 20	20	192.168.20.254	Dynamic
management	untagged	192.168.100.254	Static
virtual	N/A	192.0.2.1	Static

Part 3: Create the WLANs. Configure the WLAN to use the Personal authentication with Pre-shared Key.

Step 1: Create the first WLAN. Use the information below to create the first WLAN network. See the steps explained below.

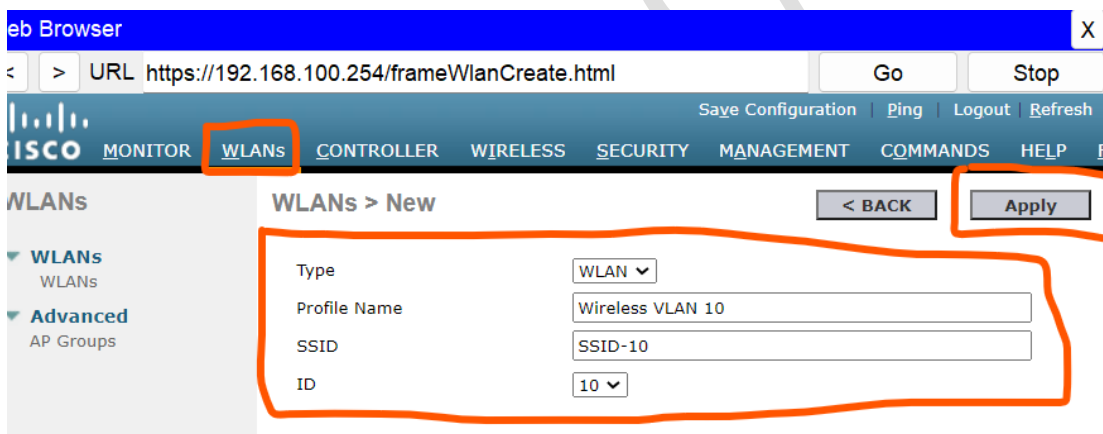
- Profile Name: **Wireless VLAN 10**
- WLAN SSID: **SSID-10**
- ID: **10**
- Interface: **WLAN 10**
- Security: **WPA2-PSK**
- Passphrase: **Pass1234**
- Under the Advanced tab, go to the FlexConnect section. Enable **FlexConnect Local Switching** and **FlexConnect Local Auth**.

1. Click on **WLANS** tab and select **Create New** and click **Go**.

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2. On the WLANs>New page, enter the information provided above and click Apply. See images for explanation.



3. On the next page, Click on the Enabled Status. Under the Interface/Interface Group, use the drop-down arrow to select WLAN 10. Click Apply.

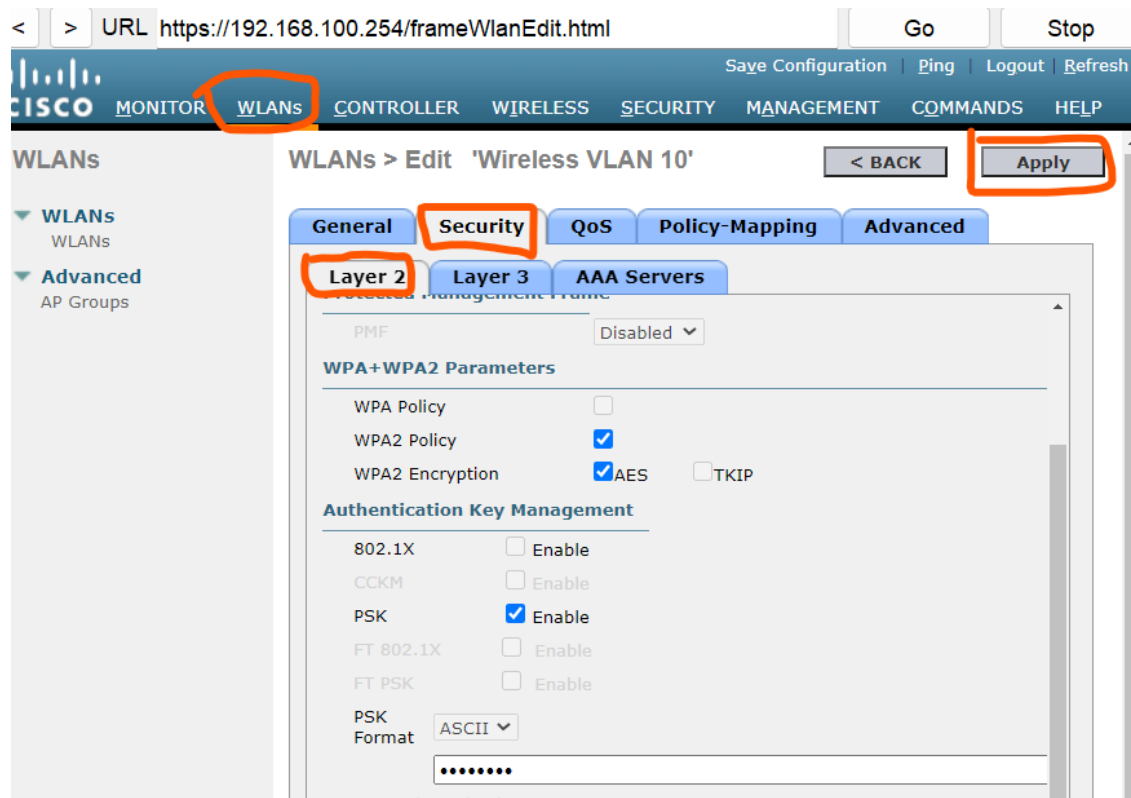
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The screenshot shows the Cisco configuration interface for "Wireless VLAN 10". The left sidebar has "WLANs" and "Advanced" sections. The main area has tabs for "General", "Security", "QoS", "Policy-Mapping", and "Advanced". The "General" tab is active, showing fields for "Profile Name" (Wireless VLAN 10), "Type" (WLAN), "SSID" (SSID-10), and "Status" (Enabled). Below these are "Security Policies" (None), "Radio Policy" (All), "Interface/Interface Group(G)" (WLAN 10), and "Multicast Vlan" (Enabled).

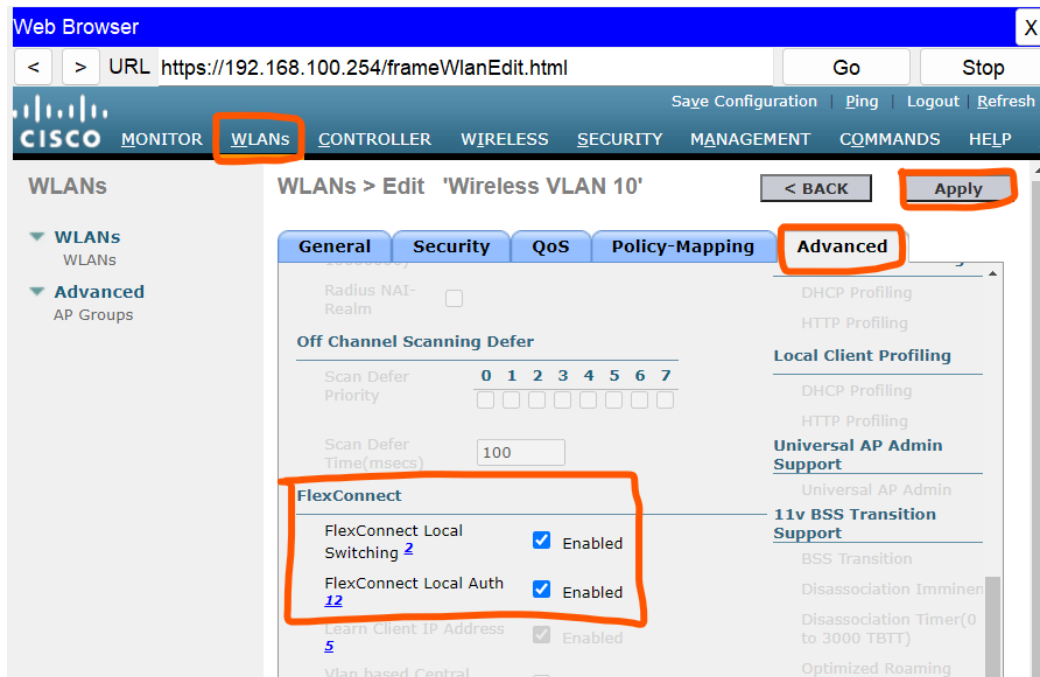
4. On the WLANs Edit page, click on Security tab and select Layer 2. Under Layer 2 Security, use the drop down to select WPA + WPA2. Under WPA + WPA2 Parameters, tick the boxes of WAP2 Policy and WPA2 Encryption. Scroll down to under Authentication Key Management and click Enable PSK (Pre-shared Key). Enter Passphrase as **Pass1234**. Then click Apply.

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5. Under the Advanced tab, scroll down to the FlexConnect section. Enable **FlexConnect Local Switching** and **FlexConnect Local Auth**.

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Part 4: Create the second WLAN. Configure the WLAN to use the RADIUS server for authentication. Use the information provided below to create the second WLAN network.

Step 1:

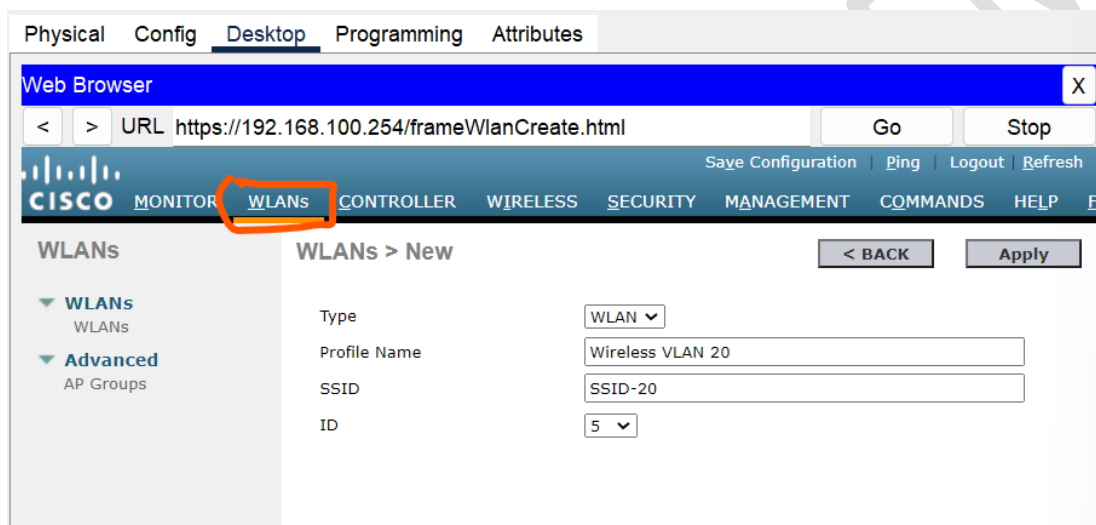
Follow the steps explained below to complete this part.

- Profile Name: **Wireless VLAN 20**
- WLAN SSID: **SSID-20**
- Interface: **WLAN 20**
- ID: **5**
- Security: **802.1x - WPA2-Enterprise**
- Make the **FlexConnect** settings as was done in Step above.

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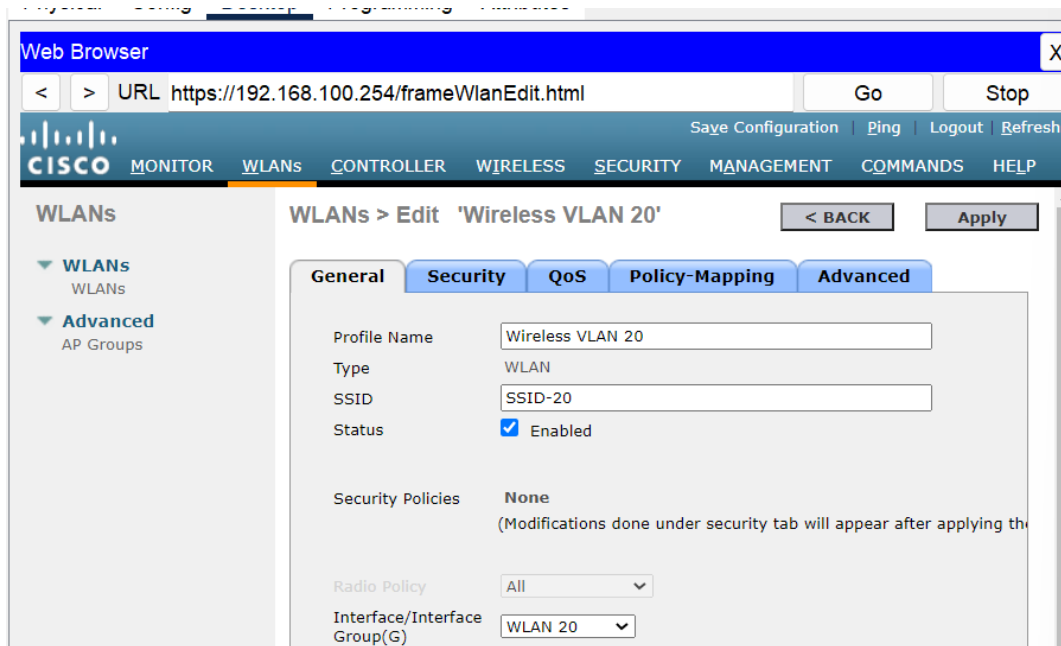


- Configure the WLAN to use the RADIUS server for authentication.
 - Radius Server IP address: **192.168.30.252**.
1. Click on **WLANs** tab and select **Create New** and click **Go**.
 2. On the WLANs>New page, enter the information provided above and click Apply. See images for explanation.



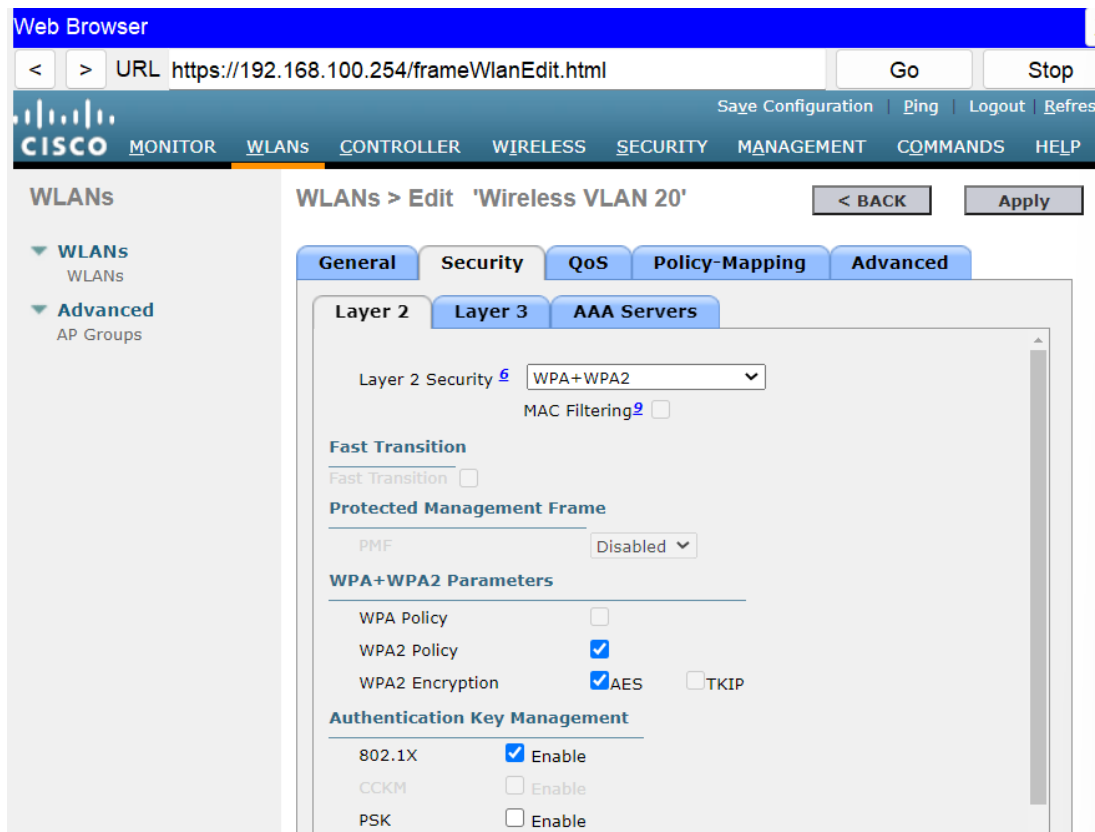
On the next page, Click on the Enabled Status. Under the Interface/Interface Group, use the drop-down arrow to select WLAN 20. Click **Apply**.

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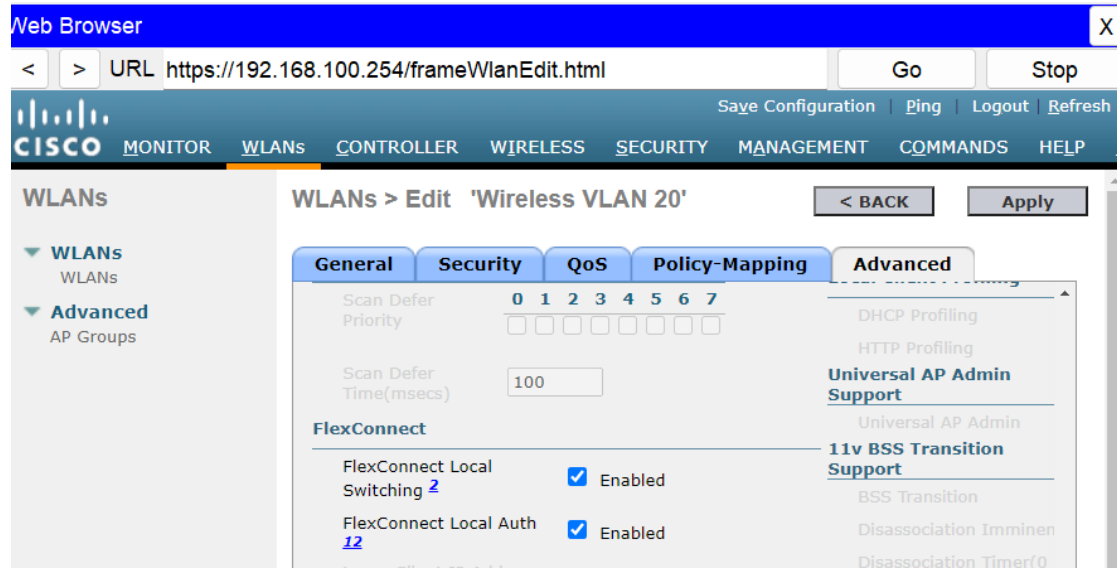
On the WLANs Edit page, click on Security tab and select Layer 2. Under Layer 2 Security, use the drop down to select WPA + WPA2. Under WPA + WPA2 Parameters, tick the boxes of WPA2 Policy and WPA2 Encryption. Scroll down to under Authentication Key Management and click Enable **802.1X**. Then click Apply.

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Under the Advanced tab, scroll down to the FlexConnect section. Enable **FlexConnect Local Switching** and **FlexConnect Local Auth**.

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Part 5: Configure the WLAN to use the RADIUS server for authentication. WPA2-Note: Enterprise uses an external RADIUS server to authenticate WLAN users. Individual user accounts with unique usernames and passwords can be configured on the RADIUS server. Before the WLC can use the services of the RADIUS server, the WLC must be configured with the server address.

1. Click the **Security** menu on the WLC.
2. Click the **New** button and enter the IP address of the RADIUS server (**192.168.30.252**) in the Server IP Address field.
3. The RADIUS server will authenticate the WLC before it will allow the WLC to access the user account information that is on the server. This requires a shared secret value. Use **Class123**. Confirm the shared secret and click **Apply**.

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Web Browser

URL <https://192.168.100.254/frameRadiusList.html> Go Stop

CISCO MONITOR WLANs CONTROLLER WIRELESS **SECURITY** MANAGEMENT COMMANDS HELP

Security Authentication Servers Apply **New...**

AAA
General
RADIUS
Authentication
Accounting
Fallback
DNS
Downloaded AVP
TACACS+
LDAP
Local Net Users
MAC Filtering

Authentication Servers

Selected Station ID IP Address

Key Wrap ☐ (Designed for FIPS customers and requires a key wrap compliant RADIUS server)

Delimiter Hyphen

MTU 1300

Management	Server Index	Server Address(Ipv4/Ipv6)	Port	IPSec	Admin Status
------------	--------------	---------------------------	------	-------	--------------

Web Browser

URL <https://192.168.100.254/frameRadiusCreate.html> Go Stop

CISCO MONITOR WLANs CONTROLLER WIRELESS **SECURITY** MANAGEMENT COMMANDS HELP

Security RADIUS Authentication Servers > New < BACK Apply

AAA
General
RADIUS
Authentication
Accounting
Fallback
DNS
Downloaded AVP
TACACS+
LDAP
Local Net Users
MAC Filtering
Disabled Clients

RADIUS Authentication Servers > New

Server Index (Priority) 1

Server IP Address(Ipv4/Ipv6) 192.168.30.252

Shared Secret Format ASCII

Shared Secret

Confirm Shared Secret

Key Wrap ☐ (Designed for FIPS customers and requires a key wrap compliant RADIUS server)

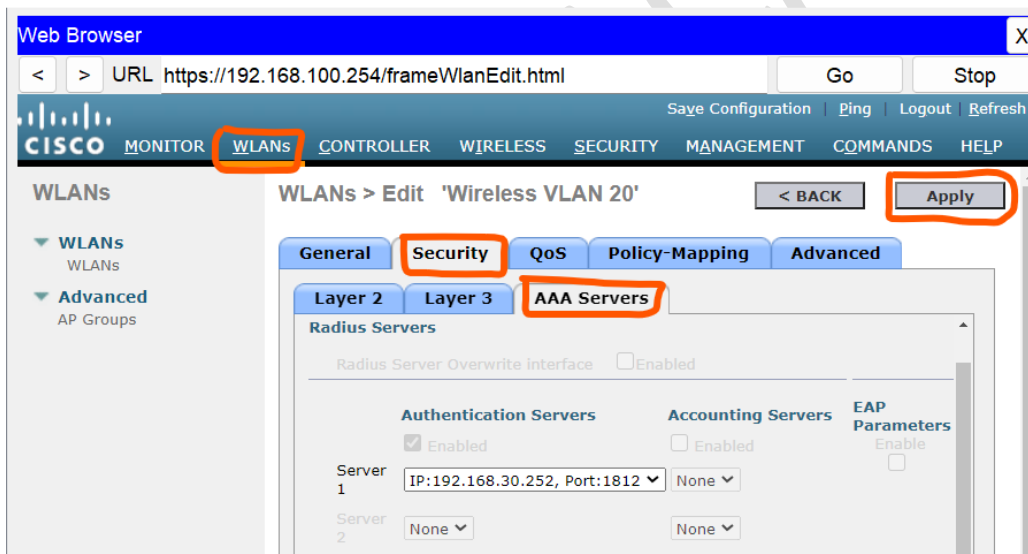
Port Number 1812

- Click the WLANs tab, click the WLAN ID 5 of the newly created WLAN to continue configuring it, if necessary.
- Under the WLANs>Edit tab, click the Security tab and select AAA Servers.
- Click the AAA Servers tab. Open the drop-down next to Server 1 in the Authentication Servers column and select the server that we configured in Step 2.

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- Click **Apply** to enact this configuration. You have now configured the WLC to use the RADIUS server to authenticate users that attempt to connect to the WLAN.



Part 6: Configure a DHCP scope for the wireless management network.

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The WLC offers its own internal DHCP server. Cisco recommends that the WLAN DHCP server not be used for high-volume DHCP services, such as that required by larger user WLANs. However, in smaller networks, the DHCP server can be used to provide IP addresses to LAPs that are connected to the wired management network. In this step, we will configure a DHCP scope on the WLC and use it to address LAP-1.

Requirement: Configure and enable an internal DHCP scope as follows:

Scope Name: **Management**

Pool Start Address: **192.168.100.235**

Pool End Address: **192.168.100.245**

Network: **192.168.100.0**

Netmask: **255.255.255.0**

Default Routers: **192.168.100.1**

DNS Server: **Optional**

Status: **Enabled**

Step 1: Click the **Controller** menu and then click **Interfaces**.

Step 2: Click the **management** Interface. Record its IP addressing information here.

IP Address: 192.168.100.254

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Netmask: 255.255.255.0

Gateway: 192.168.100.1

Step 3: We want the WLC to use its own DHCP server to provide addressing to devices on the wireless management network, such as lightweight APs. For this reason, enter the IP address of the WLC management interface as the primary DHCP server address. Click **Apply**. Click **OK** to acknowledge any warning messages that appear.

The screenshot shows a web browser window with the URL <https://192.168.100.254/frameInterfaceList.html>. The Cisco WLC interface has a navigation menu on the left with "CONTROLLER" selected. Under "Controller", the "Interfaces" link is highlighted. The main area displays a table of interfaces:

Interface Name	VLAN Identifier	IP Address	Interface Type
WLAN_10	10	192.168.10.254	Dynamic
WLAN_20	20	192.168.20.254	Dynamic
management	untagged	192.168.100.254	Static
virtual	N/A	192.0.2.1	Static

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The screenshot shows the Cisco Controller configuration page for the NAT Address section. The left-hand menu is expanded to "Internal DHCP Server". The main content area shows the following configuration:

- NAT Address**
 - Enable NAT Address: ☐
- Physical Information**
 - Port Number: 1
 - Backup Port: 0
 - Active Port: 1
 - Enable Dynamic AP Management: ☒
- Interface Address**
 - VLAN Identifier: 0
 - IP Address: 192.168.100.254
 - Netmask: 255.255.255.0
 - Gateway: 192.168.100.1
- DHCP Information**
 - Primary DHCP Server: 192.168.100.254

Step 4: In the left-hand menu, expand the **Internal DHCP Server** section. Click **DHCP Scope**.

To create a DHCP scope, click the **New...** button.

Name the scope **Management**. You will configure this DHCP scope to provide addresses to the wired infrastructure network that connects the Corp-PC, WLC, and CorpLAP1.

Click **Apply** to create the new DHCP scope.

Click the new scope in the DHCP Scopes table to configure addressing information for the scope.

Enter the information provided above.

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Click **Apply** to activate the configuration. Click **Save Configuration** in the upper-right-hand corner of the WLC interface to save your work so that it is available when the WLC restarts.

Two screenshots of the Cisco WLC web interface. The top screenshot shows the "Controller" page with the "Internal DHCP Server" option highlighted in the left sidebar and a "New..." button in the top right. The bottom screenshot shows the "DHCP Scope > New" page with a text field for "Scope Name" containing "Management" and an "Apply" button highlighted in the top right.

Web Browser

URL: <https://192.168.100.254/frameDhcpScopeList.html>

Go Stop

Save Configuration | Ping | Logout | Refresh

CISCO MONITOR WLANS CONTROLLER WIRELESS SECURITY MANAGEMENT COMMANDS HELP

Controller

General

Inventory

Interfaces

Interface Groups

Multicast

Internal DHCP Server

Mobility Management

New...

Address Pool	Lease Time	Status
192.168.1.3 - 192.168.1.14		Enabled Remove

Web Browser

URL: <https://192.168.100.254/frameDhcpScopeCreate.html>

Go Stop

Save Configuration | Ping | Logout | Refresh

CISCO MONITOR WLANS CONTROLLER WIRELESS SECURITY MANAGEMENT COMMANDS HELP

Controller

DHCP Scope > New

< Back Apply

General

Inventory

Interfaces

Interface Groups

Multicast

Internal DHCP Server

Scope Name: Management

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Web Browser

URL <https://192.168.100.254/frameDhcpScopeList.html> Go Stop

CISCO MONITOR WLANs **CONTROLLER** WIRELESS SECURITY MANAGEMENT COMMANDS HELP

Controller

General
Inventory
Interfaces
Interface Groups
Multicast
Internal DHCP Server

DHCP Scopes

Scope Name	Address Pool	Lease Time
Management	0.0.0.0 - 0.0.0.0	
day0-dhcp-mgmt	192.168.1.3 - 192.168.1.14	

Web Browser

URL <https://192.168.100.254/frameDhcpScopeEdit.html> Go Stop

CISCO MONITOR WLANs **CONTROLLER** WIRELESS SECURITY MANAGEMENT COMMANDS HELP

Controller

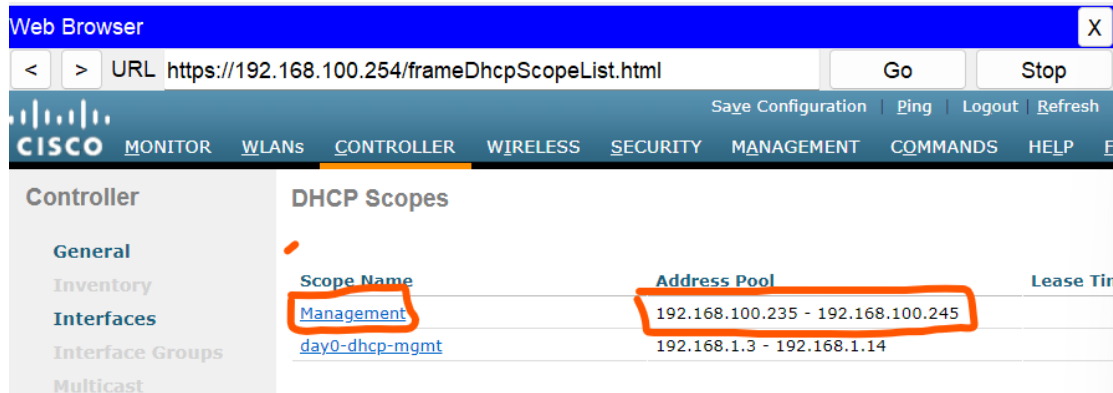
General
Inventory
Interfaces
Interface Groups
Multicast
Internal DHCP Server
Mobility Management
Ports
NTP
CDP
Tunneling
IPv6
mDNS
Advanced

DHCP Scope > Edit

< Back

Scope Name	Management		
Pool Start Address	192.168.100.235		
Pool End Address	192.168.100.245		
Network	192.168.100.0		
Netmask	255.255.255.0		
Lease Time (seconds)	86400		
Default Routers	192.168.100.1	0.0.0.0	0.0.0.0
DNS Domain Name	Not Supported		
DNS Servers	0.0.0.0	0.0.0.0	0.0.0.0
Netbios Name Servers	0.0.0.0	0.0.0.0	0.0.0.0
Status	Enabled		

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Note: The internal DHCP server will now provide an address to CorpLAP1 after a brief delay. When CorpLAP1 has its IP address, the CAPWAP tunnel will be established, and CorpLAP1 will be able to provide access to the Wireless LAN 20 (SSID-20) WLAN. If you move the mouse over CorpLAP1 in the topology, you should see its IP address, the status of the CAPWAP tunnel, and the WLAN that CorpLAP1 is providing access to.

Part 7: Configure the hosts to connect to the WLANs.

Note: In the Packet Tracer PC Wireless client app, you must configure a WLAN Profile in order to attach to a WPA2-Enterprise WLAN.

Remember to change the Laptops NIC to wireless NICs as explained in **Chapter 12**.

Requirements: CorpLaptop1 should connect to WLAN VLAN 10 and CorpLaptop2 should connect to WLAN VLAN 20.

Use the desktop PC Wireless app to configure the hosts as follows:

Step 1: Click CorpLaptop2 Host and open the **PC Wireless** app.

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Click the **Profiles** tab and then click **New** to create a new profile. Name the profile **WLAN-20-NET**.

Highlight the Wireless Network Name (**SSID-20**) for the WLAN that we created earlier and click **Advanced Setup**.

Verify that the SSID for the wireless LAN is present and then click **Next**. Wireless Host should see **SSID-20**. If it does not, move the mouse over the **CorpLAP1** to verify that it is communicating with the WLC. The popup box should indicate that **CorpLAP1** is aware of SSID-20. If it is not, check the WLC configuration. You can also manually enter the SSID.

Verify that the DHCP network setting is selected and click **Next**.

In the Security drop down box, select **WPA2-Enterprise**. Click **Next**.

Enter login name **User1** and the password **User1Pass** and click **Next**.

Verify the Profile Settings and click **Save**.

Select the **WLAN-20-NET** profile and click the **Connect to Network** button. After a brief delay, you should see the Wireless Host connect to **CorpLaptop2**.

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Corp Laptop2

Physical Config Desktop Programming Attributes

Link Information

Connect

Profiles

More Information

Infrastructure Mode

You have successfully connected to the access point



2.4GHz

Signal Strength



Link Quality



Adapter is Active

Wireless-N

Notebook Adapter

Wireless Network Monitor v1.0

Model No. WPC300N

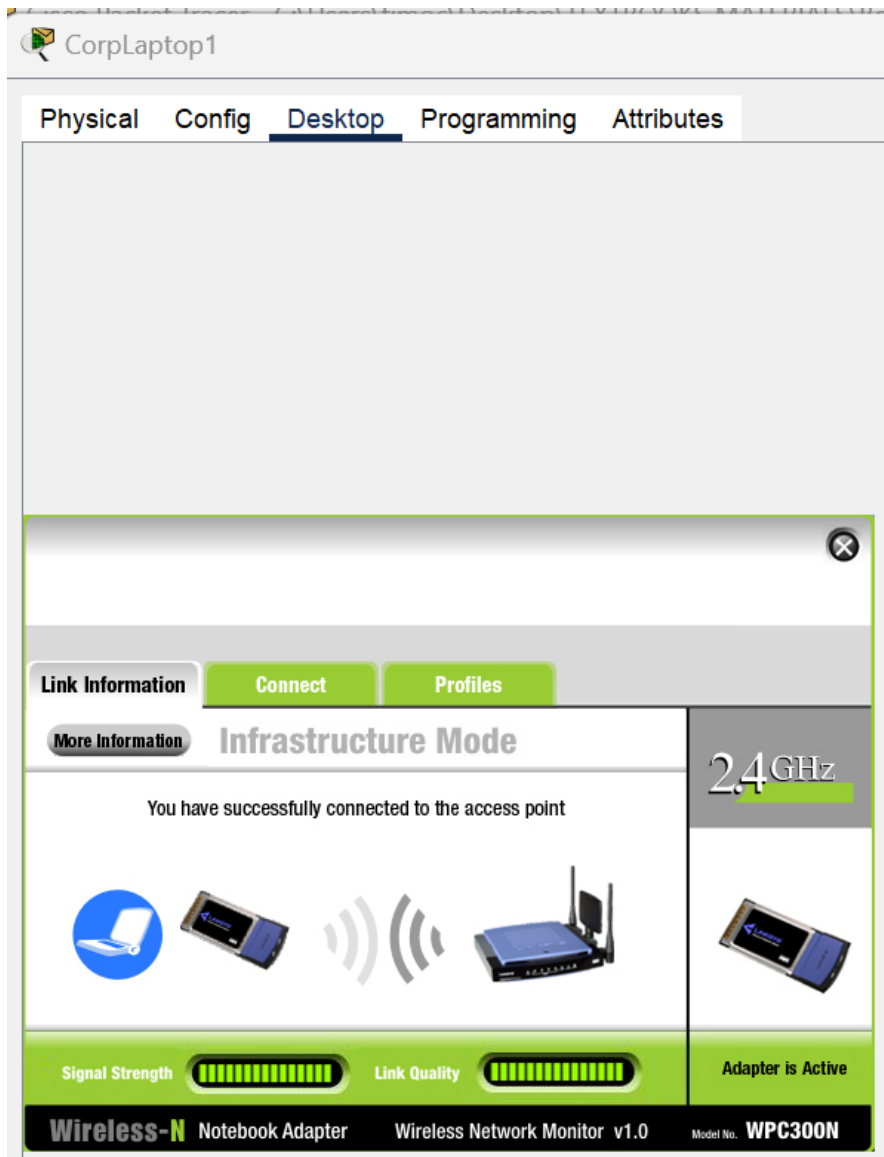
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Step 2: CorpLaptop1 will connect to Wireless WLAN 10 VLAN 10 using the credentials in the WLAN information table. CorpLaptop1 will use Personal authentication with Pre-shared Key **Pass1234**.

1. Click CorpLaptop1 Host and open the **PC Wireless** app.
2. Click Connet tab and click Refresh button.
3. On the list of Wireless Network Name, select SSID-10 and click Connect.
4. Under the WPA2-Personal, enter the Pre-shared Key as **Pass1234**.
5. Click Connect.

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Part 8: Verify that WLAN Controller is providing services to the Access Points and that Wireless Clients are connected to the Wireless networks.

Step 1: On Corp-PC, login back to WLAN Controller.

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Step 2: Click the MONITOR tab and select Access Points to the Left.

Step 3: Scroll down to the Access Point Summary, you should see the Total number of APs.

Step 4: Under the Client Summary, you should see the Total number of connected Clients.

Corp-PC

Physical Config **Desktop** Programming Attributes

Web Browser URL: <https://192.168.100.254/frameMonitor.html> Go Stop

CISCO MONITOR WLANS CONTROLLER WIRELESS SECURITY MANAGEMENT COMMANDS HELP FEEDBACK

Monitor

Summary

- Access Points** (highlighted)
- Radios
- Cisco CleanAir
- Statistics
- CDP
- Rogues
- Clients
- Sleeping Clients
- Multicast
- Applications
- Local Profiling

Up Time: 21 hours, 20 minutes, 4 seconds
System Time: Mon Nov 27 00:11:31 2023
Redundancy Mode: N/A
Internal Temperature: +31 C
802.11a Network State: Enabled
802.11b/g Network State: Enabled
Local Mobility Group: CPU(s) Usage: 0%
Individual CPU Usage: 0%/1%, 0%/0%
Memory Usage: 46%
Fan Status: 3800 rpm

Top WLANs

Profile Name # of Clients

Most Recent Traps

[View All](#)

Top Applications

Application Name	Packet Count	Byte Count
View All		

Access Point Summary

	Total	Up	Down	
802.11a/n/ac Radios	2	2	0	Detail
802.11b/g/n Radios	2	2	0	Detail
Dual-Band Radios	0	0	0	Detail
All APs	2	2	0	Detail

Client Summary

Current Clients	4	Detail
Excluded Clients	0	Detail
Disabled Clients	0	Detail

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Part 10: Test connectivity between the wireless hosts and the Web Server and also to the Remote Home Wireless Network.

Step 1: On CorpLaptop1 ping the IP address of the CorpLaptop2 (in VLAN 20 WLAN 20) and the IP address of Web Server on the ISP Cloud network.

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.2: bytes=32 time=37ms TTL=127
Reply from 192.168.20.2: bytes=32 time=39ms TTL=127
Reply from 192.168.20.2: bytes=32 time=14ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 39ms, Average = 30ms

C:\>ping 209.165.203.11

Pinging 209.165.203.11 with 32 bytes of data:

Request timed out.
Reply from 209.165.203.11: bytes=32 time=48ms TTL=125
Reply from 209.165.203.11: bytes=32 time=18ms TTL=125
Reply from 209.165.203.11: bytes=32 time=64ms TTL=125

Ping statistics for 209.165.203.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 18ms, Maximum = 64ms, Average = 43ms
  
```

Step 2: On Home-Laptop located in the Remote Home Wireless network, ping the IP address of the Corp Laptop2 (located in WLAN 20 VLAN 20) and the IP address of Web Server on the ISP Cloud network.

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```
Home Laptop
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Reply from 192.168.20.3: bytes=32 time=75ms TTL=124
Reply from 192.168.20.3: bytes=32 time=10ms TTL=124
Reply from 192.168.20.3: bytes=32 time=8ms TTL=124
Reply from 192.168.20.3: bytes=32 time=47ms TTL=124

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 75ms, Average = 35ms

C:\>ping 209.165.203.11

Pinging 209.165.203.11 with 32 bytes of data:

Reply from 209.165.203.11: bytes=32 time=38ms TTL=126
Reply from 209.165.203.11: bytes=32 time=19ms TTL=126
Reply from 209.165.203.11: bytes=32 time=5ms TTL=126
Reply from 209.165.203.11: bytes=32 time=22ms TTL=126

Ping statistics for 209.165.203.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 38ms, Average = 21ms
```

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